



Periodic Technical Report Part B

WP 1: Coordination and management

T 1.1: Project administration and coordination



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Who We Are

	Participant Name	Short Name	Country Code	Logo
1	ENGINEERING - INGEGNERIA INFORMATICA SPA	ENG	IT	
2	ETHNICON METSOVION POLYTECHNION	NTUA	EL	
3	FUNDACION CARTIF	CARTIF	ES	
4	UNIVERSITA POLITECNICA DELLE MARCHE	UNIVPM	IT	
5	ARCELIK A.S.	ARC	TR	
6	COMSENSUS	COMSENSUS	SI	
7	OURPOWER ENERGIEGENOSSENSCHAFT SCE MIT BESCHRANKTER HAFTUNG	OURPOWER	AT	
8	GREEN-POINT 62 GMBH	GREENPOINT	AT	
9	BLUEPRINT ENERGY SOLUTIONS GMBH	BLUEPRINT	AT	
10	FAELLESBO	FAELLESBO	DK	
11	NEOGRID TECHNOLOGIES APS	NEOGRID	DK	
12	EUROPEAN GREEN CITIES APS	GREEN CITIES	DK	
13	HERON SINGLE MEMBER S.A. ENERGY SERVICES	HERON	ELX	



14	DOMX IDIOTIKI KEFALAIOUCHIKI ETAIREIA	DOM-X	ELX	
15	ISTITUTO PER SERVIZI DI RICOVERO E ASSISTENZA AGLI ANZIANI	ISRAA	IT	
16	DYNAME SRL	DYNAME	IT	
17	CENTRE INTERNACIONAL DE METODES NUMERICS EN ENGINYERIA	CIMNE	ES	
18	PRODUCTORA ELECTRICA URGELLENCA	PEUSA	ES	
19	ROMUR RENOVALBES SL	ROMUR	ES	
20	CENTRICA BUSINESS SOLUTIONS BELGIUM	CENTRICA	BE	
21	UNIVERSITATEA TEHNICA CLUJ- NAPOCA	TUC	RO	
22	SMART INNOVATION NORWAY AS	SIN	NO	
23	FONDAZIONE ICONS	ICONS	IT	



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List of Acronyms

EC	European Commission
BOT	Building Topology Ontology
DLT	Distributed Ledger Technology
DR	Demand Response
DSO	District System Operator
DT	Digital Twins
EC	European Commission
ECM	Energy Conservation Measure
EO	Expected Outcomes
ESCO	Energy Service Company
EV	Electric Vehicle
GA	General Assembly
HITL	Human-In-The-Loop
HLUC	High-Level Use Cases
IAM	Identity Access Management
ICCP	Intelligent Computer Communication and Processing
ICT	Information & Communication Technologies
IDS	International Data Spaces
IEEE	Institute of Electrical and Electronics Engineers
IEC	International Electrotechnical Commission
IEQ	Indoor Environmental Quality
IISA	Information, Intelligence, Systems and Applications
IP	Intellectual Property
IPR	Intellectual Property Rights
KER	Key Exploitable Result
KPI	Key Performance Indicator
LEI	Long-term Expected Impacts
LSTM	Long Short-Term Memory



ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NILM	Non-Intrusive Load Monitoring
NGSI	Next Generation Service Interfaces
OCPP	Open Charge Point Protocol
PEP	Privacy Enforcement Point
PV	PhotoVoltaics
RES	Renewable Energy Sources
RTO	Research and Technology Organization
SDOs	Standardisation Bodies
SWOT	Strengths, Weaknesses, Opportunities, and Threats
TE	Technology Enablers
TRL	Technology Readiness Level
TSO	Transmission System Operator
TSV	Thermal Sensation Vote
UC	Use Cases
WP	Work Package



Executive summary

This document follows the template of Part B of the official Periodic Technical report and will be uploaded as a PDF document on the EC portal together with all the information required by the EU portal at the end of an official reporting period. It is intended to be the technical part of the Periodic Project Report and provides an overview of the work carried out by the beneficiaries during the first 18 months of the DEDALUS project. It starts from an explanation of the progress towards the achievement of the project objectives and also reports the effort and the costs that have been spent in the first 18 months, from May 2023 to October 2024.

DEDALUS is sticking to the plan. A continuous monitoring of all the active tasks allowed to identify potential issues and delays so that proper actions have been undertaken to keep the project on track. As reported in the previous progress report (an intermediate report), a relevant delay affected the first 7 months of the project, but all the delays have been properly managed and fixed.

Four Plenary Project Meetings (three face to face meetings, including the kick-off, and one recent remote meeting) have been held as well as many audio conferences to discuss the progress, the technical issues, and the scope of the project. The project has achieved all the objectives planned for the reporting period.

In the following chapters, the work done in the first 18 months of the project will be described in detail. As a summary of the progress, we can list:

- a proper repository has been set up together with an effective communication tool (MS Teams).
- procedures for production of documentation (presentations, reports, minutes) have been defined in a proper Project management plan (D1.1).
- periodic audio conferences and 4 plenary meetings (the kick-off in Athens, two meetings in presence in Rome and again in Athens, one remote meeting) allowed to monitor the advancement of all the running activities.
- A technical meeting in presence was held to fix many issues that were blocking some technical activities.
- The Data Management Plan (D1.2) was submitted at the end of month 10.
- An intermediate progress report, namely D1.3 DEDALUS first activity report, released as per GA.
- The transdisciplinary advisory board was set up even though not formalized yet with an NDA.
- a preliminary analysis of use cases was done for each pilot site using a bottom-up approach, identifying high-level requirements.
- 11 High-Level Use Cases (HLUCs) have been defined, and a detailed gap analysis of individual pilot sites was performed. Co-creation workshops and bilateral calls were held with pilot site partners and technical partners to ensure effective input from end-users.
- cocreation activities were designed to foster interaction between project partners and end-users, and defined user and technical requirements through questionnaires.



- Development of a set of technology enablers that will facilitate the creation of the value-added services within WP4. As a result, the first technology release of those technology enablers was delivered by the end of May 2024. The second version is under development.
- Multiple publications have been produced to describe the various achievements of this work package.
- An analysis of Technologies Enablers was done with the aim to understand how the current version of available software can be extended in the context of the Project according to the functional specification coming from the WP2.
- The first wave of DEDALUS Digital Twins and data-driven services for residential DR were released.
- The second wave of Digital Twin and data-driven services development is ongoing.
- The foundation for demonstration activities has been set up. This included evaluation of pilot infrastructure readiness for proper demonstration, understanding of the available datasets and the specific requirements of each pilot.
- A centralized data repository was created to consolidate sample datasets and documentation for API access.
- The evaluation methodology and baseline analysis were developed to assess the technological feasibility, user engagement, economic viability, and environmental impact.
- KPIs were defined.
- WP5 transitioned into the pre-pilot phase focused on finalising both physical and digital infrastructure updates across pilot sites and initiating the deployment and evaluation of the first wave of TEs. The pre-pilot phase provided actionable insights, highlighting areas for improvement and aligning technologies with the specific needs of each pilot.
- Social engagement efforts also advanced significantly, informed by earlier engagement activities, such as questionnaire surveys and co-creation workshops, which shaped the preliminary design of technical workshops.
- Detailed updates on infrastructure enhancements were provided as well as early evaluations of deployed TEs, and the progress of KPIs.
- The concept of DEDALUS Business Cases was introduced in the 1st DEDALUS Exploitation Workshop (Sep '23). Based on the HLUCs, BCs were linked with Technological Innovations (TEs) identifying the technologies that are required for each BC to materialise.
- WP6 expanded in 3 dimensions leading towards the Intermediate Exploitation Plan: Business Models & Exploitation // Validation of Business Cases and finalisations of sub-KERs, Standardisation // DEDALUS Standards Library and alignment with technology providers, Replication // Policy recommendation, bi-directional (to and from) interaction with decision makers.
- Social media channels (LinkedIn and X/Twitter) have been launched.
- The visual identity and the project's landing page (www.dedalus-horizon.eu) have been launched, followed by the first version of the website.
- The communication kit (roll-up poster; leaflet; presentation video) was created.
- DEDALUS entered the Smart Energy Cluster.
- The project participated in the technical workshop organized by DigiBUILD through the panel titled "Unlocking Building Flexibility: Blockchain-Driven DR in Electricity and District Heating Networks".



- DEDALUS also joined the BRIDGE initiative.
- The project participated in ENLIT with a stand shared with other 6 EU-funded projects within the Smart Energy Cluster.



1. Overview of the progress and work carried out by beneficiaries

1.1. Objectives

DEDALUS aims at designing, developing and demonstrating SSH-driven multi-value energy carrier-agnostic micro (home/apartment)-to macro (building & district-scale) participatory Demand Response (DR) ecosystem. Main goals are:

- (a) facilitate and scale up residential energy consumers massive participation to DR,
- (b) adapt to a variety of different mono-carrier (electricity, heat) or multi-carrier synergetic scenarios (electricity vs heat and natural gas) at building/district scale, while strengthening social interactions within respective communities.

So, the DEDALUS Consortium intends to deploy:

- 1) a Social Science Framework for multi-dimensional incentives and nudging interventions tools,
- 2) AI-based individual/building/districts consumers clustering and segmentation algorithms,
- 3) Open APIs for DR-ready smart appliances,
- 4) Energy Data Space adaptation for extended DR interoperability and privacy-preserving DLT/Blockchain Data Governance and flexibility coordination,
- 5) Digital Twins for consumers-aware DR flexibility planning,
- 6) Comfort-based flexibility models and tools for serviced apartments for elderly people,
- 7) Flexibility management tools for building and physical/virtual districts energy communities, based on pre-aggregation and shared DR assets,
- 8) Optimal aggregation of flexible energy assets for district heating and decentralized power2heat
- 9) Business Sandbox with novel sharing economy social innovation-based business models.

DEDALUS solutions is being validated by 5 Front Runners full-scale pilots and replicated by 2 Multipliers in 7 countries, deployed at the interplay of energy, and non-energy (comfort/healthcare, ageing, social innovation) sectors, and operated by different stakeholders (social housing operators, building operators, aggregators, energy communities, retailers/suppliers, DSOs/District Heating Network operators). Capacity building and blueprints will support policy makers and regulators to enable market take-up and EU-wide replication of residential DR.

To put the DEDALUS vision into reality, the project must achieve seven challenging objectives reported in Table 1: an overview of the progress towards the achievement of the project objectives in the first 18 months, spanning May 2023 up to and including October 2024. For



each objective the progress refers to the KPIs defined for that objective (reported in *italic*).

#	Brief description	WP	Progress towards the achievement of the objective
01	To make use of SSH-based tools and methodologies to facilitate the engagement of energy consumers in the residential DR	2	<i>> 1.500 stakeholders (65% consumers/35% business roles) in at least 10 co-creation sessions</i> As of M18, seven co-creation workshops have been organized via T2.2 with 64 stakeholders (residents and building managers) participating. Further workshops and engagement are planned for the duration of the project via T5.8. <i>Clustering based on an understanding of the following factors: social-economic, gender, sociocultural, and socio-political factors</i> Questionnaires were created including questions about the socio-economic, gender, sociocultural and socio-political factors. The Questionnaires were tailored for each pilot. Circulation of the questionnaires to the pilot users. From M10 to M18 the questionnaire answers were processed and fed into a Power BI dashboard. The latter was used to delineate a persona for every pilot, but also segment pilots into segment of distinct character that produce a more detailed segment-level persona. The examination of each segment persona led to insights on a tailor-made approach for more efficient outreach strategies.
02	To adapt and extend Horizon ENERSHARE Energy Data Space Interoperability and Trust layer, combining harmonisation and further extension of ongoing interoperability efforts, with privacy-preserving blockchain based enablers for trusted, sovereign secure and participatory flexibility data sharing,	3	<i>Identification of and interoperability with at least 4 Open Standards, standardisable interfaces and formal languages/ontologies, based on the Basic Application Interoperability Profiles (BAIOPs) created in accordance with the SGAM.</i> Arçelik IoT platform is integrating MQTT, which ensures efficient data exchange between distributed components, while utilizing results for web services enables interoperability with diverse platforms. Besides, the interoperability building blocks in the scope of Technology Enabler 3 are integrating a FIWARE ¹

¹ FIWARE is an open-source initiative whose mission is to build an open sustainable ecosystem around public, royalty-free and implementation driven software platform standards that will ease the creation of Smart Applications in multiple sectors,



	in order to enable DR ready user-consumption based flexibilised smart appliances		context broker and FIWARE Next Generation Service Interfaces (NGSI) IoT Agents, It is important to note that certain elements from other well-known ontologies (some of which are standards) have been incorporated to enrich the DEDALUS ontology. Additionally, this ontology will be further enhanced by integrating parts of the BIGG ontology, SAREF4CITY, and Interconnect. • <i>Open APIs and extended data Models: ≥ 10</i> Apart from the ontologies and technologies mentioned in the previous item, other open APIs and data model can be mentioned here: • The Smart Data Models used within T3.2 • The implementation of ERC721 provided by Open Zeppelin library (used in the scope of T3.5). • OAuth2 (Open Authorisation) is the authorization framework used in the scope of T3.6). • OpenID Connect (OIDC) is the identity layer built on top of the OAuth2 framework used in the scope of T3.6. • <i>15-20% increased share of aggregated flexibility mobilised by piloted assets</i> Thanks to the technology enablers being developed within WP3, the different flexibility services will be able to interact with a great amount of the devices installed in the different pilots (most of them related to flexibility issues), and this will increase the share of aggregated flexibility mobilised by pilot assets.
03	To combine H2020 eDREAM, BRIGHT, MATRYCS and HORIZON DigiBUILD building-level data-driven models, algorithms and flexible assets DTs with end-users' consumption and beyond consumption profiling models (e.g. comfort, healthcare, ageing, poverty/vulnerability) and deploy TRL6 HITL ('Human-In-The-Loop')	4	• <i>Increased predictability by 15% of DR behaviour and flexibility potential to mobilise.</i> • <i>≥ 2 Digital Twins</i> The main achievement done in this reporting period corresponds to the 1st technology development cycle completion, MS4 delivered at the beginning of July plus some further update done until M18, October 2024. It mainly consists in two Technology Enablers, TE-8 - Digital Twins for consumer-aware optimal flexibility planning and TE-5 blockchain and smart contracts integrated DT. The first one is a tool designed to optimise the management of various assets and mainly to support consumers in the optimal management and assessment of flexibility resources; it is applied across different



	flexibility assessment and valorisation DTs		scopes in pilots 2 (DK), 3 (GR), and 4 (IT). The development of TE-5, exploited in the Romanian pilot, is a digital twin model for the students' apartments building that incorporates data from heterogeneous sources with the purpose of supporting decision making in demand response programs and simulating the execution of apartment level DR actions. As DT developments, the KPI is achieved through the delivery of the first release of both TE-5 and TE-8.
04	To deliver data-driven, value stacking cross-stakeholder/cross-commodity/cross-value chain DR algorithms, models, and ICT-based services, for DR in energy-efficient residential buildings	4	<i>Number of validated data driven aggregation models and services for residential DR: ≥ 7</i> The main achievement done in this reporting period corresponds to the 1st technology development cycle completion, MS4 delivered at the beginning of July plus some further update done until M18, October 2024. It mainly consists in the following Technology Enablers: TE-10: Nudging Apps for near real-time closed-loop residential consumers activation and multivector DR TE-11: Multi-value DR service TE-12: P2P digital platform for trading pre-aggregated building flexibility TE-13: District-level DR for self-consumption and cost optimization TE-14: Aggregation of flexible energy resources for district heating management A first version of this TEs is available . The validation on pilots is in progress and will be improved with the delivery of the second version expected to be finalized soon.
05	To demonstrate, validate and replicate DEDALUS solutions through a number of different pilots, located in different geographical and climatic areas and operated by different stakeholders	5	<i>Number of consumers actively engaged in residential DR: ≥ 500</i> To achieve the goal of engaging at least 500 residential consumers in DR activities, the project has implemented several initiatives aimed at fostering awareness and participation. In the initial stages, residential consumers were informed about DEDALUS and residential DR through the circulation of questionnaires, which collected insights into their energy consumption habits and willingness to participate in DR programs. This effort resulted in responses from 236 residents. Additionally, co-creation workshops were conducted at each pilot site, involving 52 residents who provided valuable input on the barriers and drivers for active participation in DR. These insights have been instrumental in shaping the development of DEDALUS services to meet consumer needs effectively. As the project progresses and the DEDALUS solutions become accessible to end users in



		the second half of the project, the number of actively engaged residential consumers is expected to surpass the target of 500. • <i>Different types of actors engaged by DEDALUS services: ≥ 10</i> A wide array of stakeholders has been engaged, achieving a total of 13 different actor types by M18. These include consumers, building owners, building operators, facility managers, energy suppliers, energy aggregators, psychologists, healthcare staff, technical departments, distribution system operators (DSOs), battery providers, IoT providers, and city councils. Plans for the upcoming technical workshops (scheduled for M21 - 25) include inviting external stakeholders to potentially further expand the range of actors engaged. These workshops will provide a platform for stakeholders to interact with the demonstrated technologies, keeping them engaged, gaining insights and contributing feedback. • <i>Number of technologies that reached TRL6-7 by the end of the project ≥ 14</i> By M18, significant progress has been made in mapping the technologies to be demonstrated at each pilot site and establishing their technical requirements and functional specifications. While many technologies are still in preliminary development, four have already reached TRL 6, demonstrating their potential for operational deployment. These include the flexibility modelling tool (TE-4), the universal secure gateway (TE-6), mechanisms for secure and privacy-preserving data transactions (TE-7), and the district-level DR for self-consumption and cost optimization (TE-13). With continued development and iterative testing during the following two pilot cycles, additional technologies are expected to reach TRL6-7. • <i>At least 95% satisfaction accruing from pilot users.</i> A robust methodology has been created to assess and ensure high user satisfaction from pilot users with the technologies deployed. The satisfaction rate will be officially measured after the TEs that will be demonstrated in the pilot sites are fully operational. Preparations are already underway to collect user feedback systematically through scheduled technical
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			workshops (M21 - M25) which will introduce the methodologies and outputs of the TEs to pilot users. During these sessions, participants will complete pre-test social acceptance questionnaires to provide baseline data on their perceptions, including willingness to use the TEs, understanding of their functionality, and overall usefulness. Follow-up post-test surveys will assess satisfaction after the piloting activities (M26 - M30). The insights from these surveys will guide iterative improvements to ensure a high level of satisfaction, with the target of achieving at least a 95% satisfaction rate among pilot users.
06	To explore and validate new business models which combine inspiring principles from sharing economy, with multi-carrier system-level perspective and with social innovation and non-financial benefits capture for DR in energy-efficient residential buildings	6	• <i>Deliver at least 4 new business models.</i> The methodology that will deliver the business models has been defined and thoroughly analysed in D6.1. Business Cases that consist of sub-KERs were identified for each pilot in the 1st Exploitation Workshop (Sep24). Follow up Exploitation Workshops on the finalization of BCs and definition of sub-KERs (2/2024 & 9/2024). Business Models are derived from BCs with sub-KERs as their building blocks. Sub-KERs are the primary components of DEDALUS services and are related with KERs 1-8 (as mentioned in the GA). For each DEDALUS partner to have a clear contribution in the Business Models inspired by DEDALUS BCs and services it was critical that their contribution could be contained. By combining sub-KERs it is possible to build Business Models which will be exploited after DEDALUS based on bilateral agreements among the sub-KER owners guided by the exploitation plans (O7). • <i>Recommendations in each of the pilot countries and at EU level developed.</i> Effort starts on M13th, kickstarted in T6.5 and supported by 6.3 and 6.4. Interviews with politicians and relevant stakeholders revealed key positions that form the basis of policy recommendations for the adoption of DR. Workshop (9/2024) among DEDALUS consortium identified key policy and regulatory interventions required.
07	To allow effective consumer participation in DR and further market opportunities by	6,7	• <i>1 project exploitation plan, individual exploitation plans for each partner, 1 SWOT analysis and 1 sustainable plan for after project operation.</i>



providing input to regulatory bodies to stabilise and harmonise the regulatory framework for residential DR, ensuring an efficient communication and dissemination, as well as alignment and synchronisation with European relevant initiatives and projects	By M18, significant progress has been made toward the project's exploitation objectives. A structured methodology has been established, and key concepts were introduced through webinars in M3 and M10. A third workshop (M17) clarified ownership, IP strategies, and exploitation pathways for each sub-KER. Further details and preliminary exploitation strategy have been defined in collaboration with the consortium and collected in the KERS' Library. • <i>At least 7 different dissemination/communication channels set-up & maintained.</i> DEDALUS project created 2 social media channels, LinkedIn and X, for which a dedicated social media strategy has been developed. The two channels integrate the project website (started in M4) which operates as the main information gateway for stakeholders and public. Due to a technical issue related to X (not under ICONS responsibility) which led to the profile suspension, it has been evaluated the opening of a new social media channel. A project video has been released and published on YouTube channel https://youtu.be/t-ZICbyihsk. In October 2024, due to the persistence of issues related to the X account, ICONS team decided to open a DEDALUS dedicate account on Mastodon, a social media platform that is part of the "fediverse. Here is the link: https://mastodon.energy/@DEDALUS. In addition to the project's channels, the partners contribute to maximise the impact by using their own channels and re-posting/tagging the project and its activities. • <i>At least 200 organisations and 1,000 individuals following the project's channels.</i> By M18, almost 500 subjects and 24 organisations follow the project's channels • <i>Contribution to 2 standards, synergies with more than 30 relevant projects/initiatives and at least 3 cross-fertilisation activities with BRIDGE</i>
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		In the first 18 months, ICONS launched the DEDALUS clustering activities. In particular, DEDALUS became a member of the Smart Energy Cluster, managed by NTUA and IEECP, and composed by 27 Eu-funded projects operating in the Energy field. Moreover, DEDALUS started the joint dissemination and communication activities with the following EU-funded projects: DigiBUILD; BuildON; TwinEU. Finally, in January 2024 DEDALUS entered the BRIDGE initiative, allocating a several number of Consortium partners in the 4 Working Groups. Concerning the standards, initial work (D6.1) towards mapping relevant to DEDALUS standards • <i>At least 4 virtual and physical workshops with 150 trainees3 deliverables showing the results from pilot evaluation and impact assessment, integrating 1 report per pilot.</i> Not started yet • <i>At least 90% satisfaction accruing from pilot users, accruing from a transparent user validation methodology.</i> The satisfaction scores will come from the pre/post surveys linked to the technical workshops that will take place from M21 to M25 during the full pilot operational phase, when the services will be ready to be used by the pilot users. This KPI will be evaluated in the last phase of the project lifetime. • <i>Exploitation of at least 73 GBs of data/year and 200 datasets for the pilot partners.</i> During the initial 18 months of the project, installation and deployment of necessary infrastructure and systems were still underway (pre-pilot phase). Evaluation of this KPI is scheduled at the end of the full pilot operation phase (M25) and at the end of the large scale operation phase (M33), when all pilot sites are expected to be fully functional and the data exploitation can be assessed.
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1.2. Explanation of the work carried out per WP

This chapter is split in sections, one per WP. For each WP running in the first 18 months of the project a dedicated section contains a short description of the work package, a brief report on the progress, description and advancement per task, the contribution provided by all the involved partners and the effort spent to work in the WP.

1.2.1. WP1 – Coordination and management

[M1-M36] [Leader: **ENG**, Participants: all]

WP1 is responsible for ensuring that all project activities are performed efficiently, in a coordinated and integrated manner, within schedule and budget, and in a way that the high research, scientific, technological, and stakeholders' expectations are fully met. ENG, acting as Project Coordinator, brings in its experience from various previous projects to guarantee successful communication within the Consortium and the EC.

In the context of the work package and in the first 18 months of the project, a proper repository has been set up together with an effective communication tool (MS Teams). Procedures for production of documentation (presentations, reports, minutes) have been defined in a proper Project management plan (D1.1). Periodic audio conferences and 4 plenary meetings (3 in presence/hybrid and one totally remote) allowed to monitor the advancement of all the running activities. A Transdisciplinary Advisory Board has been identified thanks to the business network of ICONS and the effort they spent to find the right professionals that can give valuable contribution to the project. So, three experts:

- **Joke Kort**, senior scientist at TNO Energy and Materials Transition
- **Michael Brenner-Fließner**, scientist at Joanneum Research Life
- **Gaetano Zizzo**, associate professor at Università degli Studi di Palermo

are now discussing with the Consortium about the results achieved so far. An important output is also the Data Management Plan (D1.2) whose final version was submitted at the end of month 10 as well as the intermediate progress report (namely D1.3 DEDALUS first activity report) that collects a description of the progress at month 9 but was released at month 18 (because of a mistake in the GA to be fixed by means of a proper amendment).

1.2.1.1. Task 1.1. Project administration and coordination

[M1-M36] [Leader: **ENG**, Participants: all]

The task, started in M1, consists of the financial control and formal revision of progress reports, the supervision and information of all participants about the project progress (including sending interim reports, meetings minutes of the governing boards) and the day-to-day assistance to the overall Project Management (including administrative issues).



In particular, the coordinator, and his team have assured a first continuous support and day-to-day assistance to the project partners about any concerns or requests, on technical, financial, administrative and/or management issues.

In the first 18 months, a proper repository was set up along with an effective communication tool (MS Teams). The work was mainly dedicated to defining the procedures for producing documentation such as presentations, reports, and minutes, as well as collecting information for D1.1, which was submitted at M10. The work also involved organizing the technical committee meetings and four hybrid plenary meetings (three mainly in presence, one completely remote). However, the coordination of the project in the first months was quite intermittent and most of the activities was not properly monitored and checked. So, a relevant amount of delay has been accumulated. Starting from December 2023 the coordination planned the release of all the missing deliverables and kept the Consortium on the established plan by catching up all the delays. So, all the expected deliverables have been finalized and submitted with an irrelevant impact on the critical path of the project.

1.2.1.2. Task 1.2. Technical & scientific coordination

[M1-M36] [Leader: **ENG**, Participants: ENG, NTUA, CARTIF, UNIVPM]

The task T1.2 covers the analysis of any technical decision to be taken, to mitigate and solve technical conflicts, during dedicated video calls with the involved partners. In particular, the technical coordination of the project is carried out by the Technical Committee led by the Project coordinator in collaboration with the WP technical managers. In the context of this task, the Technical Committee therefore coordinates and assists the WP leaders on technical issues and refines technical guidance where necessary.

The task also includes collaboration with the Pilots Coordinator, to guarantee a clear and coherent integration of DEDALUS services through demonstrators, with the final aim of evaluating their impact from an economic, social and environmental perspective. Furthermore, the Scientific Manager assists the Project Coordinator and the WP leaders, and reports to the former.

ENG has monitored the overall technical progress of DEDALUS since the M1 but, unfortunately, because of internal organization problems, not in a continuous way. The first periodic calls have been held to describe, collect and solve any problems related to technological actions via cooperation among partners and WP leaders but the initial results were well behind the schedule. Starting from December 2023 (M8), ENG led many parallel conference calls and related offline work to coordinate Use Cases, User requirements elicitations, to shape properly the scope of the project. This allowed to catch up the delays in less than 3 months.

1.2.1.3. Task 1.3 Quality control, risk management and Transdisciplinary Advisory Board

[M1-M36] [Leader: **ENG**, Participants: ENG, NTUA]

This task guides the targeted research and innovation aspects of the work carried out during the project lifespan and ensures that they are produced based on high-quality standards. Moreover,



management and quality assurance procedures ensure that the project's tasks are aligned with its objectives.

The main output of this task is represented by D1.1 - Project Management Plan - which includes all necessary information related to the quality management of the project. This deliverable provides details about the quality management processes, including the distribution of peer-review activities among project partners for evaluating scientific content, language, and readability of the documents. By the end of month 9, a consolidated version of the document was circulated, and the deliverable was submitted at month 10.

Task 1.3 is also responsible for identifying potential risks to initiate timely corrective actions if necessary. These activities will be mainly performed by ENG with contributions from all consortium partners. The risk management procedure can be summarized into four phases: risk identification, risk analysis, risk evaluation, and risk treatment.

To successfully address any issues, difficulties, or problems that may arise, the task includes the establishment of a Transdisciplinary Advisory Board, encompassing experts from various fields relevant to the project scope. In the first 18 months of the project, as mentioned above, the consortium beneficiaries proposed and evaluated several candidates based on their experiences and skills to identify the most suitable members for the DEDALUS project. Two of them have been selected: Joke Kort, senior scientist at TNO Energy and Materials Transition, and Michael Brenner-Fließer, scientist at Joanneum Research Life.

1.2.1.4. Task 1.4. Open science, FAIR data and data management

[M1-M36] [Leader: **ENG**, Participants: **ENG**]

This task is continually assessing and improving the requirements of knowledge exchange, ensuring research data and outputs are managed in line with the FAIR principles. This entails development of a robust Data Management Plan (D1.2) from the onset of the project, which could be updated during the intermediate and official reporting periods. It clearly defines what data will be published, as open data might generate a wider interest towards the project. Two additional aspects are also tackled: privacy (to ensure that proprietary or personal data is not externally accessible) and ethics (to avoid the identification of individual persons from data streams, further than performing a correct use of data).

During the first 18 months of the project the Data Management plan was drafted and released (even though later than planned). This document also reports an update about the DMP in chapter 4.

1.2.1.5. Work carried out per partner per task and effort spent

Table 1 – WP1 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	Performing coordination activities, including organizing technical committee meetings and four plenary meetings.	10.93



Partner	Work carried out in the WP within the period	PMs
	- Daily coordination of ongoing activities, along with resolving any issues that arise. - The Project Management Plan (D1.1) - Finalization and submission of the Data Management Plan (D1.2) - Finalization and submission of the report D1.3 DEDALUS first activity report, to be intended as a first intermediate progress report describing the first 9 months of the project.	
2 - NTUA	- General meetings attendance. - Project administration and management. - Contribution to general project issues. - Quality review of project deliverables. - Participating in project meetings and coordination activities which mainly have been tracked on our main activities in WP5, as well as T2.3 and T3.4.	1.45
3 - CARTIF	- General meetings attendance. - Project administration and management. - Contribution to general project issues.	0.82
4 - UNIVPM	- General meetings attendance. - Project administration and management. - Contribution to general project issues.	0.60
5 - ARC	- Project administration and management - Coordination activities - General meetings attendance.	0.50
6 - COMS	- Participation in project meetings - Project administration and management	0.40
7 - OurPower	- Project administration and management - Coordination activities - Participation in project meetings	0.49
8 - GreenPoint	- Participation in project meetings - Project administration and management	0.45
9 - BLUEPRINT	General meetings attendance.	0.50



Partner	Work carried out in the WP within the period	PMs
	- Project administration and management. - Contribution to general project issues.	
10 - FællesBo	- General meetings attendance. - Project administration and management	0.21
11 - NEOGRID	- Participating in the general administration and management activities of the project, providing required input. - Review of D1.1 – Project management plan - Participating in project meetings and coordination activities which mainly have been tracked on our main activities in WP4 and WP5	0.48
12 - GREEN CITIES	- Project administration - Participation in project meetings	0.36
13 - HERON	- Project administration and management - Coordination activities mainly for WP6 and Pilot 3 activities - Participation in project meetings - Preparation activities for hosting the 4th plenary meeting (Sept 24)	0.46
14 - DOM-X	- Participation in project meetings - Contribution to general project issues	0.55
15 - ISRAA	- Participation in project administration and management activities - Participation in project meetings	0.56
16 - DYNAMIE	- Participation in project meetings - Contribution to general project issues	0.28
17 - CIMNE	- Participation in project administration and management - Coordination activities - Participation in project meetings	1.14
18 - PEUSA	Participation in project meetings	0.30



Partner	Work carried out in the WP within the period	PMs
	Contribution to general project issues	
19 - ROMUR	- Participation in project meetings - Contribution to general project issues	0.50
20 - CENTRICA	Contribute to the general administration and management activities of the project,	0.17
21 - TUC	Participation in project meetings	0.00
22 - SIN	Contribute to the general administration and management activities of the project,	0.49
23 - ICONS	Contribute to the general administration and management activities of the project,	0.04

1.2.1.6. *Deviations from the Grant Agreement*

Because of ENG's internal organization issues, the project lacked coordination in the first months. Starting from December 2024 (M8) the coordinator has properly planned the release of the missing/pending deliverable both related to WP1 as well as to other WPs. At the time of writing delays have been caught up. Concerning the WP1 deliverables and activities, the following ones are the most relevant delays:

- Submission of D1.1. Project Management Plan at M10 instead of M2
- Submission of D1.2 Data Management Plan at M12 instead of M6
- Submission of D1.3 DEDALUS first activity report at M19 (as per GA even though it reports on the first 9 months of the project)
- The Transdisciplinary Advisory Board members has been defined and the two identified members are being involved in the project.

An important deviation was the replacement of the Irish pilot, one of the two multipliers. The management of the project supported CENTRICA in the identification of an alternative pilot characterized by the same level of innovation. More details are reported in the WP5 section.

1.2.1.7. *Corrective actions taken/suggested*

As soon as the coordination team was set up, the coordinating partner, ENG, prepared a plan together with WP leaders, Technical and Scientific Manager, as well as Pilot leaders. The plan was discussed with the PO and a set of new deadlines was established to release all the pending deliverables. Many dedicated audio conferences have been organized, all the open issues discussed, all the documents drafted at that time were shared and discussed with all the involved partners. In the period between M8 and M9 all the expected deliverables were drafted.



The project started to stick on the initial plan by March 2024 (M11) and it is currently aligned with the GA.



1.2.2. WP2 – Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR

[M1-M16] [Leader: **BLUEPRINT**, Participants: all]

WP2 provides the foundation for DR in energy-efficient residential buildings, considering socio-economic enablers in the design of the DEDALUS solutions. It will establish the basis for the development of the DEDALUS socio-economic and technological framework, providing groundwork for energy market, consumers' analysis, flexibility categorisation and clustering of buildings. We ultimately achieve a co-creation and co-implementation process for the design of financial/non-financial incentives mechanisms to promote demand flexibility in energy-efficient residential buildings.

During the first 18 months of the project, four deliverables (D2.1–D2.4) were submitted, reflecting significant progress across various tasks within WP2.

D2.1 “DEDALUS gap analysis and use case capitalisation” documents the work from subtask T2.1, which focused on assessing the potential of DEDALUS High-Level Use Cases (HLUCs) at the pilot sites. The IEC 62559 standard was adopted and tailored to the specific needs of the DEDALUS project to define and describe HLUCs. Four broad domains—Data Management, Performance Assessment, Smart Control, and Market/User-Centered Interaction—were identified to guide the HLUC development process. Additionally, a list of actors essential for the harmonious description of HLUCs across partners was established. A final set of 11 HLUCs was defined, detailing their scope, objectives, preconditions, and basic steps. The deliverable also presented a gap analysis and a description of individual pilot sites, mapping HLUCs to specific demonstration sites, which formed the foundation for planning activities at each site.

D2.2 “Active societal participation, emergence, socio-economic enablers and consolidation factors of multi-value and energy carrier agnostic DR for residential buildings” presented the Social Science Framework to guide the design and implementation of DEDALUS co-creation activities. The framework provided a comprehensive understanding of consumer preferences, motivations, and barriers to participation in Demand Response (DR) programs using social science methods such as deliberative workshops, Q-methodology, and nudging interventions. It translated this framework into practical co-creation activities targeting residential DR customers and included a segmentation analysis that identified four distinct consumer clusters. Plans for workshops at each pilot site were outlined, focusing on consumer engagement and feedback for designing data-driven DR services and business models. The results from these workshops were set to feed into tasks across multiple work packages, including WP4, WP5, and WP6.

D2.3 “High-level requirements for the DEDALUS social, technological and business framework” represented a software specification document that established the high-level requirements of the DEDALUS project within social, technological, and business frameworks. The deliverable proposed a software architecture with functional and non-functional requirements and addressed ethics and security considerations. Strategies to achieve effective DR were also outlined, alongside the specification of use cases and data-driven insights. The aspiration was



to create an innovative, sustainable ecosystem for diverse stakeholders while addressing dynamic requirements in the DR domain.

D2.4 “Enabling conditions for DR in energy efficient residential buildings”, a collaborative effort among WP2 task leaders, integrated the final outputs of co-creation processes, segmentation, socio-economic enablers, and requirement refinements. Building on previous deliverables (D2.2 and D2.3), it detailed the results of the co-creation process and refined the software architecture and high-level requirements. Key findings emphasized trust, motivation, and personalized incentives as crucial for DR acceptance and effectiveness. The deliverable also demonstrated the potential of machine learning to analyse and cluster electricity load profiles and linked insights from Austrian pilot data with user questionnaire data from the Greek pilot. The findings provided a robust foundation for subsequent work in WP3, WP4, WP5, and WP6, ensuring the scalability and sustainability of DEDALUS solutions.

This comprehensive effort highlights the project's progress and the alignment of its deliverables with the overall objectives.

1.2.2.1. Task 2.1 Gap analysis and use case capitalisation for DR in the residential sector

[M1-M6] [Leader: **BLUEPRINT**, Participants: ENG, CARTIF, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAMIC, CENTRICA, TUC]

This initial task aimed to conduct a gap analysis for use case capitalization to develop easy-to-adopt innovation-based models, engaging different actors of the residential sector with varying characteristics. The DEDALUS pilots are set to be analysed and documented according to a standard methodology, ensuring a streamlined workflow of software development across the pilots throughout the project. All use cases are to be described according to the standard IEC62559 (Use Case Methodology). We aim to define and describe the use cases and requirements associated with reaching the DEDALUS project objectives.

DEDALUS project is not dealing only with smart grid or electricity field, and activities in development of D2.1 served as a preliminary analysis of the main identified use cases per each pilot site using bottom-up approach in interviews with pilot site owners, thus identifying high level requirements. We have adapted the structure of the standard UC template, suggested in IEC 62559-2, to better suit the project requirements and the partners involved in contributing to the definition and description of the High-Level UCs (HLUC), that have been developed into detailed UCs as part of D2.3.

This IEC standard provides a template that helps to define the set of (inter)actions to be developed and implemented during the course of a project. We adapted this template to suit the characteristics of the DEDALUS project better highlighting the high-level requirements and specifications for the early stage of DEDALUS development and to contribute to the basis of the



work being developed within WP2 in development of Social Toolbox and functional specifications, as well as in the other work packages (WP). Also due to the use of this methodology and before the activities of definition and description of the HLUCs, we were able to identify four broad Domains (Data management, Performance Assessment, Smart Control, Market and User-centered Interaction). These Domains have also guided the processes of defining the DEDALUS HLUCs.

In addition, we have also presented the list of the DEDALUS actors. This list of actors proved to be essential for a harmonious description of the HLUCs across the different partners involved in the process.

A final list of 11 HLUCs was defined and the main characteristics and requirements of each HLUC were presented, namely their scope and objectives, description, required DEDALUS pre-conditions and basic steps.

We also conducted a gap analysis and description of individual pilot sites in the DEDALUS project. It may be observed that while data UCs and development of DEDALUS platform support all pilot sites, other market/performance domains with multiple energy vectors will only be applied in selected pilot sites and measured with appropriate KPIs developed under T6.1.

Finally, mapping of the most important HLUCs that will be deployed in each demonstration site was conducted. The results of this preliminary exercise are fundamental for planning the activities of each DEDALUS demonstration site, mainly the selection and implementation of the most important DEDALUS services and HLUCs in each demonstration building/community.

1.2.2.2. Task 2.2. DEDALUS Community building and scenario co-creation for residential DR

[M1-M16] [Leader: **SIN**, Participants: ENG, OurPower, GreenPoint, BLUEPRINT, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAME, CIMNE, CENTRICA, TUC, ICONS]

This task defines the Social Science Framework for the engagement of residents and encouraging a design thinking approach for bringing cocreation and feedback loops with end users to the forefront in the DEDALUS project. A methodology incorporating questionnaires, qualitative research, workshops, segmentation, user feedback, etc. has been implemented to provide insights about customer needs and suggest improvements to our services.

The social science framework was developed in the first 6 months of the project through combining a literature review with past project experience from SIN. Based on the High-Performance Cycle model and the Theory of Planned behaviour, a social science framework was set up for developing effective DR-related behaviour change interventions and social engagement.



To map the local context and set-up of each pilot, including types of users, primary objectives, and engagement channels, SIN organized pilot scoping meeting and distributed pilot scoping questionnaires in M5-M7.

Leveraging the theoretical underpinnings of the social science framework and the practical learnings from the pilot scoping, in M8-12 SIN designed a co-creation workshop methodology incorporating three modular and customizable activities focusing on three distinct topics:

1. Understanding the barriers and driver to DR participation
2. Understanding relative importance of various incentives related to DR
3. Identifying effective nudging interventions

The workshop on understanding the barriers and drivers to DR participation incorporated a Q-sort methodology to investigate end user perceptions to topics such as comfort, social norms, empowerment and privacy. The DR-related incentive workshop incorporated a ranking task and group discussion to understand which types of incentive would prove most effective as a leverage for encouraging desired DR behaviours in each pilot. Finally, the nudging workshop aimed to gather end user feedback on various nudging interventions through user testing methods.

For all workshops, SIN developed detailed instructions and guides to ensure the pilot partners could carry out the workshops effectively. Additionally, SIN hosted multiple training sessions with pilot partners to introduce and explain the tools and activities; providing an understanding for them to confidently run the workshops and gather the required output and successfully engagement with their pilot community. Seven co-creation workshops were held in M12 to M14, successfully engaging a diverse group of participants across multiple pilot sites. These participants include 52 residents, building managers, and other stakeholders from various demographic backgrounds.

The social science framework and co-creation design and plan for implementation were described in detail in M12 in D2.2. In M15, the analysis and results of the co-creation workshop, including recommendations and best practices for engagement and stakeholder acceptance were described in D2.4.

1.2.2.3. Task 2.3. Consumers segmentation and clustering of buildings for optimal DR planning

[M1-M16] [Leader: **NTUA**, Participants: ENG, NTUA, ARC, OurPower, GreenPoint, BLUEPRINT, FællesBo, GREEN CITIES, HERON, DOM-X, ISRAA, CENTRICA, TUC, SIN]

This task involves conducting a comprehensive empirical analysis to understand consumers' decision preferences and the underlying cognitive, affective, and social factors in residential DR. The main objective is to create clustering and segmentation methodologies and algorithms tailor made for DR programs. Conducting data assessment based on experimental online studies to obtain representative samples. Employing AI-based clustering and market segmentation



algorithms to analyse individual consumers, with a focus on characterizing user groups by socio-demographic and psychological variables.

As mentioned above the Task can be split into two different activities, namely the data-driven clustering based on a Machine Learning-based framework for clustering electricity load profiles to enhance demand response programs and the Questionnaire-driven segmentation approach relying on questionnaire answers and comprehensive segmentation and analysis of user profiles/personas based on socio-demographic and psychological variables. The task begun in M1 and the work carried out in the first 17 months for both approaches is presented below.

1. Questionnaire-driven segmentation:

Up to M18, NTUA performed an extensive scientific literature review aiming at shedding light on the key task concepts, including principally residential DR, energy literacy, energy behaviour, and principal demographic dimensions, among others. This activity formed the basis for the initial draft of the questionnaire, where the aforementioned axes were grouped to separate sections. NTUA scheduled bilateral calls with SIN to retrieve feedback on the questionnaire development. Upon consolidating this communication, NTUA prompted pilot site leaders to prepare the questionnaire circulation. Then, questionnaires specific to each pilot were created and were shared via SharePoint. Pilot site leaders provided feedback, and each questionnaire was adjusted accordingly. As soon as questionnaires got finalized, pilot site leaders were asked to translate them into the end users' language. The translated questionnaires were distributed to the residential users through three alternative options: in-person, tablet, email. The submitted responses were collected either via google forms or excel sheets, which helped in accelerating the process. Ultimately, the responses format was aligned across all files received so as to facilitate their upcoming analysis.

From M10 to M16 NTUA processed the Questionnaire responses so as to render them suitable for data analysis and data visualisation. The processed data form was fed into Power BI where dashboards were built per pilot so as to provide the ability to investigate the data visually through custom filtering and eventually draw insights regarding their content. This visualisation procedure led to the delineation of each pilot persona; thereafter, each pilot was divided into segments of distinct character, whose persona was also analysed.

2. Machine Learning-Based Framework for Clustering Residential Electricity Load Profiles to Enhance Demand Response Programs:

During the first eighteen months of our project, NTUA developed a machine learning-based framework for achieving optimal load profiling and published a scientific article in the Applied Energy International Journal². The proposed solution was demonstrated using an open-source dataset containing information from nearly 5,000 households in London. Four widely used clustering algorithms were applied specifically K-means, K-medoids, Hierarchical Agglomerative

² <https://www.sciencedirect.com/science/article/pii/S030626192400326X>



Clustering and Density-based Spatial Clustering. An empirical analysis as well as multiple evaluation metrics were leveraged to assess those algorithms. Following that, NTUA redefined the problem as a probabilistic classification one, with the classifier emulating the behaviour of a clustering algorithm, leveraging Explainable AI (xAI) to enhance the interpretability of our solution. The scalability and versatility of our solution makes it an ideal choice for power utility companies aiming to segment their users for creating more targeted DR programs. Furthermore, the application of the methodology has been performed on the Austrian and Greek pilot and delivered in D2.2 and D2.4 respectively.

The proposed methodology is presented in Figure 1. First, a detailed description of the first phase of the pipeline that extracts the features from the given data is issued. Following, clustering algorithms and the appropriate evaluation metrics are utilized to obtain the optimal number of clusters. In the next subsection, similarity matrices between the different algorithms are used to obtain the nominal clusters produced. On top of those clusters, a probabilistic classification function and xAI part are introduced as the final step of our proposed approach.

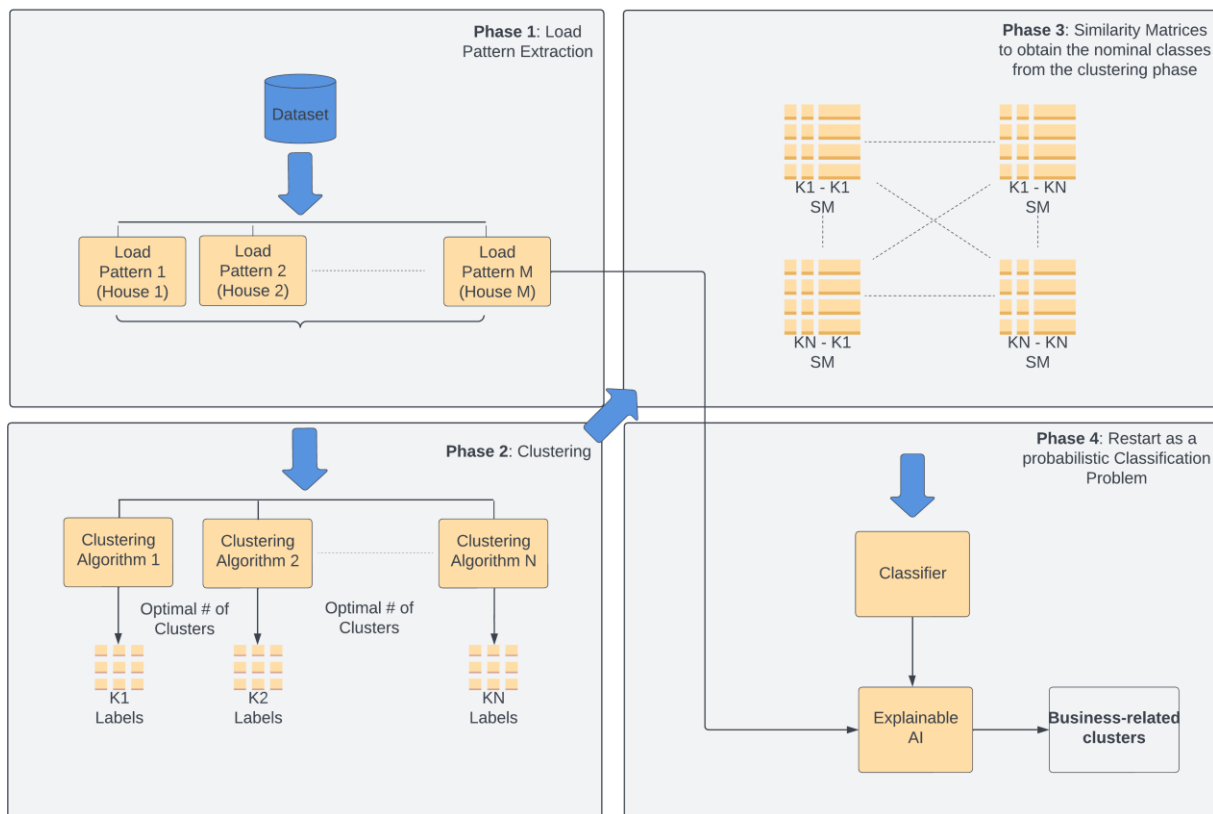


Figure 1 - Proposed clustering methodology

Finally, the combination of the two aforementioned methodologies were presented in D2.4 with the eventual comparison between the segmentation and the clustering analysis on 15 participants from the Greek pilot.

1.2.2.4. *Task 2.4. Co-designing with consumers innovative social-driven incentive mechanisms and data-driven DR services*

[M1-M16] [Leader: **DOM-X**, Participants: ENG, CARTIF, UNIVPM, ARC, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAMIE, CIMNE, PEUSA, ROMUR, CENTRICA, TUC, SIN]

Subtask 2.4.1

This subtask aimed at defining the minimum requirements of the different pilot user interfaces, eliciting user preferences in an easy and systematic way and promoting user engagement. The subtask builds on the output of the T2.1 on the requirements of each supported use case and the core identified consumer groups (T2.3), along with the output of T2.2 about the co-creation sessions with the target pilot users in a workshop format. Briefly, it aimed at organizing the co-creation workshops with the end-users for every pilot site to receive beneficial input from the real users in order to redefine the needs of every pilot site, so that to properly design and develop the features for the DR actions. The goal of this subtask was to provide an output for the activities of the WP4, specifically for upgrading the user interfaces employed in the different pilots (especially in T4.3 and T4.4).

Within Task 2.2 we have done several steps to organize the co-creation workshops that every pilot site has performed in order to receive input from the end-users for better designing the features of its pilot.

Apparently, this task is strongly associated with Task 2.2 where the societal framework has been developed by SIN, thus we have been in close collaboration with SIN, having performed several calls to understand the needs of the pilots and how to address them. Moreover, we have also organized and performed (again with SIN) several bilateral calls with all pilot sites to understand the characteristics of each one and how to organize the different features of the workshops towards fitting their needs.

Regarding the nudging activity that most pilot sites include in their scenarios, we have employed a detailed booklet by the NUDGE H2020 project that summarizes best practices as guidelines for designing nudges in mobile applications (more details can be found in D2.2). This booklet was communicated with the interested parties and bilateral calls have been realized with every partner to discuss how each pilot can augment its mobile application to also include nudging techniques to support DR strategies.

Lastly, under the umbrella of this task the co-creation workshops have been realized successfully for the Austrian, Danish, Greek, Italian the Spanish and the Romanian pilots. The workshops took place between the months M12 and M14, while the pilot leaders took significant feedback on the proposed features.



Subtask 2.4.2

This subtask engages residential consumers and stakeholders to define needs and requirements for a comfort-based flexibility model. It assesses Indoor Environmental Quality (IEQ) in terms of thermal, acoustic, and visual comfort, and indoor air quality. Deliberative Futures Workshops (Task 2.2) will identify these needs, which will be translated into technical requirements for Task 2.5. The results will inform the implementation of comfort-flexible demand response (DR) models in Task 4.2, with a specific focus on the comfort needs of the elderly. From M1 to M16 of the project, we defined user and technical requirements through questionnaires completed by occupants and facility managers.

Subtask 2.4.3

This subtask aimed at analysing the financial and non-financial (e.g. social) incentives mechanisms to increase consumers' engagement in residential DR, studying the impact of diverse classic and behavioural interventions. It is based on outputs from T2.2 and T2.3, regarding consumers segmentation results coming from the segmentation questionnaires. Furthermore, the task examined how different interventions impact the behaviour of key user clusters (T2.3), to develop tailored group-specific interventions.

In close collaboration with SIN that is responsible for T2.2, the actions foreseen in the tasks have been developed as follows: first, a literature review has been conducted, to understand existing theories, frameworks, and findings related to monetary and non-monetary incentives for consumer engagement in residential Demand Response. The first part of literature review was reported in D2.2. This allowed also to identify previous projects that have implemented incentive mechanisms and analyse their outcomes, considering both successful and unsuccessful interventions to gain insights into what works and what does not.

The second step is to use the co-creation workshops planned for WP2 as a valuable tool to investigate and test the perception, feasibility and effectiveness of financial and non-financial incentives in the context of demand response and energy conservation practices. In collaboration with SIN, a specific activity for the co-creation workshop was developed and it is focused on a ranking exercise of different types of incentives and participants are asked to rank them in order of how important they are. This provides insight and data in which incentives are the most valued.

As a third step, the findings from the literature review, the gaps identification and the results from the workshops activity will be combined to understand what incentives are most effective according to type of users, type of buildings and other relevant parameters. As final outcome of the subtask, a *Toolkit for a successful implementation of incentives* is planned to be developed. A general outline of the content has been developed and it includes guidelines on how to choose, develop and implement successful incentives schemes for DR programs.



1.2.2.5. Task 2.5. Residential consumers business/privacy/cyber-security requirements and functional specifications

[M3-M16] [Leader: **ENG**, Participants: **ENG**, **NTUA**, **CARTIF**, **UNIVPM**, **ARC**, **COMS**]

The primary aim of Task 2.5 is to develop comprehensive functional specifications for the DEDALUS business, privacy, and cyber-security framework. This framework addresses the standardized access and exchange of data, ensure the security and privacy of this data, and considers any potential ethical issues. The expected outcome of these specifications will facilitate the integration of the existing technology enablers (TE) with the requirements and functionalities needed to support the DEDALUS framework.

In the first 18 months of the project, significant progress has been made in coordinating task activities and advancing project goals. Regular conference calls at the task level have been instrumental in ensuring effective collaboration among team members.

An in-depth analysis of the existing Technology Enablers (TE) was conducted, building upon the preliminary assessment from the initial months. This analysis led to the development of detailed Use Cases, which were organized by categories to streamline implementation and facilitate understanding among stakeholders.

The team also revisited the legislative frameworks concerning Privacy, Cybersecurity, and Ethics in the demand response scenario. This review ensured that the project remains compliant with current regulations and addresses any new considerations that have emerged.

A new version of the system architecture was released, featuring more detailed specifications and enhancements based on the latest findings and analyses. This updated architecture provides a robust foundation for the next phases of the project.

The document D2.3 was written and completed. This comprehensive deliverable consolidates all the efforts made in analysing technology enablers, categorizing use cases, reviewing legislative frameworks, and updating the system architecture. A contribution was made to Deliverable D2.4, where refinements and details of the software architecture were developed, accompanied by a completed list of refined high-level requirements.

1.2.2.6. Work carried out per partner per task and effort spent

Table 2 - WP2 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	- Participation in the regular meetings for WP2 (Bi-weekly) and Task2.5 (Monthly). - Contribution to Deliverable D2.1, D2.2 and D2.4 - Participation in physical plenary meetings - Writing Deliverable D2.3	8.75



Partner	Work carried out in the WP within the period	PMs
	• Deep Analysis of TE • Identified requirements for the definition of the Use Cases • Definition of the DEDALUS architecture. • Revisited the legislative frameworks concerning Privacy, Cybersecurity, and Ethics in the demand response scenario	
2 - NTUA	General: • WP2 meetings attendance • Contribution to D2.1, D2.2, D2.3 and D2.4 • Participation in WP2 physical plenary meetings Task 2.3 • Conducted extensive scientific literature review aiming at shedding light on the key task concepts, including principally residential DR, energy literacy, energy behaviour, and principal demographic dimensions, among others. • Drafted the first version of the questionnaire. • Held bilateral calls with SIN to finalize the questionnaire • Adjusted the questionnaire after receiving feedback from the pilot leaders. • Guided the pilot site leaders into the process of translating and handing over the questionnaires. • Aligned questionnaire answers for further analysis • Processed the questionnaire answers to render them into a form suitable for data analysis and visualisation • Built Power BI dashboards on the questionnaire answers and utilised its filtering capacities to draw insights • Used the Power BI insights to characterise the pilot persona, segment pilots into further segments and derive their personas as well • Concluded on the personas' DR prospects and key elements for an efficient outreach strategy • Applied clustering algorithms into an open dataset to showcase the Machine Learning aspect of the segmentation and clustering problem • An empirical analysis as well as multiple evaluation metrics are leveraged to assess those algorithms.	12.92



Partner	Work carried out in the WP within the period	PMs
	Leveraged Explainable AI (xAI) to enhance the interpretability of our solution - Received different questionnaire answers and analysed them, creating the segmentation methodology and user personas for each country involved. - Delivered the combination of the two methodologies developed in D2.4, demonstrating the advantages in the Greek pilot, Task 2.5 - Contribution to the definition of the architecture - Bilateral calls with ENG to define the technical architecture	
3 - CARTIF	- WP2 meetings attendance. - Contribution to the definition of the DEDALUS architecture. - Contributions to D2.1, D2.3 and D2.4.	2.99
4 - UNIVPM	- WP2 meetings attendance. - Contribution to the definition of user requirements and technical requirements - Contributions to D2.1, D2.3 and D2.4	6.00
5 - ARC	- WP2 meetings attendance. - Contribution to the definition of technical requirements	4.00
6 - COMS	- Attending WP2 meetings - Contributing to definition of DEDALUS architecture - Contributing to data privacy and cyber security - Contributing to deliverables D2.1 and D2.3	2.49
7 - OurPower	- Gap analysis and high-level use case identification/definition contributing to D2.1 - Concept development for user engagement and co-creation - Pilot scoping regarding Austrian pilot - Consumer survey at Austrian pilot	4.17



Partner	Work carried out in the WP within the period	PMs
	• Organization, execution and evaluation of co-creation workshop with residential stakeholders • Concept creation and redesign of OurPower nudging application • Participation in meetings • Contributions and/or input to D2.1, D2.2, D2.3 and D2.4	
8 - GreenPoint	• Implementation out of Austrian pilot • Development of energy monitoring and distribution system • Consumer Survey at Austrian pilot • Participation in meetings	3.59
9 - BLUEPRINT	• Leading WP2 (reviewing and coordinating tasks, project management, scheduling meetings) • Adopted the IEC 62559 standard as a methodology to define and describe the DEDALUS Use Cases (UCs). • Highlighted high-level requirements and specifications for the early stage of DEDALUS development. • Identified four broad domains: Data Management, Performance Assessment, Smart Control, Market, and User-Centered Interaction. • Presented the list of DEDALUS actors. • Conducted workshops with pilot site partners and technical partners to ensure a harmonious description of the HLUCs across different stakeholders involved in the process. • Defined a final list of 11 HLUCs for the DEDALUS project. • Presented the main characteristics and requirements of each HLUC, including their scope and objectives, description, required DEDALUS pre-conditions, and basic steps. • Conducted a gap analysis and description of individual pilot sites in the DEDALUS project. • Conducted mapping of the most important HLUCs to be deployed at each demonstration site.	12.00



Partner	Work carried out in the WP within the period	PMs
	- Responsible of the submission of the last WP2 deliverable D2.4 including reviewing and writing parts of it. - Organizing inputs from task leaders into D2.4	
10 - FællesBo	Involvement of tenants (T2.3), including co-creation workshop (T2.4), and preparing pilot demonstration activities together with NEOGRID.	2.99
11 - NEOGRID	- Gap analysis and high-level use case identification/definition contributing to D2.1 defining and developing HLUC's - Input provided for D2.2 and D2.3 - Concept development for user engagement and co-creation - Preparation of co-creation workshops to be held in the spring 2024 - Pilot scoping regarding Danish pilot	1.14
12 - GREEN CITIES	- Participation in meetings and keeping track of the project developments. - Co-designing with consumers (Faellesbo)(T2.4) innovative social-driven incentive mechanisms and data-driven DR services - Concept development for user engagement - Layout and preparations of D2.4	3.44
13 - HERON	- WP2 meetings attendance. - Participation in defining activities for the use cases and requirements - Questionnaires preparation and delivery - Contribution to the definition of user requirements and technical requirements - Concept and scope development for Greek pilot site along with users engagement methodology - Organization of 1 co-creation workshop with end-users (Greek pilot site) focusing on drivers and barriers of the DR service and it nudging interventions	7.00



Partner	Work carried out in the WP within the period	PMs
	Collection and analysis of Greek pilot sites outputs contributing to D2.1, D2.3 and D2.4	
14 – DOM-X	- Organization of the co-creation workshops - Organization of bilateral calls with all pilot sites for the co-creation workshops - Synch with SIN for the co-creation workshops - Communicating the NUDGE H2020 project booklet with best practices on how to implement nudging features in a mobile application for supporting DR schemes - Organize and execute the Greek pilot co-creation workshop (gas pilot site) - Writing in the D2.2 - Compiling input from other partners for the D2.2 - Participation in WP2 calls	12.46
15 - ISRAA	- Attendance to WP2 meetings - Contribution to use cases definition with input provided for D2.1 - Contribution to T2.2 and T2.4 with incentives online study. Contribution to D2.2 through literature review and definition of the methodology for incentives online study. - Pilot scoping regarding Italian pilot, engagement of end-users - Questionnaires preparation and delivery - Preparation of the co-creation workshops 2 workshops carried out with the residents to discuss barriers and drivers and incentives to foster engagement - Outline of the Toolkit for a successful implementation of incentives as outcome of subtask 2.4.3.	7.98
16 - DYNAMIE	- Participation in project meetings WP2 - Contribution to use case - Contribution to pilot scoping, engagement of user - Contribution to energy issues - contribution to incentives related to energy market	1.51



Partner	Work carried out in the WP within the period	PMs
	Contribution to user's relevant arguments for engagement	
17 - CIMNE	- Participation in project meetings WP2 - Contribution to pilot scoping, engagement of users - Questionnaires preparation and delivery - Preparation and organization of co-creation workshops with building end-users - Direct contact and interaction with end-users.	3.01
18 - PEUSA	- Participation in project meetings - Questionnaires preparation and delivery - Preparation and organization of co-creation workshops with building end-users	0.50
19 - ROMUR	- Participation in project meetings - Contribution to pilot scoping, engagement of users - Questionnaires preparation and delivery	1.00
20 - CENTRICA	- Attend the meetings for WP2 - Contributing to deliverable D2.1	0.59
21 - TUC	In T2.1 TUC has contributed to the HLUCs specifications by defining HLUC10: < Building-level decentralized flexibility aggregation and tokenization for DR> and the specification of TE-5: DLT/Blockchain/Smart contracts solution. Also, TUC has provided the scope, context and existing infrastructure & flexible assets specifications for Pilot 7 multiplier (RO) and contributed to the overall DEDALUS architectural design. In T2.2 TUC has organized the co-creation workshop in Romania to explore how NFTs can be used to incentivize the participation in DR considering the housing context and collect inputs on NFT visual nudges. Finally, in T2.3 TUC has allocated efforts for collecting pilot 7 DR participant's socio-economic information, motivation and awareness regarding energy issues using DEDALUS questionnaires. In the reporting period TUC	9.00



Partner	Work carried out in the WP within the period	PMs
	contributed to three WP2 deliverables: D2.1, D2.2 and D2.4.	
22- SIN	• Internal coordination within WP2; WP meetings, bilateral meetings, emails • Attending kick-off meeting & GA • Developing the overall SSH framework integrating several WP2 tasks and overall project and pilot needs • Literature review & collecting information from previous projects to develop project activities on nudging, incentives, motivations, community building • Pilot scoping meetings & questionnaires to map pilots' setups, types of users, main goals and possibilities for co-creation workshops and other user engagement • Designing co-creation workshop methodology. This includes three distinct co-creation activities that link to ST2.4.1 (nudging UI and messaging), ST2.4.2 (user preferences for comfort-based flexibility modes), ST2.4.3 (financial and non-financial incentives) • Preparing discussion guides, note taking templates, and other facilitator materials for co-creation workshops • Writing D2.2 • Developing the SSH aspect of the segmentation questionnaire • Integrating segmentation results into co-creation workshop design • Integrating segmentation results in D2.2 • Collaboration in co-creation workshop planning and developing nudging elements	9.24
23 - ICONS	• Attending WP2 meetings.	0.34

1.2.2.7. Deviations from the Grant Agreement

As already reported in WP1, the project lacked coordination in the first months and this affected all WPs, especially the ones started at M1, like WP2. Starting from December 2024 (M8) the coordinator has properly planned the release of the missing/pending WP2 deliverables. At the time of writing delays have been caught up. Concerning the WP2 deliverables and activities, the following ones are the most relevant delays:



- Submission of D2.1. at M10 instead of M6
- Submission of D2.2 at M11 instead of month 8
- Submission of D2.3 at M13 instead of month 9.
- Submission of D2.4 at M17 instead of month 16.

1.2.2.8. *Corrective actions taken/suggested*

As soon as the coordination team was set up, the coordinating partner, ENG, prepared a plan together with WP2 leader. The plan was discussed with the PO and new deadline was established to release the pending documents. Many dedicated audio conferences have been organized, all the open issues discussed, all the documents drafted at that time were shared and discussed with all the involved partners. In the period between M8 and M9 all the expected deliverables were finalized and submitted. The last deliverable was only a couple of days late.

1.2.3. **WP3 – Technology enablers for residential energy consumer data-driven ‘activation’**

[M4-M28] [Leader: **CARTIF**, Participants: ENG, NTUA, CARTIF, ARC, COMS, OurPower, GREENPOINT, BLUEPRINT, FællesBo, NEOGRID, HERON, DOM-X, ISRAA, DYNAMIE, CIMNE, PEUSA, ROMUR, CENTRICA, TUC]

WP3 aims at delivering: i) DR-ready smart appliance IoT platform Energy Data Space compliant interoperability layer for residential flexibility mapping among stakeholders; ii) Data/metadata representations for flexibility and DR services characterisation applying a semantic interoperability framework; iii) A suite of ML models for multi-energy flexibility assessment and forecasting in residential buildings with installed smart devices; iv) blockchain-based data exchange and flexibility tokenisation with DR settlement using smart contracts; and v) Privacy, security and data protection framework for sovereign data sharing and smart devices integration in the DR services.

This work package is delivering a set of technology enablers that are being used by the different services under development in the scope of the scope of WP4.

During the first eighteen months of the project, WP3 partners have worked on the various tasks to develop a set of technology enablers that will facilitate the creation of the value-added services within WP4. As a result, the first technology release of those technology enablers was delivered by the end of May 2024 (MS4), and the second technology release is currently under development, with completion targeted for the end of December 2024 (MS7). Besides, deliverable D3.1 (delivered at M12) was completed, and D3.2 is expected to be finalized soon (at the time of writing it has been already submitted at month 19, as per GA).

It is worth mentioning that monthly WP3 follow-up meetings were established from the very beginning of the project. In these meetings, the partners (specially task leaders) can update CARTIF (as WP3 leader) on the new achievements of their tasks.

Lastly, it is important to highlight that multiple publications have been produced to describe the various achievements of this work package.



1.2.3.1. Task 3.1 DR-ready smart device & flexible resource data collection

[M4-M28] [Leader: **ARC**, Participants: ENG, CARTIF, ARC, OurPower, GreenPoint, BLUEPRINT, NEOGRID, HERON, DOMX, ISRAA, DYNAMO, CIMNE, PEUSA, ROMUR, CENTRICA]

This task aims to ensure technical interoperability between smart devices through a DR-ready IoT platform (commercially known as Arçelik HomeWhiz and internally as TE-2). The Arçelik IoT Platform, branded as HomeWhiz, is detailed in D3.1 and serves as a transformative solution for advancing connected and intelligent living spaces. The platform will provide open APIs, service enablers, and tools for integration across various use cases. Integration of legacy devices will be facilitated using smart controllers with protocols such as BACnet, Modbus, LonWorks, Opentherm, and OCPP. As part of the DEDALUS project, a customization has been implemented with an MQTT broker interfacing with HomeWhiz's IoT Core to ensure seamless data flow for the project-specific requirements. Secure edge gateways will aggregate data from devices using Wi-Fi, ZigBee, and LoRa, exposing vendor-specific APIs via frameworks like OpenHAB and HomeAssistant.

Data from appliances such as washing machines and refrigerators use built-in Wi-Fi for direct connectivity, while smaller devices connect through Bluetooth Low Energy (BLE) gateways. The data includes metrics such as power consumption, voltage, and current collected from smart plugs and environmental readings like temperature and humidity from gateways, enabling detailed appliance usage monitoring. Data from sources including smart meters, thermostats, sensors, smart appliances, and EV chargers will be harmonized and integrated using open connectors over data brokers (e.g., MQTT), REST APIs, and repositories such as Microsoft SQL Server and European Data Spaces.

For example, data packets from smart plugs are structured as follows:

```
{
  "plug_id": "S909923555334062637262",
  "power": 0.31,
  "voltage": 221.37,
  "current": 0.01
}
```

Gateways provide complementary environmental data:

```
{
  "gateway_id": "S90991410259867202126",
  "temperature": 28.4,
  "humidity": 43.3
}
```

In addition to data collection, commands can be issued to devices, such as:



```
{
  "plug_id": "S909923555334062637262",
  "gw_id": "S90991410259867202126",
  "state": "ON/OFF"
}
```

Additionally, possible AC models for use in demand response (DR) pilots have been evaluated, and the development of data sharing capabilities from smart plugs and gateways to pilot systems has commenced.

The HomeWhiz platform leverages Amazon Web Services (AWS), including AWS IoT Core for device connectivity, AWS Lambda for microservice management, and Amazon RDS for data storage, ensuring scalability and robustness. Devices communicate via MQTT for secure and efficient data exchange.

HomeWhiz is a smart home platform that integrates various appliances (e.g., ovens, refrigerators, washing machines) for remote control and management. It utilizes AWS Cloud services to ensure scalability and robustness, with devices connecting via IoT Core over MQTT. The data collected includes power consumption, temperature, and humidity from smart plugs and gateways.

The platform is developed to support flexibility services, web services, and MQTT integration, ensuring real-time data processing, scalability, and interoperability.

In Greece, 47 smart plugs and 12 gateways will be deployed across 4 units, with plans to expand to integrate air conditioners (ACs) for DR readiness. In Spain, 34 smart plugs and 10 gateways have been deployed for pilot testing, allowing evaluation and refinement of integration protocols. Alongside these deployments, possible AC models for demand response (DR) pilots have been evaluated, and the development of data-sharing capabilities from smart plugs and gateways to pilot systems has been initiated.

The DEDALUS Ontology, based on the Distributed Collaborative Ontology (dCO), is being implemented to support energy generation, distribution, consumption, and optimization. This builds on frameworks such as SAREF, BOT, and IFC, ensuring a standardized approach to data payloads.

1.2.3.2. Task 3.2 Energy Data Space compliant interoperability layer

[M4-M28] [Leader: **ENG**, Participants: **ENG**, **CARTIF**, **ARC**, **DOM-X**]

The main objectives of Task 3.2 are to map the full DR chain from the DR service consumer request to the control of the flexibility resources (mainly smart devices) on the field, to manage the exchange of information between the business roles (e.g. prosumers or residential buildings - the source of flexibility, and energy market stakeholders, such as aggregators). The exchange of information will be based on functions (e.g., flexibility offers of prosumers and aggregated



flexibility) and interfaces (i.e. the connector to the roles) that will represent the generic business process covered by the proposed set of DR services. This task will provide technical and syntactic interoperability towards the smart appliances infrastructure (T3.1), as well as and semantic interoperability in services and systems (T3.3),

In the referred period, potential solutions to data-driven Energy Data Space compliant interoperability have been analysed and proposed. Data Space is a decentralised infrastructure for trustworthy data sharing and exchange in data ecosystems, based on commonly agreed principles. A Data Space represents a data sharing concept without a central storage. Thus, data remains at its source and is only shared when needed. This enables data providers to be sovereign over the data as long as data is needed, and when it is needed IDS have to ensure to attach usage policies to the data, which systems and users can follow. In this regard the IDS aims at meeting the following goals. A data space ensures Trust, Security, Data sovereignty (a data owner can define usage restriction to their data, before sharing it with data consumers. Data consumers must accept the usage restrictions). Standardized interoperability is needed to build up data space, because different data ecosystems will exchange different kinds of data in different formats and protocols. Only when the interoperability is standardized, every system can be interoperable in a data space. Data connectors are the core of the data space ecosystem; it is the gateway to connect existing systems and their data to an IDS ecosystem. The IDS Connector allows to exchange data and enrich it with metadata. In the referred period, different types of connectors have been examined. The connectors for data space connect existing systems and their data to a data space ecosystem. Different types of connectors have been evaluated.

The OneNet Connector will be installed at pilot sites. It combines the IDS principles with the advantages of the FIWARE ecosystem and ensures a seamless and secure data exchange in a completely end-to-end decentralized approach. The OneNet Connector is ready-to-go, ready to be installed in any environment and integrated with existing platforms via application programming interfaces (APIs). It is fully integrated with the FIWARE Context Broker (in the NGSI-LD version), facilitates data exchange process through the graphical user Interface (GUI) or via API and extends the interaction between data producer/consumer by providing handshake and publish/subscribe mechanisms for the offered services. The OneNet Connector supports more than 60 different harmonized services and data profiles in the smart grid and energy field and can be easily extended with additional services also in other domains.

Furthermore, a FIWARE based solution for technical and semantic interoperability have been analysed and proposed to the Consortium. FIWARE is an open-source initiative that promotes open standards, open-source software, and open data to endorse collaboration, interoperability, and reusability. FIWARE provides a common platform and a set of APIs that ensure interoperability between different systems and components. FIWARE Next Generation Service Interface (NGSI) is the API exported by a FIWARE Context Broker, the core component of FIWARE based platform, used to update or consume context information. FIWARE NGSI Agents allow to simplify the management and the integration of devices by collecting data from devices through heterogeneous protocols and translating them into the standard platform language. FIWARE Context Broker allows to publish, consume, and subscribe to data coming from multiple sources.



As part of the deliverable D3.1 “DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (1st technology release)”, a Proof of Concept (PoC) was designed and implemented. The main purpose of the PoC was to demonstrate the feasibility of the proposed solution and its functionalities. It has been assumed that two temperature sensors are connected to electric devices to monitor their working status and report anomalies. The FIWARE platform is connected to the two sensors, receiving the appropriate MQTT topics from the sensors, by means of the Node-RED IoT Agent configured for the PoC. The Agent works as a wrapper between the sensors and the FIWARE Context Broker. Two versions of the Context Broker have been deployed, namely Orion-v2 and Orion-LD. The FIWARE Context Broker receives continuously NGSI entities, provided by the Node-RED IoT Agent on the basis of the received measurements.

In the next PoC refinement phase, which led to the PoC released as D3.2 “DEDALUS technological enablers for multi-value and energy carrier-agnostic residential DR (2nd technology release)”, the system was integrated with live data from the Danish pilot. The integration is built upon the previous PoC by refining the data processing steps and ensuring real-time data exchange with other components of the DEDALUS framework. In the refined PoC, semantic interoperability has been addressed enabling different components to interact within a common platform, using all the same common data by calling them unambiguously and precisely, but at the same time allowing them to continue to use their own local data by calling them differently and independently of others. For example, the concept of temperature can be associated with the agreed URI [http://www.w3.org/2011/02/safer/temperature](#), while one component may internally call it 'temperature' (in English), another 'temperatura' (in Italian), and another 'temperatur' (in Danish). To this end, the vocabulary of the DEDALUS ontology has been used.

The Danish IoT platform sends to the DEDALUS interoperability layer information about temperature, humidity, CO₂ emission, and battery voltage level, gathered through the devices it manages. This information is specifically named by each device, and the platform exposes it via an appropriate API. An ETL (Extract, Transform, Load) module converts the information from the device into a standardized, agreed-upon JSON format. The information is, then, sent via an MQTT queue, following the publish/subscribe pattern, to an IoT Agent. The IoT Agent, which is suitably configured and parameterized, performs a transformation of all specific terms related to the measurements sent by that platform into a common vocabulary chosen to be used within the NGSI-LD Context Broker.

1.2.3.3. Task 3.3 DEDALUS flexibility and DR data models and standards

[M4-M28] [Leader: **CARTIF**, Participants: CARTIF, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, HERON, DOM-X, ISRAA, DYNNAME, CIMNE, PEUSA, ROMUR]

The main objective of this task is to define the ontology that will be used within the DEDALUS project to represent the information to be managed by other DEDALUS developments. The main idea is to adapt and/or extend existing ontologies such as SAREF, OpenADR, Brick.



Multiple activities have been conducted in the scope of T3.3 in order to create an ontology that considers the information requirements of the various services, pilots, and other aspects of the DEDALUS project.

As there is a broad range of data requirements (coming mainly from the technology enablers (WP3), the services to be developed (WP4) and the pilots (WP5)), a detailed analysis of all of this information requirements is key in order to know all the entities, attributes and relationships that would be needed within the final DEDALUS ontology.

Considering this matter, the following process was defined as the methodology to develop the final ontology:

- First of all, a SoTA review should be done in order to choose the most appropriate ontology to be used as a starting point taking into account DEDALUS high level needs.
- Then, all the partners (mainly technology enablers (WP3), services to be developed (WP4) and the pilots (WP5)) should be asked to specify their specific data requirements from their specific point of view.
- Once the aforementioned specific data requirements were available, a matching between them and the selected ontology should be conducted.
- Based on the results of the matching, a proposal will be done on how to extend the starting ontology with entities/attributes/relationships from other known (and public) ontologies.

Ontologies are able to provide a standardized way to represent knowledge and concepts within a specific domain. They help developers in modelling complex systems, defining relationships between entities, and establishing a common understanding of domain-specific terminology.

It is important to note that one of the most important practices in ontology development is based on reusing, so a set of five ontologies have been finally explored to determine the most suitable one to serve as the foundation of the DEDALUS ontology. In this sense, the set of ontologies that were reviewed are: Smart Applications REFERENCE Ontology (SAREF³), Open Automated Demand Response ontology (OpenADR⁴), Brick⁵, domOS Common Ontology (dCO⁶) and the BIGG Project Ontology⁷.

Once the aforementioned ontologies were analysed and compared, it was dCO the one selected to be used as the starting point for the DEDALUS ontology, but, as there was some information needed that were not included in dCO, it was decided to extend it with some parts from other well-known ontologies as BIGG, Interconnect and SAREF4City.

As guidance should be provided to the partners in order to help them using the ontology, some documentation was prepared in order to give them a starting point to start working on it. In this sense, an example on how to populate the ontology (and how to do it using Protegé) were delivered, and a set of SPARQL queries to let the partners realize about the potential of extracting knowledge from a populated ontology. This information will be included in D3.2.

³ <https://saref.etsi.org/>

⁴ <https://albaizq.github.io/OpenADRontology/OnToology/ontology/openADRontology.owl/documentation/index-en.html>

⁵ <https://brickschema.org/ontology/>

⁶ <https://www.w3id.org/dco>

⁷ <https://github.com/biggproject/Ontology>



Continuing with the efforts to help the partners on the use of ontologies (specifically, the DEDALUS one), an interoperability workshop was organized in September 2024. During this workshop, both ENG and CARTIF gave the partners a lot of information on:

How to use the interoperability layers from T3.2.

How to populate the DEDALUS ontology using Protégé.

How to work with ontologies from Python (mainly using two libraries: OwlReady2 and RDFLib).

Once all this information was provided to the partners, the pilots and services began considering how to use the ontology. To support this process, several audio meetings are being held to discuss and address the matter.

In the case of the Spanish pilot, CIMNE has generated a v2 of the ontology BIGG. Several classes of the former BIGG ontology are already included in the DEDALUS ontology, and both are based on the SAREF family. This new version(v2) of the BIGG ontology includes a new class named s4bldg:ElectricFlowStorageDevice, corresponding to the district-level battery already installed in the Spanish pilot. This new class was taken from s4bldg, which makes it interoperable and integrable in the DEDALUS Ontology. Furthermore, CIMNE, in collaboration with CARTIF, is developing a procedure to make both ontologies more interoperable.

Regarding D3.2, CARTIF has created the ToC and has provided contents to most of the sections devoted to CARTIF's activities. Besides, CARTIF is reviewing and supervising all the partner's contributions.

It is worth mentioning that two publications, associated with attendance at the corresponding conferences, have been produced in collaboration with ENG to disseminate the achievements of T3.2 and T3.3.

Last but not least, CARTIF has started to study how to generate a Smart Data Model based on the DEDALUS ontology aiming to prepare the ontology to be used within T3.2 developments.

1.2.3.4. Task 3.4 Multi-energy flexibility modelling for data-driven residential DR

[M4-M28] [Leader: **NTUA**, Participants: NTUA, CARTIF, COMS, GreenPoint, FællesBo, NEOGRID, HERON, DOM-X, ISRAA, DYNAME, CIMNE, CENTRICA]

The main objective of this task is to develop a methodology for estimating the potential flexibility that may be offered through heterogeneous multi-energy flexible loads, in the context of a building or district. This methodology relies on the estimated, predicted or real (if known) abundance of different flexible loads but also can calculate the estimated flexibility based on building, apartment or dwelling demand time series. Moreover, the task focuses on investigating how the output of this methodology will be used in real residential DR scenarios.

The task is divided into two distinct approaches: aggregated building flexibility and disaggregation-based flexibility at the appliance level. Initiated in month 1, the work conducted over the first 18 months for the first approach can be seen below.



In the first eighteen months, our efforts focused on the construction of a building's flexibility baseline load, approximating its electricity consumption and production in the absence of specific demand response (DR) events. Deviations in energy consumption and production from this baseline, within predetermined margins, were identified as indicators of potential flexibility. To accurately forecast next-day load variations, we employed Long Short-Term Memory (LSTM) networks, optimizing their parameters to enhance performance in predicting energy consumption and production for all weekdays. The proposed solution was validated through real-world experiments using prosumer data collected from a building of the Austrian Pilot. Towards a better understanding and quantification of flexibility, specific Energy Flexibility Indicators (EFIs) have been defined, addressing both energy consumption (consumers) and production (prosumers). Furthermore, a scientific article has been published in the 15th International Conference on Information, Intelligence, Systems and Applications (IISA 2024) that was hosted in Chania, Greece. The article gained the Best Student Paper Award of the conference as well.

Adding to the previous, the disaggregated flexibility assessment methodology has been created. In more detail, disaggregation is the process of separating total energy consumption data into its individual components such as lighting, heating, air conditioning, EV chargers, and other household appliances. The primary goal of this methodology is to create a clearer picture of each energy load's flexibility potential. The development of this methodology includes the implementation of advanced machine learning models for disaggregation, such as Non-Intrusive Load Monitoring (NILM) techniques, and the integration of device-level predictive models. NTUA introduced a Denoising Autoencoder architecture for disaggregating household loads and tested on open datasets. This effort and the produced results have been documented in D3.2.

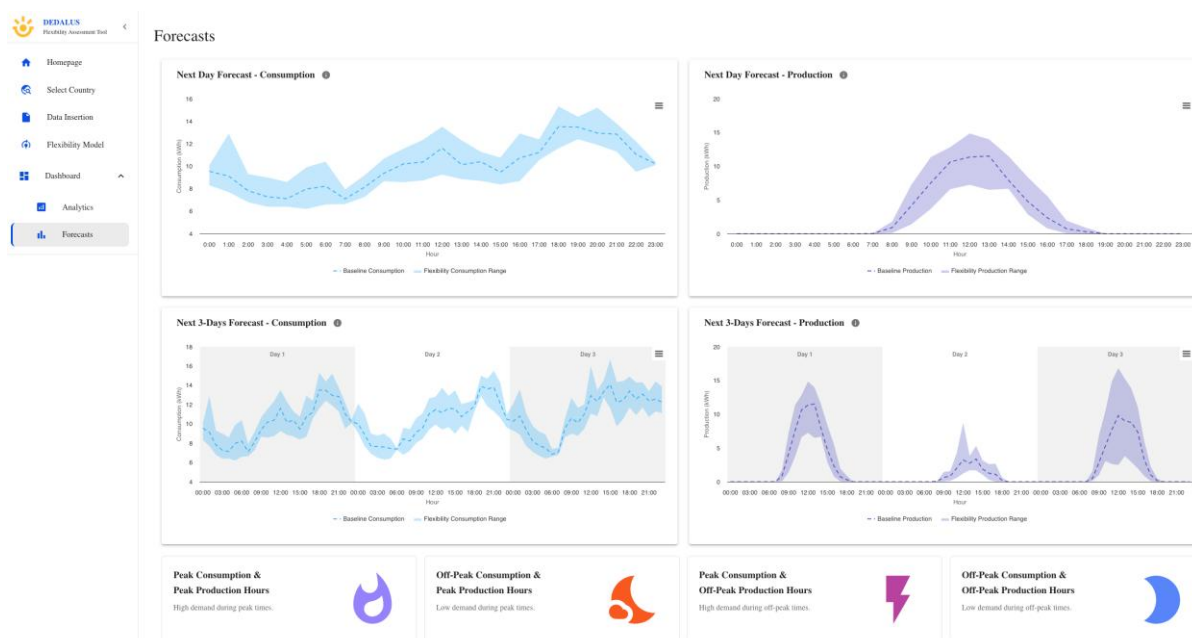


Figure 2 - Flexibility Assessment Tool

Following the work in the aggregated flexibility assessment, the DEDALUS Flexibility Tool has been created. It is a platform designed for energy stakeholders from 7 different EU countries –



namely Greece, Spain, Italy, Denmark, Austria, Romania and Ireland. Building managers, operators, ESCOs and Utility Companies can access this platform to analyse and manage energy flexibility in buildings or apartments, especially with integrated Photovoltaic systems. Based on the flexibility assessment, forecasting and Non-Intrusive Load Monitoring (NILM) capabilities provided, detailed suggestions for energy management (up to 7 days in advance) are received by the users.

1.2.3.5. Task 3.5 Blockchain-based interoperability for flexibility tokenisation & DR settlement

[M4-M28] [Leader: TUC, Participants: ENG, COMS, TUC]

The main objective of task T3.5 is to develop and deploy a blockchain-based solution for privacy-preserving flexibility assessment, P2P flexibility tokenisation and DR settlement using smart contracts (TE-5). The solution is based on a scalable blockchain based energy ledger that enables secured, trusted, and flexible data/models sharing among buildings and residents. It does innovatively consider the integration of cryptographic techniques, such as holomorphic encryption or zero knowledge proofs to assure the privacy of energy and non-energy data. This helps in assuring the data integrity and transparency due to blockchain integration needed for data sovereignty in the context of Energy Data Spaces, and the secured data exchange in DR settlement. The blockchain infrastructure is being utilized for decentralised parametrisation of prediction models in a way of processing the data within the stakeholder premise (e.g., prosumer, building, or community level) and sharing in a tamper-proof manner the model via the blockchain network.

We have started by carrying out an analysis of blockchain based solutions in previous projects to understand the adaptations needed for in the context of buildings and DR as foreseen in DEDALUS. In parallel, we have conducted an analysis of the currently available approaches for blockchain-based flexibility tokenization and decentralized DR. We have then defined an internal architecture for TE-5 technological enabler for blockchain-based flexibility tokenization & DR settlement and implemented the TE-5 data model to be compatible the DEDALUS ontology model. TE-5 consists of several interconnected components that facilitate decentralized DR and building flexibility aggregation. We have employed a blockchain-based distributed ledger to securely store and digitally fingerprint energy consumption data. We have developed components to integrate energy data from IoT devices, to digitally fingerprint it using hashes and store it on chain on hourly bases. A forecasting component has been developed that utilizes monitored data from the database to predict building energy demand and flexibility for the day-ahead. Using the predicted values, the building manager can define a bid for energy flexibility in a building-level flexibility market. This bid aims to aggregate the building's flexibility needs from residents' apartments (i.e. offers of flexibility) to provide services to the local grid. The entire trading process is decentralized and implemented using self-enforcing smart contracts. A DR settlement component uses monitored data to trigger the appropriate remuneration for flexibility providers, based on their energy delivery commitment. Smart contracts compare the monitored data against these commitments, and commitments and generate the actual settlement using flexibility tokenization (based on ERC20, ERC721). The first version of TE-5 has been delivered



in D3.1 while the refined and improved 2nd version will be reported in D3.2 at M19 (new token implemented based on ERC1155, new forecasting algorithms integrated, new dashboards developed). The 2nd version of TE-5 will be next evaluated in the full-pilot cycle phase (M17-M25) according to the pilots' implementation plan in DEDALUS Pilot 7 from Romania and the results of the evaluation will guide the refinements and new improvements of the 3rd version of the technology (i.e. homomorphic encryption integration).

1.2.3.6. Task 3.6 Privacy, security and data protection framework for trusted data sharing and AI [COMS]

[M4-M28] [Leader: **COMS**, Participants: ENG, ARC, COMS]

The goal of this task is to develop tools that can ensure compliance with privacy, security and legal requirements defined in the project. The tools enable the pilots trusted data sharing among DEDALUS services and potential external data spaces. The focus of task is fine-grained identity and access management (IAM) with mechanisms for data registration, consent exchange, policy-based data exchange and auditing.

This task has started in month 4 and in the first 5 months of activities, the technology enablers such as Universal Secure Gateway (TE-6) and Privacy Protection Enforcement (TE-7) with identity and access management were investigated for the integration. The overview of privacy, security and legal compliance requirements defined in T2.5 was done and extended to include also technical aspects of framework developed in T3.6. A solution to protect an interoperability tool from T3.2 with identity and access management based on Envoy Proxy and Keycloak was proposed and outlined.

From M9 until M18 the initial proposal for the API Gateway was implemented. Fine-grained access management was implemented using Keycloak as local identity provider with support for OpenID Connect, OAuth2 and UMA 2 authentication and authorization mechanisms. Envoy proxy was integrated to provide mechanisms of authentication and transport encryption while separating the security and privacy protection mechanisms from the business logic of DR applications. Successful establishment of encrypted mTLS connection at the API Gateway entry point results in redirection to the custom policy enforcement point (PEP) proxy implementing the User-Managed Access (UMA) 2.0 protocol for fine-grained user and client-based authorization. The solution is currently being evaluated in Austrian pilot during the full-pilot cycle phase (M17-M25) and the results of evaluation will guide us to the implementation of improvements.



1.2.3.7. Work carried out per partner per task and effort spent

Table 3 – WP3 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	In the referred period, ENG has analysed and proposed the Consortium potential solutions to data-driven Energy Data Space compliant interoperability: • A solution based on the FIWARE Context broker and Next Generation Service Interface (NGSI) Agents which allows to model, manage, and gather context information at large scale enabling context-aware applications. • A solution based on Connectors for Data Space connect existing systems and their data to a data space ecosystem; different types of connectors have been evaluated. ENG has participated to periodic WP3 remote meetings. Both the solutions will be applied in DEDALUS. A preliminary PoC for the FIWARE solution has been developed and released as D3.1. A refined version of the PoC has been released as D3.2. ENG has participated to periodic WP3 remote meetings and, moreover, to periodic call of WP5 in order to understand pilot interoperability needs.	9.18
2 - NTUA	General: • WP3 meetings attendance. • Contribution in D3.1 and D3.2 • Participation in WP3 physical plenary meetings Task 3.4: • Extensive scientific literature on aggregated energy flexibility. • Definition of aggregated (building) flexibility baseline load. • Definition of aggregated (building) maximum flexibility load. • Definition of aggregated (building) minimum flexibility load. • Introduction of Energy Flexibility Indices (EFIs)	5.58



Partner	Work carried out in the WP within the period	PMs
	• Long Short-Term Memory (LSTM) model development for prediction of energy consumption and production for all weekdays. • Austrian Pilot Data pre-processing pipelines. • Evaluation of the methodology on the Austrian Pilot. • Introduction of disaggregated (device-level) flexibility methodology • Non-intrusive load monitoring (NILM) techniques using a Denoising Autoencoder • Design definition of the Flexibility assessment tool • Technical Architecture of the Flexibility assessment tool creation • Created detailed mockups using Figma • Defined Restful APIs for the backend side of the application incorporating all functionalities • Defined the database including schema, deletion and indexing pipelines • Build the front-end upon React framework • Bug fixes and development for the finalization of the first version of the Flexibility Assessment Tool	
3 - CARTIF	• Review of the initial set of ontologies to be checked in order to decide about the one to be used as starting point. • Information gathering (both from services and pilots) about data requirements. • Follow-up meetings organization and WP3. • Definition of the table of contents for D3.1. • Creation of a proposal of distribution of the work to be done within D3.1 among all the involved partners. • Contribution to D3.1, including the supervision and review of all partners' contributions. Management of the review process and preparation of the final version of the document. • Complete development of the DEDALUS ontology. • Creation of an example on how to use the ontology (populating the ontology, creation of SPARQL queries, etc.) • Interoperability workshop.	14.20



Partner	Work carried out in the WP within the period	PMs
	• Creation of the ToC and contribution to D3.2, including the supervision and review of all partners' contributions. • Definition of the integration between the interoperability layers from T3.2 and the ontology from T3.3. • CARTIF has been working on analysing the data provided by CIMNE (Spanish Pilot) and on understanding the structure of the demo-site to develop a disaggregation algorithm and compare its results to the ones obtained by NTUA in the scope of T3.4.	
5 - ARC	• WP3 meetings attendance. • Contribution in D3.1 • Participation in WP3 physical plenary meetings. • Proposed an architecture for DR-ready smart device and flexible resource data collection. • Conducted meetings with pilots in Greece and Spain. • Sent smart plugs and gateways to pilots in Greece and Spain. • Evaluated possible AC models to utilize in pilots for DR. • Started the development of sharing data from smart plugs and gateways to pilots.	10.50
6 - COMS	• Coordination of activities and meetings on T3.6 • Reviewing the technology enablers in T3.6 • Privacy, security and data protection framework definition • Privacy enforcement point API Gateway solution definition to provide identity and access management to the data owner • Keycloak integration into API Gateway solution (T3.6) • Envoy proxy integration into API Gateway solution (T3.6) • Implementation of custom Policy Enforcement Point (PEP) proxy to support fine-grained access management	9.67



Partner	Work carried out in the WP within the period	PMs
	- Participation in activities related to T3.4 and T3.5 - Contributing to D3.1 and D3.2	
7 - OurPower	- Implementation and refinement of data collection tools - Implementation and refinement of data processing pipelines - Data model requirements definition - Consultations with stakeholders to gather data model requirements - Participation in meetings and workshops - Contributions to and discussions on service integration	1.08
8 - GreenPoint	- Development of data collection method - Implementation of data collection tools - Legal consultations for data use and collection - Data monitoring - Participation in meetings	2.24
9 - BLUEPRINT	- Analysis for pilot input for data model requirements and contributions to discussions on service integration - Participation in meetings	0.80
10 - FællesBo	Preparing pilot demonstration activities (T3.3)	0.99
11 - NEOGRID	- Data model requirements definition - Proposed and worked with the DomOS Ontology, providing answers to DomOS questions. - Provide input for data model requirements - Participation in meetings - Developed input and prototypes for nudging activities - Ontology discussions and requirements with respect to Pilot 2 compliance	0.63
13 - HERON	- Participation in meetings for Information gathering on data requirements - Data provision and required information for the development of DEDALUS services	3.81



Partner	Work carried out in the WP within the period	PMs
	- Information gathering for data models requirements - Preparation in planning and deployment activities for data sharing solutions - Participation in activities for adapting proposed ontology for DEDALUS services - General WP3 meetings attendance.	
14 - DOM-X	- Data provision and required information for the development of Dedalus services - Development of specific APIs for data sharing among the DEDALUS consortium - Information gathering for data models requirements - Participation in WP3 meetings	4.66
15 - ISRAA	- Attendance to WP3 meetings - Preparing pilot demonstration activities - Coordination with technical partners contributing to the Italian pilot (ENG, UNIVPM) concerning the use of ontology - Attendance to the workshop	1.90
16 - DYNAME	- Participation in project meetings - Contribution to Pilot activities with reference to requirement of services, multi energy, residential scenarios - Contribution to collect and analyse data available for pilot	0.92
17 - CIMNE	- Data model requirements definition - Participation in coordination meetings - Participating to develop an architecture for DR-ready smart device and flexible resource data collection. T3.1 - Test smart plugs and gateways from Arcelik. T3.1 - Implementation of data collection tools, from smart plugs and smart appliances. T3.1 - Test of sharing data platform from smart plugs and gateways from Arcelik. T3.1	4.00



Partner	Work carried out in the WP within the period	PMs
	- Revision of BIGGv2 to include heatpumps, PV, smartplugs, and battery. T3.3 - Definition of new battery element in the ontology. T3.3 - Identification of the procedures to make BIGGv2 and DEDALUS ontologies more interoperable. - Integration of the data inside BIGGv2. T.3.3	
18 - PEUSA	- Data model requirements definition - Meeting and networking with stakeholders to gather and harmonize data - Participation in coordination meetings	1.00
19 - ROMUR	- Participation in coordination meetings - Participating to develop an architecture for DR-ready smart device and flexible resource data collection. - Implementation of data collection tools - Data model requirements definition	2.74
20 - CENTRICA	- Attend WP3 meetings - Review the state-of-the art on flexibility forecast	1.98
21 - TUC	In T3.5 TUC has designed and developed the 1st and 2nd versions of TE-5 technological enabler for blockchain-based interoperability for flexibility tokenization & DR settlement. Also, TUC has worked on integrating the 2nd version of TE-5 in DEDALUS Pilot 7 from Romania that will be evaluated in the full-pilot cycle phase (M17-M25).	5.00

1.2.3.8. *Deviations from the Grant Agreement*

Up to now, there have been no deviations from the original roadmap, ensuring that all scheduled milestones are met according to the predefined timeline.

1.2.3.9. *Corrective actions taken/suggested*

Not applicable.



1.2.4. WP4 – Value added energy and non-energy data-driven services for DR in energy-efficient residential buildings

[M5-M29] [Leader: **ENG**, Participants: ENG, NTUA, CARTIF, UNIVPM, COMS, OurPower, BLUEPRINT, NEOGRID, HERON, DOM-X, ISRAA, CIMNE, ROMUR, CENTRICA, TUC]

WP4 is delivering value-added, data-driven, energy and non-energy services for DR in energy-efficient residential buildings, addressing: Comfort-based flexibility models & DR optimal management tools; Flexibility assessment and pre-aggregation for energy communities/districts; Optimal aggregation of flexible energy resources for district heating management.

This WP started at M5, and the main progress achieved in the first months are mainly related to organisation of activities at task level focusing on the main Task's outcome and the active participation of involved partners. Moreover, an analysis of Technologies Enablers involved in this WP was done with the aim to understand how the current version of available software can be extended in the context of the Project according to the functional specification coming from the WP2. Indeed, a strict collaboration with this WP was established as requirement for the incoming development activity. In parallel, conference calls have been organized at WP and Task level in order to coordinate the work among participants and monitor the development activity. The main achievement of this WP in this reporting period is D4.1. This deliverable reports about the first wave of DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the 1st technology development cycle completion, MS4 due at M13.

D4.1 has a twofold objective: to describe the software artifacts released, as DEDALUS Technology Enablers (TE) and services, while also providing the technical specifications of TEs and more in general of services that are outcome of WP4. Indeed, a GitHub repository has been set up for allowing technical partners to commit their software code. Moreover, D4.1 has been conceived as technical specification, addressing as intended audience, mainly services developers. After the submission of D4.1, participants focused on D4.2 "Second wave" of DEDALUS Digital Twins and data-driven services for residential DR, corresponding to the second technical release MS7 due at M20. This deliverable is currently being written.

1.2.4.1. Task 4.1 Digital Twins for optimal assets vs consumers flexibility planning

[M5-M29] [Leader: **ENG**, Participants: ENG, HERON, CENTRICA]

This task is related to the development of Digital Twins for optimal assets vs consumers flexibility planning. The Technological Enablers (TE) that are involved are:

- TE-8: Digital Twins for consumer-aware optimal flexibility planning: tool designed to optimise the management of various assets and mainly to support consumers in the optimal management and assessment of flexibility resources
- TE-5 blockchain and smart contracts integrated DT: a DT model for the students' apartments building with the purpose of supporting decision making in demand response programs and simulating the execution of apartment level DR actions.



Both TEs will be extended in the context of the project according to the functional requirements defined in the context of the WP2.

Task 4.1 started at M5, and the work carried out from M5 to M18 can be summarised as follows:

- Kick off the task was arranged to formally start the task activity and involve partners.
- The scope of the task was defined underling the main outcomes and the plan to follow to achieve them.
- The role of each partner was defined in terms of development activity and support to the pilot in which TEs are expected to be exploited.
- The list of TEs under the scope of this task was refined together with the mapping with pilots in which they will be exploited.
- Development of first wave of TE-8 and TE-5 formally provided in D4.1

TE-8 is applied across different scopes in pilots 2 (DK), 3 (GR), and 4 (IT). In the first two pilots, MS4 consists in a data analysis process as prerequisite for developing the Simulation Engine component; regarding the pilot 4, a first version of Middleware API is developed, The first version of TE-5, mainly exploited in the Romanian pilot, consists in the initial implementation of Analytics component, of Apartment models and the DR Optimisation Decision Making algorithm.

1.2.4.2. Task 4.2 Optimal trade-off among comfort, energy and building performance

[M5-M29] [Leader: **UNIVPM**, Participants: ENG, UNIVPM, ISRAA]

The task aims to develop a comfort-based flexibility model for Demand Response (DR) services. It integrates sensors for indoor environmental conditions and weather data, using machine learning to personalize actions based on user behaviour. This model optimizes energy use while maintaining comfort, enabling dynamic adjustments to building energy demand in response to grid conditions.

In the first 18 months of the project, significant progress was achieved in the following aspects related to the development of the comfort-based flexibility service:

1. **Concept Creation and Model Definition:** The project focused on defining a robust concept for the "Optimal trade-off among comfort, energy, and building performance" service. This involved creating strategies to ensure indoor comfort by harmonizing factors of Indoor Environmental Quality (IEQ) such as thermal comfort with energy consumption and costs. The service aimed to facilitate Demand Response (DR) actions while prioritizing occupant health and well-being during their implementation. A specific architecture has been developed for the comfort-based flexibility model.
2. **Participation in Coordination Meetings (WP4 and WP3):** Regular coordination meetings within WP4 were integral to advancing the development of innovative services. In addition, coordination meetings have been made to define a strategy to include the energy flexibility model of T3.4 into this task.



3. Utilization of AI-based Algorithms: the service leveraged AI-based algorithms to quantify and predict comfort-related factors such as Thermal Sensation Vote (TSV), predicted Temperature and Humidity parameters. Historical data from pilot buildings and real-time sensor data were used to develop predictive models. These models provided insights to stakeholders for managing buildings efficiently while maintaining optimal comfort conditions.
4. Thermal Sensation Vote questionnaire development.
5. DR strategy based on comfort-based flexibility model

Overall, efforts were aimed at laying the foundation for implementing a comprehensive comfort-based flexibility model within the project framework.

1.2.4.3. Task 4.3 Optimal DR management strategies (implicit and explicit) adaptable to different segments of residential energy consumers

[M5-M29] [Leader: **DOM-X**, Participants: ENG, NTUA, OurPower, BLUEPRINT, HERON, DOM-X, CENTRICA]

This task aims at designing and implementing optimal DR strategies for the Greek, the Austrian and the Irish pilot sites. It receives input by several other tasks, like the T2.4, the T3.1 and the T3.4, while it builds the Technology Enablers TE-10 and TE-11. Specifically, we foresaw to implement nudging services by augmenting the existing mobile applications of the Greek and the Austrian pilot sites. The services/tools use the historical activity of the household and provide recommendations to shift their demand and offer flexibility to the grid. Moreover, the aim is to identify the optimal incentives so as to enable the consumers to proceed with the suggested DR actions.

In the first 18 months of the project, we have identified the exact activity for every partner, in order to deliver an optimal DR management service that regards to the involved pilot sites. The workload is divided into different services, while most of them are already in an advanced stage of development.

Below we list the identified services along with the responsible partner(s) defined on each activity:

Service A (DOM-X): Design of optimal DR strategies for the gas pilot:

In this service we aim at identifying the gas pilot households with high energy-savings potential in order to apply DR strategies. To implement this service DOM-X has already enhanced the DOMX dynamic load simulator which is trained with the datasets collected by the different households to calculate their energy under different configuration settings. The simulator will be used as proof of concept prior to applying nudging notifications to the selected households. Moreover, DOM-X is developing a clustering methodology using deep learning technique for identifying/selecting homes with high energy-savings potential in order to apply DR schemes. For the aforementioned clustering methodology, we are using several different parameters for identifying the desired households. Notably, several steps have been made for the development of the clustering methodology while some preliminary results have been produced.



Service B (NTUA & DOM-X): Building performance and end-user profiling for the gas pilot:

The objective of this service is to unveil concealed intricacies within heating load profiles, affording a more distinctive comprehension of energy consumption dynamics and heating system operations focusing on identifying houses with the highest estimated energy savings, paving the way for sustainable Demand Response (DR) programs. A first version of the software and results with the identified clusters are available by the NTUA team. Notably, in this task DOM-X provides the data for analysis by the Greek pilot and guidance in the meaning of the data towards developing the overall service.

Service C (NTUA & HERON): Energy disaggregation for the electricity pilot:

The goal of this service is to utilize Non-Intrusive Load Monitoring (NILM) techniques to disaggregate energy consumption data at the appliance level. A first version of the software is available as well as some first results from the households that have been analysed. NTUA is developing the disaggregation techniques while HERON is providing the data by the Greek pilot.

Service D (HERON): Integration of tariff system for implicit and explicit electricity DR services:

This service aims at generating a tariff system based on multiple conditions and their combination, so that the end-consumers can view when RES production is available and shift their loads accordingly. A first version of the proposed system is available in the recently launched mobile application HERON for the end consumers.

Service E (OurPower, BluePrint): Augmenting OurPower nudging application to support nudging features: The goal is to enhance OurPower's mobile application to provide personalized suggestions and to enable users to optimize their energy consumption based on the generated PV power. In the reporting period, the concept and design for the nudging application, including a comprehensive redesign and expanded feature scope, have been finalized. The planned application features include:

- Individual view with data and statistics on energy consumption and production, including forecasts and impact evaluations.
- Community view with aggregated statistics on energy consumption and production, shared forecasts, impact evaluations, and access to documents and invoices.
- Daily nudges with actionable suggestions, a feedback mechanism, and calendar integration.
- Community feed enabling interaction between managers and members of the energy community, acting as a nudging mechanism.
- Education and quiz features with interactive content as nudging mechanism to promote energy awareness.
- Personalization options, such as customizable profiles, notifications, and app design, which act as nudging mechanisms.

Service F (Heron & ENG): Multi-value Demand Response for virtual clusters of buildings:


In this service HERON is analyzing the flexibility potential of the Greek's households as well as of the individual home appliances highlighting the energy-savings potential. An analysis of some first datasets has been already realized, while it will be completed with all available Greek pilot's data. The ultimate goal is to provide DR actions as a stand-alone service that can be integrated in third-party applications and systems.

Service G (ENG): Integration of Implicit Demand Response service - Recommendation in the DT:

This service aims at integrating the outcome of the demand response services within the context of the Digital Twin TE-8 in Task 4.1. The work on this service will take place when the first results from service F are available. ENG is responsible to integrate the proposed recommendations into the DT.

Finally, we would like to note that one more service for automated DR will be defined for the Irish pilot that will be led by CENTRICA.

1.2.4.4. Task 4.4 Local flexibility management for energy communities and self-consumption/cost optimisation

[M5-M29] [Leader: **OurPower**, Participants: ENG, CARTIF, COMS, OurPower, BLUEPRINT, CIMNE]

The task "Local flexibility management for energy communities and self-consumption/cost optimisation" aims to explore and implement strategies for optimizing and utilizing flexibility options at the community or district level, balancing economic, social, and environmental benefits.

In the first nine months of the project, our activities for the task primarily focused on conceptual development, laying the foundation for implementation in the upcoming project periods. We investigated various possibilities for optimizing and utilizing flexibility options at both community and district levels, aligning these strategies with economic, social, and environmental benefits. In doing so, we considered results from other work packages, primarily WP2 and WP3. We implemented data collection and processing tools and began developing a pre-aggregation framework to interface with local assets and systems, aimed at exposing flexibility opportunities to the nudging app. Additionally, we worked on integrating various services and systems, including the Universal Secure Gateway for resource virtualization and real-time management.

Between months 9 and 18, the focus shifted from conceptual development to the implementation of services within the task. This period also involved significant refinement of the initial concepts, incorporating insights gained during earlier phases and aligning them more closely with practical implementation needs. For instance, the concepts were refined through Blueprint's research, which evaluated flexibility options by identifying and categorizing opportunities, assessing their impacts, and establishing KPIs to measure their effectiveness. The economic, social, and environmental benefits of each option were also analysed.

COMS worked on the implementation and integration of the Universal Secure Gateway (USG) to provide building-level demand response management by integrating and virtualizing local DR



resources. Digital twins of individual local DR resources are being implemented and will be utilized to provide means for local energy optimization (model-in-the-loop for evaluation of different DR optimization strategies). Once the majority of local DR resources are fully integrated into the pre-aggregation platform (supported by the USG), further development will focus on implementing and integrating algorithms for local flexibility management and optimization.

OurPower focused on the development of the energy community platform, especially data integration. Integration efforts included integration with EDA, the Austrian energy data exchange system, to request 15-minute interval data from the grid operator. Additionally, OurPower and COMS implemented integration with the building energy management system at the pilot site to enable near real-time data exchange. This data is now incorporated into the OurPower energy community platform and the COMS Universal Secure Gateway, providing critical insights for operational decision-making.

OurPower began developing and partially implementing a concept for the management of a virtual balancing group, designed to request flexibility from the energy community based on the situation within the virtual balancing group. The energy community shall then respond and coordinate flexibility measures internally. The concept is still under refinement and its full implementation is planned for the next reporting period. In collaboration with ENG, forecasting capabilities for energy production from photovoltaic and wind power plants are under development. These forecasts will support the virtual balancing group's management, contributing to the optimization of energy distribution and consumption within the energy community.

CARTIF focused on strategies for managing the flexibility of shared renewable generation, storage, and shifting loads in energy communities. A first version of the software associated with the service was successfully completed and is intended for deployment during the demonstration phase. The developed algorithm is highly adaptable to various community configurations, depending on the characteristics of the involved assets, and primarily focuses on maximizing economic benefits by minimizing energy costs. Preliminary stand-alone tests were carried out using the Austrian demo as a target community. The results confirmed that the algorithm performs as expected, with shared solar production utilized to meet the building's demand, and any surplus energy stored. Further performance tests are planned, including studies on the grid impact of different charging requests for electric vehicles and defining software requirements for the deployment phase.

CIMNE has been developing data-driven algorithms to optimize energy sharing, storage, and flexibility in the Spanish pilot, which features a centralized battery, photovoltaic panels, and multiple consumers. The focus has been on creating tools tailored to the unique Spanish regulatory framework, where energy is shared before consumption. A Python-based tool was developed to optimize energy sharing, maximizing savings and minimizing surplus. Preliminary tests using historical data were conducted, though further validation with real data is required. Additionally, a second Python-based tool addresses daily battery operation, optimizing its use while considering energy-sharing schemes. Tests revealed that incorporating future days into the optimization improves savings but requires considering prediction errors in consumption and energy costs. In summary, preliminary tools for operating and managing energy communities



under the Spanish framework have been created. Further improvements and validations are planned to fully optimize the system in the Spanish pilot.

1.2.4.5. Task 4.5 Optimal Aggregation of flexible energy resources for heat vs electrical flexibility valorisation

[M5-M29] [Leader: **NEOGRID**, Participants: ENG, NEOGRID]

The services developed under the headline Aggregation of flexible energy resources for district heating management extend the NEOGRID Platform with new services for optimal aggregation of thermal flexibility at individual buildings and at district level, where flexibility in how an energy-efficient building is heated can be used towards the district heating system.

The development activities in the timeframe M1-M9 of the project, have focused primarily on conceptualizing the service and planning the integration of its various components, setting a solid conceptual foundation for future development and implementation phases.

A lot of time has been spent in coordination meetings within WP4 and WP3 to secure the right level of integration and also to secure that commercial platforms can fit into the DEDALUS architecture.

Furthermore, other services have been supported with respect to providing input. High level use cases have been developed and documented in D2.1.

Concept development on nudging application for tenants in Danish Demonstrator together with WP2

In the timeframe M10-M18 work have been invested in braking down the project goals of UC14, UC16 and UC17 into development tasks. The main development activities have been focusing on extending the Neogrid platform from focusing on the building as a whole to dividing the building up into staircases, apartments and rooms creating the basic infrastructure required. This is achieved by implementing a zone based hierarchy within the building model and backend enabling the control and frontend to view devices in the right information context. Furthermore, it is required to have the tenants into the control loop providing feedback to the controllers via setpoints and a slider-based flexibility option. On top of the hierarchy we have been focusing on implementing a permission structure for zones, restricting access to data, so users only can view and control their designated zones and related sub-zones. In order to have the possibility to push behavioural nudges within the system, targeting each apartment/tenant, work on a push notification framework have been started.

Finally, implementation have started on first iteration of control algorithm (UC14+UC17), where DH flow temperature is optimized based on radiator thermostat valve feedback.



1.2.4.6. Work carried out per partner per task and effort spent

Table 4 – WP4 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	- Addressed the coordination of all tasks under WP4 with the organization of regular WP4 monthly meeting and regular communication via emails. - was in contact with WP2 and WP5 via emails and audio conferences. - Attended conference calls organised per pilot to collect relevant information about users' needs and expectation on the development to be done - In the context of Task4.1, ENG developed the first wave of TE-8 and start working on the second wave due at the end of December - In the context of Task4.3, ENG developed a first version of middleware API related to service <i>G Integration of Implicit Demand Response service - Recommendation in the DT</i> - As responsible for D4.1, ENG monitored and coordinated the preparation of this deliverable until its final submission. - Coordinated the activity related to the deployment of services	15.92
2 - NTUA	- Meetings attendance. - Data collection mechanisms from the Greek pilot. - Input to D4.1 - Conceptualization of heating time series segmentation on the heating side of the Greek pilot based on a multi-dimensional approach. - Experiments on heating boiler data, applying the clustering methodology for the heating, building, user, temperature and boiler dimensions. - Paper preparation titled, "A multi-dimensional unsupervised machine learning framework for clustering residential heat load profiles" which has been submitted in Energy International Journal - Non-intrusive load monitoring (NILM) techniques on the electricity side of the Greek pilot.	1.87
3 - CARTIF	- Meetings attendance. - Requirements to be fulfilled by the flexibility service to be developed by CARTIF. - Software design of the service architecture - Optimization algorithm programming	4.64



Partner	Work carried out in the WP within the period	PMs
	Validation of the service using synthetic data	
4 - UNIVPM	- Meetings attendance. - Concept creation and model definition for the Optimal trade-off among comfort, energy and building performance service that includes data analysis on real data from pilots. - Participating in coordination meetings within WP4 and with technology enablers in WP3 with respect to defining services integration into DEDALUS architecture - Contributing to D4.1	4.10
6 - COMS	- Participating in WP4 and T4.4 meetings - Defining the pre-aggregation platform based on Universal Secure Gateway (T3.6) - Preparing data collection tools for pre-aggregation platform - Pre-aggregation platform implementation - USG-supported data collection implementation - Building-level monitoring implementation	3.83
7 - OurPower	- Development of data exchange interfaces - Integration of data collection tools - Integration of data processing pipelines - Data collection, management and processing - Concept creation on community onboarding - Concept development on nudging application and local flexibility management - Development of the energy community platform with a focus on data integration (EDA, EUDT) - Concept development and partly implementation on virtual balancing group management and flexibility activation - Organization of task 4.4 activities - Participation of meetings and workshops - Contribution to D4.1	8.08
9 - BLUEPRINT	Participated in WP4 meetings	9.50



Partner	Work carried out in the WP within the period	PMs
	T4.3 - Conducted research to explore potential nudging mechanisms for application integration. - Reviewed multiple meta-analyses to gain a comprehensive overview of: - Behavioural research on nudging mechanisms. - Psychological studies related to nudging. - Identified a subset of nudging mechanisms suitable for digital contexts: - Filtered and prioritized mechanisms based on literature review findings. - Analysed nudging mechanisms to determine: - Their frequency of use. - Their alignment with specific nudging principles. - Selected 10 nudging mechanisms based on analysis and review in meetings with OurPower. - Created practical implementation examples for each of the 10 selected mechanisms. T4.4 - Conducted analysis of flexibility options at the community or district level. - Categorized flexibility options into: - Economic benefits. - Social benefits. - Environmental benefits. - Explored the potential impacts of each flexibility option category. - Researched & Developed 21 Key Performance Indicators (KPIs) for evaluating flexibility options: - Defined metrics for assessing benefits and feasibility. - Aligned KPIs with community/district-level goals	
11 - NEOGRID	Participating in coordination meetings within WP4 and with technology enablers in WP3 with respect to defining interfaces and securing that existing commercial platforms can fit into DEDALUS architecture.	7.21



Partner	Work carried out in the WP within the period	PMs
	• Scoping the service with respect to developing the use cases for the services which will be developed to fulfil the requirement of Task 4.5 • Concept development on nudging application for tenants in Danish Demonstrator together with WP2 • Contributing to D4.1 • Breaking down requirements into development tasks and internal coordination • Starting up development activities focusing on preparing the Neogrid platform do be able to deliver the specified use cases by: • Started Implementation of hierarchy within building model and backend so backend control and the frontend can view devices in the right information context (building->technical room->Apartments->rooms) • Started Implementation of a permission structure for zones, restricting access to data so that users can only view and control their designated zones and related sub-zones • Started Implementation of push notifications for e.g. Behavioural nudges within the system targeting each apartment/tenant • Started Implementation of backend framework enabling control on multiple zones in apartment building • Implementation started on first iteration of control algorithm (UC14+UC17), where DH flow temperature is optimized based on radiator thermostat valve feedback	
13 - HERON	• Meetings attendance. • Collaboration with NTUA for data collection activities to disaggregate energy consumption data at the appliance level. • Preparation initial data sharing and requirements for future implementation. • Development of technical requirements and infrastructure for DT and DR service.	7.99



Partner	Work carried out in the WP within the period	PMs
	- Literature review for DR flexibility concepts in residential buildings. - Initial model development activities (integration of formulars and calculations). - Initial analysis of general Service F requirements (interface, privacy etc.). - Survey preparation for the constraint definition of optimisation strategies. - Starting preparation (hosting, deployment etc.) for integration activities of services to DT. - Analysis and development of dynamic tariffs and especially RES-driven strategies integration in HERON's EnergiQ mobile app.	
14 – DOM-X	- Organization of the T4.3, including calls and communication with the partners for identifying their roles within the Task and monitoring their progress - Enhancing the domx dynamic load simulator that is able to simulate the energy consumption performance of the Greek pilot households - Developing a clustering algorithm to identify households with high energy-savings potential to be part of the Greek DR scenarios - Releasing the new domx mobile application to support nudges - Paper preparation - Participation in WP4 calls	5.65
15 - ISRAA	- Attendance to WP4 meetings - Concept definition regarding comfort flexibility model and Digital twin (T4.2) to be deployed in the Italian pilot. - Definition of Digital Twin and comfort flexibility model characteristics (technical, interface and privacy requirements) - Collaboration with UNIVPM to develop the questionnaire on thermic sensation to be answered periodically by the residents.	3.80



Partner	Work carried out in the WP within the period	PMs
17 - CIMNE	- Meetings attendance. - Initial definition of the flexibility service - First version of the flexibility service to be developed - First test of the service in an offline environment - Coordination with Spanish pilot partners, specially ROMUR and PEUSA	3.00
19 - ROMUR	Initial adaptation of the storage system to receive services inputs	1.51
20 - CENTRICA	- Attend WP4 meetings - Initial study on Digital twins	0.29
21 - TUC	- TUC has started the work for the design and development of an apartments building DT model for the students' apartments building in Pilot 7 that incorporates data from heterogeneous sources. - The DT purpose is to support the decision-making process in DR programs and to simulate the execution of apartment level DR actions. - The DT model will be integrated with TE-5 DLT/Blockchain/Smart contracts solution allowing for communication and data exchange between the DT components and TE-5 smart contracts. - The design of the DT has been reported in deliverable D4.1 and its further developments will be reported in deliverable D4.2 and D4.3.	2.00

1.2.4.7. Deviations from the Grant Agreement

TUC has shifted the 6 PM from task T4.4 to T4.1 for developing an apartment building Digital Twin to facilitate the simulation of DR actions regarding appliance activation in the context of DR settlement smart contracts. The building DT model integrates a diverse range of data sources, such as the DEDALUS self-reporting questionnaire and energy metering data. The objective is to identify apartments that should adjust their energy consumption profile to meet the building's target energy profile while optimizing appliance usage patterns within those apartments. The DT will be evaluated in Pilot 7 Romania in the last evaluation phase together with TE-5. The remaining efforts will be allocated for developing the DT core business logic and dashboards in the second reporting period.



1.2.4.8. *Corrective actions taken/suggested*

No corrective action has been undertaken.

D4.1 does not provide information about the development of services of CENTRICA due to the issue with the Irish pilot.



1.2.5. WP5 – Demonstration of multi-value and energy carrier agnostic residential DR, and social acceptance

[M1-M34] [Leader: NTUA, Participants: all]

WP5 is responsible for the overall planning, management, and evaluation of the DEDALUS pilots. We aim at replicating and validating DEDALUS solutions in 7 pilots (5 Front Runners and 2 Multipliers), having different system needs, consumer needs, economic conditions or different climates. We aim to assess the impact as well as the potential to scale and replicate the results achieved in the pilots. The focus of this WP is also given to evaluate the acceptance of the DEDALUS concepts and tools by the residential consumers (and energy stakeholders in general) involved in the project.

During the first 13 months, WP5 activities were centered on Task 5.1, which provided the foundation for demonstration activities. This included evaluation of pilot infrastructure readiness for proper demonstration of the DEDALUS solutions, understanding of the available datasets, and planned upgrades, as well as the specific requirements of each pilot. Data Gathering and Pilot Infrastructure Templates were developed and distributed to standardize the collection of information and evaluate infrastructure needs. A centralized data repository was created to consolidate sample datasets and documentation for API access.

The evaluation methodology and baseline analysis were developed to assess the technological feasibility, user engagement, economic viability, and environmental impact of DEDALUS solutions. KPIs were defined in collaboration with pilot site leaders, who provided baseline values through detailed templates. These activities culminated in the development of pilot-specific implementation plans, including Gantt charts outlining timelines for activities, from initial assessments to technology deployment and evaluation.

D5.1, submitted at the end of M13, consolidated these efforts into a comprehensive roadmap. It detailed the evaluation methodology, the baseline analysis, and structured implementation plans for each pilot, providing stakeholders with a clear understanding of the progress and expected outcomes. This document acts as a critical resource for the demonstration and evaluation phases of DEDALUS.

From M14 onwards, WP5 transitioned into the pre-pilot phase under Tasks T5.2 through T5.8. This phase focused on finalising both physical and digital infrastructure updates across pilot sites and initiating the deployment and evaluation of the first wave of TEs. TEs developed under WP3 and WP4 were implemented in select front-runner pilot sites with adequate infrastructure readiness, enabling real-world testing of their functionality, integration capabilities, and user acceptance. The pre-pilot phase provided actionable insights, highlighting areas for improvement and aligning technologies with the specific needs of each pilot. These findings were critical for guiding iterative refinements and optimizing future deployments.

Social engagement efforts also advanced significantly, informed by earlier engagement activities, such as questionnaire surveys and co-creation workshops, which shaped the preliminary design of technical workshops. These workshops, planned for M21 – M25, aim to



assess user perceptions, attitudes, and satisfaction with the TEs, utilizing pre- and post-test questionnaires based on the Technology Acceptance Model (TAM).

D5.2, submitted at the end of M18, captured the progress made during the pre-pilot phase. It provided detailed updates on infrastructure enhancements, early evaluations of deployed TEs, and the progress of KPIs. Using the evaluation framework outlined in D5.1, the deliverable thoroughly assessed each TE's performance, including its inputs, outputs, and end-user impact, and offered recommendations for improvement. Additionally, D5.2 included revised pilot-specific implementation plans extending to Month 25, featuring updated Gantt charts and outlining schedules for upcoming TE implementations and evaluations.

Throughout the first 18 months, NTUA organized 16 monthly WP5 plenary meetings to address pilot-specific issues, track task and deliverable progress, and facilitate collaboration between technology providers and pilot site leaders. These meetings were essential for ensuring the proper deployment of digital and physical infrastructure and guiding demonstration activities. Detailed minutes are uploaded to the project SharePoint to maintain transparency and enable ongoing reference.

1.2.5.1. Task 5.1 Evaluation framework, baseline analysis and pilots' preparation

[M1-M13] [Leader: **NTUA**, Participants: ENG, NTUA, CARTIF, UNIVPM, ARC, COMS, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, HERON, DOM-X, ISRAA, DYNAMO, CIMNE, PEUSA, ROMUR, CENTRICA, TUC]

The task started with the actual engagement of end users and the installation of required digital systems/platforms/applications per pilot to assure interoperability and integration of the energy facilities and assets and to start the pilot monitoring period. We also analysed the high-level use cases (HLUCs, according to WP2) that comprise the pilots and the methodology to be followed for their evaluation, which is closely related to the evaluation of the technology enablers demonstrated in each pilot site. The evaluation methodology (developed by M10) is centred on the elaboration of the appropriate performance indicators for the DEDALUS social, technological, environmental (including thermal comfort aspects) and business framework (resulting in the DEDALUS baseline analysis as part of D5.1), as well as the user perspective, in terms of compliance with the respective requirements specified in WP2.

Following is the timeline describing the work carried out in the first 18 months for T5.1 (M1 – M13):

M1 – M3: In the first 3 months of the project NTUA initiated the task by conducting bilateral meetings with each of the pilot site leaders to better understand the scope of each pilot site, their infrastructure (digital and physical), their beneficiaries and actors, the current available datasets, and what are the plans for future upgrades in the context of DEDALUS. In the first round of the bilateral meetings, BLUEPRINT also participated in order to facilitate the discussion around the use cases that would be identified under WP2 and specifically Task 2.1.



After carefully examining the gathered information from each pilot site, NTUA proceeded with the creation of information templates that should be filled out by the responsible partner in each pilot site, which includes either the pilot site leader (i.e., building owner) or the technology provider that currently provides the digital platform used by the residents in the pilot site. These information templates consisted of two templates, described below, and were used in order to register the available data sources, as well as to evaluate the existing and foreseen infrastructure in the scope of the project. These ledgers trace the sufficiency of resources and any lack thereof.

- The Data Gathering Template was used to consolidate the data provision information from pilots and relevant information; sets the reference of the data availability that needs to be reported both on volume and variety for technical WPs and service developments.
- The Pilot Infrastructure Template presented technical information regarding the pilot's infrastructure; including the location site details, the installed equipment together with the new equipment anticipated.

Afterwards, to facilitate the provision and completion of the templates a second round of bilateral calls was conducted, explaining to the partners how to fill out the templates and the deadline. Although the initial completion of the templates was finished by the end of M3, refinements were made in the following months following the updates in the infrastructure of the pilot sites and the data availability.

M4 – M6: Following the completion of the information templates and to facilitate the development of the different services, NTUA created a data repository that was filled out by each pilot site partner with samples of the current relevant datasets (mentioned in the data gathering template), or detailed documentation regarding API accessibility to the pilot specific digital platform database. In parallel, the user engagement of the residents was initiated through the provision of questionnaires, explaining the scope of DEDALUS and demand response programmes in general, which were circulated either by e-mail or in person. The circulation of the questionnaires was also part of T2.3 which focuses on the profiling and segmentation of the residential consumers and was mostly completed by M9. Another social engagement activity that is also planned to be organised in the following months is the co-creation workshops with the involvement of the pilot site leaders and the residential consumers under the guidance of SIN and in the context of T2.2.

In addition, based on the discussions that were held in the previous months and the initiation of WP3 and WP4 activities, an updated mapping of the technology enablers to be demonstrated in each pilot site was deemed necessary. This was due to the unavailability of necessary infrastructure in pilot sites that made it impossible for the demonstration of specific technology enablers which was not taken into consideration during the writing of the proposal. NTUA facilitated this process by conducting an exercise in the second plenary meeting held in Rome (M8) which the technology provider and pilot site leaders were present and provided their feedback.

M7 – M9: After the proper mapping of the technology enablers, WP5 activities in the final three months focused mainly on the elaboration of the appropriate key performance indicators (KPIs) for the DEDALUS social, technological, environmental, and business framework. Based on the



KPIs that are described in the grant agreement, and the infrastructure and scope of each pilot site, NTUA conducted bilateral calls with the pilot site leaders to identify which KPIs could be addressed in each case. In addition, NTUA circulated a template containing detailed description of each KPI, and a list of proposed means of verification for each one of them to provide sufficient information to the pilot site leaders to understand how to tackle the KPIs that will be tracked in their demo site. Currently, pilot site leaders have started the process of filling out the template with the formula for measuring the KPIs, and their baseline values.

M10 – M12: Following the plenary meeting in Rome, pilot site leaders completed the selection of the KPIs that will be addressed. They also worked on identifying the most suitable formulas for measuring these KPIs and defining the baseline analysis. Bilateral calls were organized with each pilot site to support these efforts, and to facilitate the planning of technical workshops. Based on the discussions that were held, technical workshops will be scheduled during the second pilot cycle, once the TEs reached a more mature stage. These workshops will provide an opportunity for technology providers to engage directly with the related end users (residents/technology experts), presenting the methodologies, outputs, and benefits of their developments. Additionally, another series of meetings was conducted with each pilot site to refine the requirements for each TE, providing detailed clarifications. The focus was to update the general requirements outlined in D2.1 and specify the functional requirements for each TE to be demonstrated. These discussions involved pilot site leaders and technology providers, who jointly defined the key functionalities for the services being developed. A pilot-specific implementation plan was also developed to establish a detailed roadmap for each pilot site. This included the creation of Gantt charts for each pilot, outlining all completed, ongoing, and planned activities. These charts also specified the implementation timelines for each TE at the respective pilot sites. Lastly, the ToC for D5.1 was created, and initial feedback from the pilot site leaders started.

M13: During this month, technology providers contributed by completing the Gantt Charts, which outlined the timelines for TEs and specified those to be evaluated during the upcoming pre-pilot phase. To support the evaluation process of the TEs, a comprehensive template was created and included in deliverable D5.1. Additionally, pilot site leaders finalized their input for the baseline analysis by providing initial values for the KPIs, establishing a critical reference point for tracking progress during the demonstration phase. On the social engagement front, an approach to assess the social acceptance of DEDALUS solutions was developed, leveraging the Technology Acceptance Model (TAM). This approach incorporates pre- and post-test surveys designed to evaluate user perceptions, attitudes, and behavioural intentions toward the TEs. The surveys cover dimensions such as perceived usefulness, ease of use, trust, and compatibility, providing insights into user satisfaction and adoption potential. By the end of Month 13, deliverable D5.1 was successfully submitted, consolidating these elements into a strategic roadmap for pilot activities.

1.2.5.2. Task 5.2 Flexibility pre-aggregation for building-level energy community (Austria)

[M14-M34] [Leader: **OurPower**, Participants: ENG, NTUA, COMS, OurPower, GreenPoint, BLUEPRINT]



Task 5.2 officially began in M14; however, the installation of the relevant infrastructure had already started earlier. This task focuses on the demonstration activities related to Pilot 1 in Austria (targeting residential consumers from Wiener Neustadt, Lower Austria), including the involvement of OurPower, GreenPoint, BLUEPRINT and COMS. The demonstration showcases and evaluates services developed or enhanced in WP4, including, e.g., the nudging application and the P2P pre-aggregation flexibility platform. The work carried out from M14 to M18 can be summarized as follows:

- Continued installation of necessary infrastructure, e.g., improvement of energy management system and development of new functions to follow price-based charging strategies.
- Deployment of a new battery storage system with a capacity of 232 kWh, which can also serve as a booster for an EV charging station with 150 kW power and charge or discharge according to price signals
- Integration of EUDT API, which is the building energy management system API, into the OurPower platform and COMS pre-aggregation platform
- Preparation of necessary systems, e.g. local Mini PC, for integrating local systems, such as Sonnen storage and EcoFactor EV charging station platform
- Integration of price signals of Austrian electricity market and OurPower marketplace, e.g., as signals for charging strategies

1.2.5.3. Task 5.3 Energy efficiency and DR optimal interactions between district heating and electricity networks in social housing (Denmark)

[M14-M34] [Leader: **FællesBo**, Participants: NTUA, FællesBo, NEOGRID, GREEN CITIES]

This task focuses on the demonstration activities related to Pilot 2 in Denmark (targeting residential consumers from Herning), including the involvement of FællesBo, NEOGRID, ECG. The link between electricity and heat via the smart ventilation control will be tested, providing flexibility at the building level to the electricity grid. However, it should be noted that there is no heating surface in ventilation system and therefore the two energy systems are uncorrelated and cannot substitute each other. Electricity consumption for ventilation systems is very low, as they only are providing air change, this challenges the business case, but as a stacked business case it can be relevant for FRR services towards the grid. NEOGRID will enable the tool to monitor and control the flexibility possibilities on the two energy carriers (electricity and district heating). For district heating it is mainly peak shaving and for electricity it could be the FRR services.

Pre-pilot activities concerning energy control have been started M14.

Timeline of infrastructure updates and technological implementations at the Danish Pilot Site:



#	Delivery	Timeline
1	Heating control in homes by smart thermostats on all radiators (MClimate Vicki LaRaWAN)	Q1-Q2 2024 - Done
2	Monitoring of indoor climate parameters in homes with (Elsys ERS CO2 Lite)	Q2 2024 - Done
3	Monitoring of decentralized residential ventilation systems via the manufacturer's cloud solution	Q3 2024 - Ongoing
4	Start-up of predictive central heating control via Danfoss ECL regulator and district heating main meter (NeoGrid)	Q3 2024 - Started
5	Monitoring of bi-meters for cold and hot domestic water and district heating in homes	Q3 2024 - Started
6	Web-app updates for data analysis (Neogrid Preheat App V2)	Q1 2025 - Planned
7	Web-app interface for residents with the possibility of interactive choice of indoor climate comfort and financial savings.	Q1 2025 - Planned

Task 5.4 Multi-value DR for multi-segments of residential energy (electricity and gas) consumers (Greece)

[M14-M34] [Leader: **HERON**, Participants: NTUA, ARC, HERON, DOM-X]

This task focuses on the demonstration activities related to Pilot 3 in Greece, targeting residential consumers in collaboration with HERON and DOMX. The task demonstrates multi-value DR activities for electricity and natural gas consumers, integrating and expanding pilot assets to provide flexibility and energy management at the household level.

It should be noted that according to the GA, it was expected that 100 households would be equipped with smart metering infrastructure provided by HERON (for electrical appliances) and DomX (for natural gas and electricity HVAC) to enable implicit and explicit DR services. Significant efforts during the pilot recruitment phase to achieve a broader impact of the DEDALUS outcomes led to an increase in participants for the Greek pilot, reaching 200 residential buildings. These participants were allocated, as outlined in the GA (100 by HERON and 100 by DomX), across multiple clusters based on factors such as the energy vector monitored, appliance usage, heating preferences, and other criteria. This expansion further advances the DEDALUS vision by ensuring the demonstration of its services to a higher-than-promised number of actively engaged consumers within the project (Objective 5, Table 1).

Progress and infrastructure expansion:

- HERON has integrated 90 households into the DEDALUS ecosystem, equipping them with 92 smart meters and 10 smart plugs to enable energy monitoring and management.
- 48 households are equipped with 93 flexible assets, such as washing machines, air conditioners, dishwashers, dryers and electric boilers
- Plans to expand to 100 households, with 10-15 designated as “super” households to be equipped with up to 5 Arcelik smart plugs each.
- Targeted surveys for electricity consumption habits for the “super” households
- Integration of advanced DR-ready AC systems in 20 households



- HERON energy Control has been updated through API changes with the introduction of “home” which enables the integration of multiple assets under a unified concept for scalability and efficient energy management across a larger network of interconnected devices.
- Flexibility potential assessment for initial households and installed appliances (Washing Machines & Dishwashers).
- Development of APIs to facilitate real-time data access and enable advanced DR services
- HERON introduced dynamic tariffs in its EnergiQ mobile app, encouraging energy demand shifts to periods with high Renewable Energy Sources (RES) availability.
- Initial design and development of nudging features in the EnergiQ mobile app to support DR actions, leveraging behavioural insights from pilot participants.
- DOMX has installed 35 new DOMX heating controllers in the households of the Greek pilot for controlling gas boilers.
- Moreover, 10 more DOMX heating controllers has been installed in heat-pumps for the collection of meaningful data for analysis and further development of services.
- The plan is to exceed the number of 80 heating controllers in total (both for gas and heat-pumps).
- 30 Energy meters has been deployed in the households of the Greek pilot, 10 of which are installed in the pilot sites that are equipped with heat-pumps in order to correlate the behaviour of the heat-pump with the corresponding energy consumption data. The plan is to reach to 50 installed meters and that all heat-pumps are accompanied with an energy meter as well.
- Lastly, 10 DOMX indoor air quality devices have been installed in the households of the Greek pilot, while more will be deployed in the near future, having as goal to deploy 30-50 sensors.
- Regarding the software development DOMX has developed corresponding dashboard for visualizing the collected measurements and all device types have been integrated with the developed web portal.
- Additionally, the DOMX mobile application has been enhanced to support all device types, while a new release will be launched soon that will contain DR capabilities based on the nudging mechanisms already resulted by the work done in WP2.

1.2.5.4. Task 5.5. Trading off comfort with energy consumption in DR programs for elderly people living in serviced apartments (Italy)

[M14-M34] [Leader: **ISRAA**, Participants: **ENG**, **NTUA**, **UNIVPM**, **ISRAA**, **DYNAME**]

This task deals with the demonstration activities related to the Italian pilot, involving two buildings (Casa Tito Garzoni and Casa Maddalena) of ISRAA's Borgo Mazzini Smart Co-housing. With the involvement of UNIVPM, ENG, DYNAME, the task will demonstrate the comfort flexibility model developed by UNIVPM (T4.2), that will combine the feedback provided by residents with the real-time data of energy consumption, IAQ and light intensity sensors for the optimal trade-off for elderly people. On the other side, the centralised management of supplied energy will be



supported by the Digital twin developed by ENG, that will help the building manager by delivering recommendations on the most effective management of energy. The work done so far has focused on the definition of the pilot actions, the installations of the required technological systems and platform and analysis of the best approach to guarantee the engagement of residents:

- Installation of n° 16 3-phase Energy Meters (one per apartment) to measure real time energy consumption and n° 16 IAQ and Light intensity sensors (one per apartment) that will provide real time information on environmental conditions (temperature, humidity, CO2, PM10, etc..) within the apartments.
- ISRAA' s technical department, responsible for the centralised management, now has direct access to the monitoring platform, enabling them to continuously monitor energy and environmental conditions.
- DYNAMES has been performing an analysis on historical data of energy bills, to contribute to the measurement of the KPI and to provide insights on costs of contracts, including energy generation, transmission, distribution, operations and maintenance.
- Regarding the Digital Twin, ENG presented a draft version of the User Interface.
- As first step for the development of the comfort flexibility model, the system has started to collect feedback on thermic sensation (TSV) from the residents, through an online user-friendly questionnaire sent via Telegram link, in order to combine it with the environmental data.
- Concerning the engagement of residents, activities foreseen in T2.2, T2.3, T2.4, such as segmentation questionnaires, interviews and a workshop were carried out; moreover, a second additional workshop to foster residents' engagement was organised at the beginning of November. This second workshop capitalized on the outcomes on another ongoing European funded project on age-friendly and eco-friendly environment.
- The discussion about strategy to involve residents and building manager in the DR actions is ongoing both internally (through the engagement of psychologists and BMSC coordinators) and externally (through cooperation with SIN and Dedalus' partners).

1.2.5.5. Task 5.6. District-level DR for increasing community-level self-consumption while providing flexibility to local DSO (Spain)

[M14-M34] [Leader: CIMNE, Participants: ENG, NTUA, UNIVPM, ISRAA, DYNAMES]

This task focuses on the activities of Pilot 5 to validate demand response (DR) and electricity flexibility at the district level, considering the integration of shared storage resources. Validation will occur using a district-based electricity storage system, distributed rooftop photovoltaic (PV) generation plants, and 12 residential dwellings. No validation has been performed because the necessary services have not yet been developed. However, the infrastructure required to deploy these services has been deployed. More specifically, the following work has been done:

- The actors involved in the demonstration have been engaged, including the DSO, the battery provider, the IoT provider, and the city council of La Seu.



- The different residents have been contacted, and they have shown interest in participating in DR activities by participating in a workshop T2.2. and T2.4. and answering a questionnaire.
- ROMUR has deployed a 120KWh Battery in the PEUSA grid, controllable through an API.
- Advanced metering for the different dwellings has been deployed to provide higher frequency and cadency lectures.
- The installation of Smart plugs and gateways for the end users' homes has started. Moreover, the infrastructure for communicating with these devices has been established through an API, T3.1.

Start designing the nudging inputs to the end user. In the first months of the project an extra effort has been devoted to this task, as there was a change in the city hall government, and it was necessary to repeat meetings with new responsible. Likewise, more documents and administrative forms had to be prepared as well. Moreover, some technical requirements had to be covered in order to integrate the district battery. However, this has not affected the objectives of the pilot, but it was necessary to put more PM at the beginning than the ones expected initially.

1.2.5.6. Task 5.7. DEDALUS replication across Front Runners and Multipliers

[M14-M34] [Leader: **TUC**, Participants: ENG, NTUA, CARTIF, UNIVPM, ARC, COMS, OurPower, BLUEPRINT, NEOGRID, HERON, DOM-X, CIMNE, CENTRICA]

This task deals with the replication of part of the services developed in WP4 in the pilots acting as a Replication Multiplier (CENTRICA and TUC). The objective is to deal with replication weaknesses for increasing residential DR in Europe by carefully examining the parameters that may affect the seamless integration of pilot results in other real/business environments to ensure that the activity is realized effectively.

The task started at M14 and the first 4 months from the reporting period TUC has started to work on the pilot specific implementation plan and allocated efforts to test and prepare the available logical business platforms for piloting activities. TUC has developed components for extracting and disseminate data from the pilot facilitating the integration with the local business platform of the technology enablers (i.e. TE-5) that will be tested in the pilot. The data has been used by TE-4 Flexibility modelling tool for residential DR to allow training of prediction algorithms that have been implemented in the pre-pilot cycle. In the same cycle TUC has collaborated with DEDALUS partners for evaluating TE-1 Consumers clustering and segmentation models in Pilot 7 where specific questionnaires have been used for collecting information. During M17 and M18 TUC has started the work for preparing the second evaluation cycle (i.e. second evaluation iteration for TE-4 and first iteration for TE-3 Residential DR/flexibility interoperability building blocks (DEDALUS ontology) and TE-5).



1.2.5.7. Task 5.8. Consumers engagement, social acceptance assessment and impact analysis

[M14-M34] [Leader: **SIN**, Participants: ENG, NTUA, CARTIF, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAME, CIMNE, PEUSA, ROMUR, CENTRICA, TUC]

This task deals with the engagement of the energy consumers and stakeholders from the energy value chain on the usage of the DEDALUS services (WPs 3-4) considering the particularities of various market roles, as defined in T2.1. Evaluation of the social acceptance and impact assessment will be provided.

As this task started in M14, it is still in its early phases. Work in the first four months has focused on analysing the insights generated from WP2, such as the output from the co-creation workshops from T2.2 and the segmentation analysis from T2.3. These insights form the foundation for defining the engagement strategy for each pilot. As part of the process for defining the engagement strategy, engagement planning calls for each pilot site be conducted by SIN in collaboration with NTUA in M19 and M20. The insights from WP2 also feed into the technical workshops planned in related to the social acceptance and impact assessment.

The technical workshops will serve as platforms for technology providers to exhibit and demonstrate their advancements of their technology enablers to energy stakeholders and residential consumers at each pilot site. They will also include mechanisms for collecting qualitative feedback from both residential and professional end users. In M17-M18, SIN and NTUA have collaborated on defining a workshop structure and methodology as well as pre/post questionnaire assessment tools. The approach to the technical workshops has been written up and included in D5.2.

1.2.5.8. Work carried out per partner per task and effort spent

Table 5 – WP5 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	- Participation in the bi-monthly meetings and conference calls organised per pilot - Analysis of the different options that have been found to replace the Irish Pilot	4.24
2 - NTUA	General: Coordinated monthly WP5 plenary meetings to discuss progress, address challenges, and ensure alignment across tasks.	10.58



Partner	Work carried out in the WP within the period	PMs
	- Conducted bilateral meetings with pilot site leaders to monitor and facilitate updates on pilot-specific activities and upgrades in the infrastructure. Task 5.1: - Developed and distributed Data Gathering Templates and Pilot Infrastructure Templates to register available data sources and evaluate the existing and planned infrastructure at each site. - Created a centralized data repository containing samples of current datasets and documentation regarding API access to pilot-specific platforms. - Updated the mapping of TEs to reflect the technical and contextual realities of each pilot site. - Developed the evaluation framework based on the identification of proper KPIs for the social, technological, environmental, and business aspects of DEDALUS. - Circulated KPI templates for pilot site leaders, enabling them to define measurement formulas and baseline values, resulting in a comprehensive baseline analysis. - Developed pilot-specific implementation plans, including detailed Gantt charts outlining all completed, ongoing, and planned activities for each pilot. - Initiated the planning of the technical workshops and developed a user acceptance framework, introducing pre- and post-test questionnaires to assess user acceptance of DEDALUS solutions. - Drafted and successfully submitted Deliverable D5.1, consolidating the baseline analysis, evaluation framework, and implementation plan. Tasks 5.2–5.7: - Oversaw updates to both physical and digital infrastructure during the pre-pilot cycle, ensuring readiness for technology demonstrations. - Monitored and supported the demonstration of TEs at pilot sites, including their evaluation during the pre-pilot phase. - Facilitated updates and reporting on KPIs and implementation plans, preparing for the next pilot cycle. - Contributed to the drafting and review of Deliverable D5.2, documenting progress in the pre-pilot phase and plans for the full demonstration phase.	



Partner	Work carried out in the WP within the period	PMs
	Task 5.8: - Refined and finalized the format for technical workshops, specifying target end-users, timelines, and workshop objectives. - Enhanced the pre- and post-test questionnaires, defining their structure and tailoring the questions to align with the Technology Acceptance Model (TAM). - Provided support in planning the overall social engagement strategy across all pilot sites, ensuring alignment with project objectives and end-user needs.	
3 - CARTIF	- Meetings attendance - Collaboration on T5.1 activities for baseline analysis. - Connection of the service with data repositories of the frontrunner and multiplier pilots.	1.42
4 - UNIVPM	- Meeting attendance - Elaborating appropriate KPIs for indoor environmental quality assessment - Calculation of defined KPIs for baseline assessment - Contributing to D5.1 and D5.2	0.90
5 - ARC	- Meetings attendance and collaboration on T5.1 activities. - Worked with Greek and Spanish pilots for energy consumption data collection from smart plugs.	2.10
6 - COMS	- Attending meetings - Contributing to discussions and specifications in T5.2 - Pre-aggregation platform deployment - USG software deployment	1.24
7 - OurPower	- Creation of Austrian pilot evaluation framework including KPI definition - Development of evaluation concept on flexibility pre-aggregation - Description and definition of technical architecture regarding Austrian pilot - Integration specification for systems in Austrian pilot	4.64



Partner	Work carried out in the WP within the period	PMs
	- Integration of price signals of Austrian electricity market and OurPower marketplace - Preparation of necessary systems for local system integration - Participation in meetings	
8 - GreenPoint	- Installation of required digital systems in Austrian pilot - Evaluation methodology of data collection Demonstration of the multi-value DR activities for electricity consumers - Contacts to residential consumers from Wiener Neustadt	3.67
9 - BLUEPRINT	- Creation of Austrian pilot evaluation framework including KPI definition - Investigated and analysed options for interconnecting the Demand Response (DR) platform, which comprises a data aggregation platform (Comsensus) and a trading platform (OurPower). - Aimed for flexibility in the DR platform to accommodate various interconnection options. - Recommended standardized solutions suitable for the majority of cases, minimizing the need for customizations. - Participating meetings	4.50
10 - FællesBo	In cooperation with Neogrid: - Creation of Danish pilot evaluation framework including KPI definition. - Equipping the buildings with required infrastructure and sensors required for WP3 and WP4 tasks. - Coordinating regarding building up the demonstrator site.	4.76
11 - NEOGRID	- Creation of Danish pilot evaluation framework including KPI definition - Description and definition of technical architecture regarding Danish pilot	4.45



Partner	Work carried out in the WP within the period	PMs
	• Participation in meetings • Coordinating with Fællesbo regarding building up the demonstrator site • Equipping the buildings with required infrastructure and sensors required for WP3 and WP4 tasks • Configuring and setting up sensors for datalogging in context of building layout • Providing access to data for WP3 and WP4 tasks • Finalizing the sensor installation and configuration	
12 - GREEN CITIES	• Participating in meetings • Support of the Danish pilot site (FB) • Making leaflet – F to send out • Participating in KLIMAFOLKEMØDET in DK (see 2.1.6.6)	0.43
13 - HERON	• WP5 meetings attendance to ensure alignment with pilot activities, KPIs, and timeline • Contribution and input to D5.1 and D5.2. • Contribution and input to meeting presentations regarding the Greek pilot site • Participation in bilateral meetings organised by NTUA to discuss the scope, infrastructure, beneficiaries, datasets, and upgrade plans. • Input in the Data Gathering and Pilot Infrastructure Templates created by NTUA. Discussions and clarifications in order to register data sources and evaluate infrastructure. • Installed 7 Arcelik smart plugs and 1 gateway in a prepilot household • Collaboration with Arcelik for energy consumption data collection from smart plugs and temperature from the gateway. • Recruitment and integration of 92 HERON smart meters into the DEDALUS pilot, supporting data acquisition and baseline analysis for electricity consumption across 90 households • KPIs definition for baseline assessment focusing on energy efficiency, RES integration, flexibility and environmental impact	5.18



Partner	Work carried out in the WP within the period	PMs
	Design and develop users' engagement strategy given their requirements and needs in order to promote their real engagement and increase social acceptance impact	
14 – DOM-X	- Installation of 45new devices (for monitoring gas boilers and heat-pumps) - Installation of 30 energy meters, 10 of which are installed in the same households that feature heat-pumps - Installation of 10 indoor air quality sensors - Development of DOMX dashboard for controlling and visualizing the various assets - All device types are integrated with the DOMX dashboard - Development of the new DOMX mobile application with support for DR, to be released soon - Pilot preparation for being able to measure and report the required KPIs - Participation in WP5 meetings	5.75
15 - ISRAA	- Meetings attendance - Creation of Italian pilot evaluation framework, through engagement of end-users and preparation of the pilot with the installation of required digital systems/platform. - Collection of information regarding pilot infrastructure and data gathering systems - Definition of KPI for pilot actions - Collaboration with UNIVPM, ENG, DYNNAME for the actions to be implemented in the Italian pilot (T5.5) - Engagement of residents for the collection of thermic sensation data through hourly provided online questionnaires - Engagement of ISRAA technical department - Monitoring of KPI for pilot actions	8.32
16 - DYNNAME	- Participation in project meetings - Contribution to evaluation framework	2.34



Partner	Work carried out in the WP within the period	PMs
	• Analysis of data and info gathering market/performance energy • Analysis and contribution to KPIs • Contribution to actions for pilot • Contribution to business case qualification • Contribution to energy market analysis • Analysis consumption energy of buildings	
17 - CIMNE	• Meetings attendance and collaboration on T5.1 activities. • Conducted bilateral meetings with NTUA. • Participated in the data repository filling with samples of current datasets of the Spanish pilot. • Coordinating the Spanish pilot bilateral meetings. • Interaction with end-users at the pilot site to analyse the inputs for the nudging app. • Develop the data acquisition tools. • Developing and integrating communications of the battery and the smart meters deployed in the Spanish pilot. • Developing communications for the acquisition of the data of smart plugs and integrating it into the CIMNE infrastructure. • Start deploying Smart plugs and gateways with cooperation with PEUSA. • Defining the market and legal framework of the Spanish pilot. • Defining the KPIs for the Spanish pilot. • Defining the Local Area Business platform for the Spanish pilot. • Started developing the data acquisition for the Local Area Business. • Coordinating the Spanish pilot activities • Contributed to the drafting and review of Deliverable D5.2, documenting progress in the pre-pilot phase and plans for the full demonstration phase.	15.29
18 - PEUSA	• Participation in project meetings	6.50



Partner	Work carried out in the WP within the period	PMs
	- Description and definition of technical architecture regarding Spanish pilot - In cooperation with ROMUR: - Battery installation - Installation of required equipment in Spanish pilot	
19 - ROMUR	- Participation in project meetings - Description and definition of technical architecture regarding Spanish pilot - Battery installation - Developing specific EMS for Spanish pilot	12.75
20 - CENTRICA	- Attend WP5 meetings - Coordinate Irish pilot - Addressing the issue with Irish pilot and looking for new pilot	1.66
21 - TUC	In WP5 TUC has provided detailed infrastructure and data specifications for Replication Multiplier in Pilot 7 (RO). Also, TUC has contributed to analysing the DEDALUS KPIs and mapping the fitting ones onto Pilot 7 specifications. TUC has allocated efforts for defining the evaluation plan for Pilot 7 Romania contributing to deliverables D5.1 and D5.2. Finally, TUC has carried out activities for integrating and evaluating the TEs in the first evaluation cycle.	2.53
22- SIN	- Attend WP5 meetings - Collaborated with NTUA to define the framework for the technical workshops (timeline, participants, structure, content), - Developed the pre- and post-test questionnaires for assessing social acceptance of TEs. - Planned the overall social engagement strategy across all pilot sites, ensuring alignment with project objectives, pilot leader's goals and end-user needs.	1.39
23 - ICONS	N/A	0.00



1.2.5.9. *Deviations from the Grant Agreement*

Three main deviations from the Grant Agreement have been identified in the first 18 months of the project:

Firstly, the mapping of the technology enablers to the pilot sites mentioned in the GA required adjustments, as some of the TEs cannot be demonstrated in the mentioned pilot sites due to the unavailability of the required physical and/or digital equipment. For example, district heating is only present in the Danish pilot site, which means that TE-14 'Aggregation of flexible energy resources for district heating management' cannot be demonstrated in the Austrian Pilot Site as a multiplier like it was mentioned in the GA.

Secondly, the KPIs that are described in the GA cannot be addressed by all the pilot sites as they lack the necessary infrastructure to track them, or they are not relevant to the activities or the scope of the pilot demonstration. In addition, the KPI "Reduced share of pollution (NO_x and SO_x) at local level assessed through pilots' validation" has been removed because it does not apply to dwellings and thus cannot be measured.

Regarding the pilots:

Pilot 6 in Ireland has faced issues with the heat pump provider in the Irish demonstration. CENTRICA, as the pilot leader, lost this pilot and had to begin searching for an alternative pilot that could meet the following prerequisites for participation:

- Heat pumps should already be installed.
- Social housing units located in the Republic of Ireland, with ideally around 30 housing units involved in the pilot.
- A willingness to share heat pump sensor data with CENTRICA and potentially other project partners according to appropriate data sharing agreements.
- Any additional data, such as the number of residents in each household or smart meter data, would also be helpful but not strictly necessary.
- It would be advantageous if the site also had solar PV panels installed.

Concerning the WP5 deliverables and activities, the submission of D5.2, which was due by the end of September, was delayed by one month. This was required due to the tasks contributing to D5.2 (T5.2 - T5.8) commenced in June, leaving just four months to complete the deliverable, including the review process, which typically takes two weeks. Furthermore, during these four months, many partners were unavailable due to the summer holidays, which impacted their ability to provide timely feedback.

Additionally, due to unresolved issues in Pilot 6 located in Ireland, D5.1 and D5.2 were submitted with incomplete input from this pilot site. Once a new pilot site is finalized, the Project Coordinator plans to request the rejection of these deliverables to allow for their resubmission with the updated and relevant information.



1.2.5.10. *Corrective actions taken/suggested*

Corrective actions have been taken to address the mentioned deviations:

- Regarding the first point, an updated mapping of the technology enablers to each pilot site has been created with the support of the technology providers and the pilot site leaders. This should provide clear understanding and guidance to the service providers which will be the pilot sites that will demonstrate their solution.
- Regarding the second point, the mapping of the KPIs mentioned in the GA has been achieved through the provision of templates which clearly describe each KPI and ways to verify them, which were filled out by each pilot site.
- Regarding the Irish pilot (Pilot 6), CENTRICA has explored different alternatives to find a replacement pilot. Since May 2024, CENTRICA has actively engaged with 38 organizations, including heat pump providers, social housing associations, energy providers, and other stakeholders. Through over 170 interactions, CENTRICA identified 11 potential pilot sites in Ireland and Belgium (as it was allowed by the project officer). However, five sites declined participation, and four ceased communications despite follow-up attempts.

To ensure a viable solution, CENTRICA proposed adding a sub-contractor to the project. This new partner would provide customer engagement, hardware installation (if required), and maintenance support until the project's conclusion. The proposed pilot will target individual end-users in Ireland who already have installed heat pumps. This plan also involves a budget reallocation within CENTRICA's allocation in the Grant Agreement, shifting funds from Equipment Costs (Category C2) to Sub-contracting. Feedback from the Project Officer regarding this proposal has been positive. Following the first periodic report, an amendment will be initiated to formalize these changes. Additionally, once the new pilot site is finalized, activities that were not conducted due to these challenges will resume, and the relevant deliverables with missing inputs from CENTRICA will be updated and resubmitted with complete information. This ensures alignment with the project's objectives and reinstates the pilot's full operational capacity.



1.2.6. WP6 – People-powered Go2Market and exploitation strategies towards EU replication and market scalability

[M1-M36] [Leader: **HERON**, Participants: all]

WP6 aims at identifying the market impact of DEDALUS results, develop the instruments that will be used to define specific Go2Market pathways, by creating exploitation and replication strategies based on an exploitation framework which consists of market analysis. It aims to do so by developing multi-energy Demand Response Business Models that bring together aspects of sharing economy, social innovation and socially and environmentally driven non-financial benefits. WP6 will validate derived Business Models through a Business Sandbox, identifying main barriers and opportunities and value propositions. In this context, special focus is given in scaling-up consumer adoption of DR products and how to create market opportunities through policy interventions at EU and National level.

Finally, exploitation plans at project level and for the individual partners are drawn. This process is facilitated by the methodology developed in Tasks 6.1, 6.3 & 6.4, which identify the building blocks of DEDALUS TEs and related services and reconfigures them in Business Cases from which Business Models are derived.

Main effort of WP6 core partners with PMs in M1-M9 (HERON, ICONS, CENTRICA) focused on the successful completion of the Primary Exploitation Plan (Deliverable D6.1). Through M10-M18, WP6 expanded in 3 dimensions leading towards the Intermediate Exploitation Plan (D6.2):

1. Business Models & Exploitation // Validation of Business Cases and finalisations of sub-KERs
2. Standardisation // DEDALUS Standards Library and alignment with technology providers
3. Replication // Policy recommendation, bi-directional (to and from) interaction with decision makers

Effort at Task level is analysed below.

1.2.6.1. Task 6.1 DEDALUS sharing economy and social innovation-based business models

[M1-M36] [Leader: **HERON**, Participants: ENG, ARC, COMS, OurPower, BLUEPRINT, NEOGRID, HERON, DOM-X, DYNAM, PEUSA, CENTRICA]

Task 6.1 develops easy-to-adopt innovation-based business models for residential DR, engaging different actors of the energy value chain with different characteristics and apply/evaluate them through the practical application of DEDALUS pilots. Different business models capture, propagate, illustrate and share the flows of energy, payments, flexibility services, and system balancing and non-financial segments (e.g. environmental, social cohesion, job creation, improved efficiency of business activities), through the development of single-component diagrams, showing the way that each pilot project works and facilitating the optimization of procedures and results.

The concept of DEDALUS Business Cases was introduced in the 1st DEDALUS Exploitation Workshop (Sep '23). All project partners were asked to identify the Business Cases associated with their pilot sites as it is expected that each pilot stakeholder anticipates deriving value from



their involvement in DEDALUS irrespective of their role (commercial entity or non-profit). To this end, Business Cases were characterized by their primary owner, a detailed description and the relevant High-Level Use Cases (HLUCs) as they were identified in D2.1. Business Cases are restricted to each pilot and utilize multiple TEs and hence they break down DEDALUS KERs into sub-KERs aligned with DEDALUS Ambitions and Objectives. Based on the HLUCs, BCs were linked with Technological Innovations (TEs) identifying the technologies that are required for each BC to materialise. BCs will be the foundation blocks of the Business Models and sub-KERs the foundation blocks of the Business Models. Business Models are analysed in T6.3 and Sub-KERs are utilised in T6.4. Task 6.1 effort in the period M1- M9 was mainly documented in Section 6 in D6.1. During the period M10-M18 the concept of sub-KERs initially presented in D6.1 was further developed. KERs identified in the proposal preparation stage and listed in the GA, could not capture the effort of individual of each partner toward the development of DEDALUS TEs and services and therefore had to be broken down (an example of KER 1 and its sub-KERs in Table 6), In doing so, it is possible to identify the appropriate IR ownership, which is essential for effective exploitation, especially for commercial results. Two workshops (described in Task 6.4 in more detail) compiled DEDALUS sub-KER Library which defined IP foreground and clarified ownership and rights for using the results beyond the project. This ongoing work is part of the two-dimensional strategy regarding sub-KER contribution in DEDALUS Business Model development as the building blocks of DEDALUS Go2Market products and services:

1. Business Case and Business Model development: identify partner contribution towards a market ready product (T6.1 & T6.3)
2. Exploitation Strategy: background and foreground IP contribution for exploitation of market ready products (T6.4)

Table 6 - Indicative breakdown of a KER into sub-KERs.

KER 1. Social Science Framework, innovative multidimensional incentives, and nudging interventions for residential DR)	IP Foreground owner/s
1.1 SSH-based tools and methodologies	SIN
1.2a Methodologies for multidimensional incentive mechanisms	SIN
1.2b Toolkit for multidimensional incentive mechanisms	ISRAA
1.3a Nudging app intervention (electricity)	OURPOWER
1.3b Nudging app intervention (gas heating)	DOMX

1.2.6.2. Task 6.2 Contribution to standardisation

[M1-M36] [Leader: **CENTRICA**, Participants: ENG, CARTIF, ARC, COMS, BLUEPRINT, CENTRICA]



Task 6.2 contributes to relevant Standardisation Bodies (SDOs) at the interplay among ICT, smart energy grids, DR and Energy Data Spaces (assessment of existing standards on both national and EU level). Creation of synergies with energy regulatory bodies and network operators' associations, to boost data sharing friendly regulatory/business frameworks.

As part of the Preliminary Exploitation Plan DEDALUS standardization advice to policymakers and standard development organizations was outlined. Task 6.2 was initiated in M1-M9 and is expected to progress as DEDALUS data infrastructure and DEDALUS technical services are further developed.

For M1-M9, Task 6.2 effort was documented in Sections 5 and 7 in D6.1.

Following the conclusion of D6.1, during M10-M18, a DEDALUS standards library has been under development. Existing standards are under review and through interactions with WP3 & WP4 partners, their adoption in DEDALUS is under consideration. In this context, standards related to the DEDALUS ontology were reviewed as a part of WP3 and the results have been reported in Chapter 5 of D3.1. Moreover, Centrica reviewed OpenADR2.0 standard and identified its gap. The results of this review will be included in D6.2.

1.2.6.3. Task 6.3 Go2Market strategy: Market impact

[M1-M36] [Leader: **HERON**, Participants: ENG, NTUA, UNIVPM, ARC, COMS, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, HERON(4), DOM-X, ISRAA, DYNAM, CIMNE, PEUSA, ROMUR, CENTRICA, TUC, ICONS]

Task 6.3 evaluates the market impact of DEDALUS outcomes, identifies sub-KERs, characterises partners as “technical providers” or “service users” and calculates financial indicators associated with DEDALUS KERs through feasibility studies. Following T6.1 and T6.4 by grouping sub-KERs, Business Cases and Business Models are being drawn and for each of them a thorough market analysis is being conducted, consisting of a stakeholder analysis, target market assessment and competition and regulation overview (if regulatory barriers exist) with the results captured through a Business Model Canvas. Based on the Market Assessment and Business Model Canvas, a detailed Value Proposition for each of the commercially exploitable DEDALUS outcomes is being drawn.

The effort undertaken as part of Task 6.3 introduced the Business Model Methodology. From the proposed instruments (Market Analysis, Business Model Canvas & SWOT) T6.3 presented Market Analysis. The general definition of Market Analysis included Target Market segmentation, types of competition and legislation assessment. In addition, T6.3 introduced a methodology for the Business Cases market impact. Task 6.3 effort in the period M1- M9 was mainly documented in Sections 5 & 6 in D6.1.

During the period M10-M18, market assessment of Business Cases has been initiated based on the role of DEDALUS partners (technical providers, research organisation or service users) in the sub-KERs they either own or contribute significantly. Having defined the market analysis methodological tools in D6.1, this is the first step of the Go2Market strategy towards the assessment of the market impact of DEDALUS Business Models.



1.2.6.4. *Task 6.4 Go2Market strategy: Innovation management and exploitation*

[M1-M32] [Leader: **ICONS**, Participants: ENG, NTUA, CARTIF, UNIVPM, ARC, COMS, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAME, CIMNE, PEUSA, ROMUR, CENTRICA, TUC, ICONS, SIN]

This task focuses on implementing exploitation activities and consolidating the final exploitation plan by identifying DEDALUS Key Exploitable Results (KERs) and analysing their market impact. An IPR clinic will assess IP foregrounds and define IPR measures. Different workshops are being held throughout the project to refine exploitation strategies for each identified result, evaluate business potential, monitor market trends, and carry out SWOT analysis. Continuous feedback with KER owners will ensure consensus, culminating in a final exploitation workshop to consolidate the plan.

The task kicked off at the project's start, with initial activities focused on establishing a methodology for innovation management and exploitation of DEDALUS results. The methodology includes the following steps:

- 1- KER Identification Phase: Conducted until M16 for the initial identification of KERs and sub-KERs and extending to M36 to integrate feedback from subsequent activities, fine-tune KERs and sub-KERs, and adjust IP and IPR strategies as needed.
- 2- KERs Characterization Phase: Conducted until M24 for continuous monitoring of the identified KERs and sub-KERs from the project's first year, with ongoing updates until the project's end if new KERs emerge.
- 3- Consolidation of Business Cases: Concluding by M36, based on the outcomes from Steps 1 and 2.

To execute the first step, a preliminary exploitation webinar with the DEDALUS Consortium was conducted in M3, followed by a second session in M10. These webinars covered critical topics such as exploitation in Horizon Europe, the fundamental elements of exploitation, the exploitation process and the key components specific to DEDALUS, Key Exploitable Results (KERs), intellectual property (IP) foreground ownership, exploitation strategies, and stakeholder mapping. Additionally, the second webinar introduced the KERs Library, the tool developed to identify, collect and map all the KERs developed throughout the project.

The third workshop, held during the Plenary Meeting in Athens (M17), further analysed the sub-KERs derived from each result, confirming the IP background and collaboratively examining the IP foreground with all partners. This process identified single ownership or co-ownership of the results. This workshop marked the start of the second phase, laying the groundwork for building the KERs Library.

Through the involvement of all partners and one-on-one calls where necessary, the KERs have been detailed, and possible exploitation strategies have been defined for each one. Currently, the ICONS team is analysing the information collected in the KERs Library to refine the exploitation strategies and start defining the exploitation roadmaps for each KER.



1.2.6.5. Task 6.5 Roadmaps for replication strategies beyond DEDALUS

[M13-M36] [Leader: **GREEN CITIES**, Participants: ENG, NTUA, GREEN CITIES, ICONS, SIN]

This task has as goal to replicate and upscale the project's methods and solutions, towards implementation in other regions in the pilot countries and within Europe. This will consider where possible national, regional and local needs, as well as legislative, regulatory and institutional conditions and the energy market. The outcomes of this task will be blueprints and policy recommendations.

EGC has prepared an initial plan for developing the replication and policy recommendations (T6.5 and writing of D6.4).

The work with the policy recommendations and the roadmap started after the first year of the project (M13), and the roadmap will be completed by M25, so that it can be widely disseminated and presented until the end of the project, as well as after its completion. In this context, the first policy event took place by M16 (see in the following). The next will be in M25, and a final policy event is envisaged for M36 (to be included in D6.2)

The first two steps in the plan involved the engagement with politicians at the **KLIMAFOLKEMØDE** in Denmark (Peoples' Climate Meeting), and the workshop at the GA Athens to start getting input from all partners.

The Danish partner, EGC, had the opportunity to participate in the **KLIMAFOLKEMØDE** in Denmark with more than 30.000 participants. It was an outstanding opportunity to present DEDALUS to a larger audience and to get some feedback, especially from politicians. EGC invited 22 politicians and conducted 11 interviews. The results are gathered in an article, which is to be published on social media, on the website and to be mentioned in newsletters. Conclusions from the article were among others:

"Currently, not all stakeholders are fully aligned on the potential of DR systems, highlighting the need for open dialogue to integrate these technologies on a larger scale across Europe. Public awareness campaigns and access to smart technologies are crucial to ensure citizens actively participate in the transition. With well-defined roles across government, industry, and communities, Europe can transform its building sector into a driver of energy efficiency, grid stability, and carbon reduction, paving the way for a sustainable and resilient future."

Furthermore, EGC organised a workshop at the GA in Athens. We gained 12 insights about replication and policy from partners, such as:

- Replication will have to highlight in particular the issue of INTEROPERABILITY, and secondly the need for ENGAGEMENT with end-users through educational workshops, nudging and a user-friendly approach
- Expected challenges of replication are primarily lack of user acceptance and engagement



- Majority of partners are not, or to some extent, aware of current policies on DR systems in their country
- To implement DR systems, countries need to primarily implement proper standardisation and regulation, aside of clear value proposition for stakeholders and incentives
- Progress in our project has been blocked by lack of incentives and end-user awareness, non-flexible contracts and regulations, inefficiency of local government, and problems retrieving data from grid operators.

EGC is preparing a reach out to all partners to collect insights from the implementation of the project so far and follow up in relation to the policy events – the next to be organized in M25.

1.2.6.6. *Work carried out per partner per task and effort spent*

Table 7 – WP6 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	• Participation in the periodic meetings. • Attendance at exploitation workshops • Support in elaborating DEDALUS KERs and sub-KERs • Completion of the required definitions for each sub-KER, including owner, contributors, users, TEs, and type. • Peer review of D6.1. Active participation in T6.2 KoM.	2.41
2 - NTUA	• Participation in the periodic WP6 meetings. • Participation in the first exploitation workshop and support on elaboration of DEDALUS KERs and providing input to D6.1 • Participation in the second exploitation workshop in order to identify for each business use case specific sub-KERs • Participation in the third exploitation workshop (plenary meeting in Athens) providing validation of the second Exploitation workshop results and partial definition of the identified sub-KERs • Complete for each sub-KER the required definition including owner, contributors, user, TEs, and type • Contribution to the Go2Market strategy (T6.3-T6.4)	0.95



Partner	Work carried out in the WP within the period	PMs
3 - CARTIF	- WP6 meetings and exploitation workshops attendance. - KERs definition - Contributions to T6.2 (regarding the different standards addressed within WP3) and T6.4.	0.76
4 - UNIVPM	- WP6 meetings - Contributing to T6.3 and T6.4	0.70
5 - ARC	- Participation in the periodic meetings. - Participation to the exploitation workshop	2.00
6 - COMS	- Participating in meetings - Providing related KERs information	1.47
7 - OurPower	- Provide and review information on KER identification, stakeholder analysis and exploitation strategies regarding Austrian pilot; contributing to D6.1 - Participation in meetings and exploitation workshops - Gather relevant information and provide related KERs information - Contribution to deliverables	1.25
8 - GreenPoint	- Customer contacts and analysis and exploitation strategies in Austrian pilot - Participation in meetings and workshops	0.60
9 - BLUEPRINT	Participation of joint workshops with pilot sites, reviewed D6.1 and coordinated and aligned HLUC with business models and TEs.	1.30
10 - FællesBo	Contributing to Go2Market strategy (T6.3-T6.4)	0.45
11 - NEOGRID	Participating in First Exploitation workshop	0.00
12 - GREEN CITIES	- Lead T6.5 - Participation in meetings - Planning, organizing and implementing participation in KLIMAFOLKEMØDET (Peoples' Climate Assembly) in DK (WP7)	0.81



Partner	Work carried out in the WP within the period	PMs
	- Conducting 11 interviews with politicians (WP7) - Participating in workshops - Organizing workshop on policy recommendations in Athens	
13 - HERON	- Coordination of WP6 periodic meetings - Co-organisation of 1st Exploitation Workshop - Contribution on D6.1	4.34
14 - DOM-X	- Participation in the exploitation workshop - Definition of the sub-KERs - Participation in WP6 meetings	0.82
15 - ISRAA	- Meetings attendance - Participation in First Exploitation workshop, with inputs provided on KERs regarding the Italian pilot - Definition of Sub-KERs and discussion with interested partners	1.00
16 - DYNAME	- Participation in project meetings - Contribution to workshop - Contribute to KER analysis - Participation to market and exploitation strategies - Contribution to the sub-KER discussion	0.92
17 - CIMNE	- Participation in work package meetings - Contribution to workshop. - Contribute to KER analysis.	0.99
18 - PEUSA	- Participation in work package meetings - Contribution to workshop.	1.50
19 - ROMUR	Contribution to workshop	1.01
20 - CENTRICA	- Attend WP6 meetings - Contribute to D6.1 - Coordinate T6.2 - Working on the feedback regarding OpenADR2.0	0.16
21 - TUC	N/A	0.00



Partner	Work carried out in the WP within the period	PMs
22 - SIN	Meeting attendance	0.01
23 - ICONS	- Lead beneficiary of D6.1 - Co-organisation of 1st Exploitation Workshop - Co-organisation of the 2nd Exploitation Workshop - Co-organization of the 3rd Exploitation Workshop (Plenary Meeting in Athens) - Participation in the WP6 periodic meetings - Coordination of T6.4 initial activities: creation of the Key Exploitable Results (KERs) Library by collecting information from all partners; validation of the information and definition of IP ownership through one-on-one calls to establish the basis for final exploitation roadmaps; continuous monitoring and updates of KERs to refine exploitation strategies throughout the project.	5.28

1.2.6.7. *Deviations from the Grant Agreement*

As already reported in other WPs, the project lacked coordination in the first months and this affected all WPs, especially the ones started at M1, like WP6. Starting from December 2024 (M8) the coordinator has properly planned the release of the missing/pending WP6 deliverables, in particular, D6.1 Preliminary exploitation plan, whose official release was expected at month 6. At the time of writing delays have been caught up. Concerning the WP6 deliverables, the following one is the most relevant delay:

- Submission of D6.1. at M10 instead of M6

Following the submission of D6.1, there were no other deviations from the GA timeline.

1.2.6.8. *Corrective actions taken/suggested*

As soon as the coordination team was set up, the coordinating partner, ENG, prepared a plan together with WP6 leader. The plan was discussed with the PO and a new deadline was established to release the pending document (D6.1). Many dedicated audio conferences have been organized, all the open issues discussed, the document was drafted at that time and was shared and discussed with all the involved partners. The deliverable was released by the end of February 2024 (M10).



1.2.7. WP7 – Share, connect and attract

[M1-M36] [Leader: **ICONS**, Participants: all]

WP7 aims to design and implement a highly effective dissemination and communication strategy (according to the general principles outlined in Section 2.2.2) to maximise the impact of DEDALUS. There is an increasing need to fill the gap between technical/scientific ‘complex’ dissemination to experts (e.g. technical, science and policy experts) and ‘simplified’ communication to general public target audiences. Towards this direction, this WP will organise a constant dialogue between the project team and target stakeholder groups, turning the project key findings and lessons-learned into shorter and easy-to-access material using common language rather than specialist terms that speak to citizens. More specifically, the strategy will be built around a set of targeted tools and activities, conceived as a way to: (1) Raise awareness among the broader public and stakeholders on the potential benefits and challenges behind the residential DR solutions developed by the project; (2) Facilitate knowledge transfer around the innovative pool of DR assets provided by the project and tested in 5 pilot sites and 2 multipliers; (3) Engage with key stakeholders, market players and end-users in the EU residential sector to build a multilateral community; (4) Implement public outreach and engagement activities to raise understanding and social acceptance. DEDALUS will contribute, upon invitation by the CINEA, to common information and dissemination activities to increase the visibility and synergies between Horizon Europe supported actions.

By M18, the following communication activities have been implemented:

- June 2023: Launch of the DEDALUS social media channels (LinkedIn and X/Twitter)
- August 2023: Launch of the visual identity and the project’s landing page (www.dedalus-horizon.eu)
- November 2023: Launch of the project website
- December 2023; Production of communication kit (roll-up poster; leaflet; presentation video)

January 2024 – October 2024: Ongoing activities in the field of the editorial production, through the collaboration of a several number of partners, as well. Production of 7 editorial contents and their distribution on the project’s social media channels and the main European media multipliers (such as AlphaGalileo; BuildUP; X Science).

Regarding the clustering activities:

- September 2023: DEDALUS enters the Smart Energy Cluster, managed by IEECP and NTUA
- October 2023: DEDALUS participates in the technical workshop organized by DigiBUILD through the panel titled "Unlocking Building Flexibility: Blockchain-Driven DR in Electricity and District Heating Networks " attended by Dr. Tudor Cioara, from University of Clu-Napoca, as a panelist.



- Moreover, at the end of 2023, another joint technical workshop was organized by Blueprint Energy Solutions (Blueprint Energy Solutions) and OurPower Energiegenossenschaft SCE mbH (OurPower Energiegenossenschaft SCE mbH) with the EU-funded project ENPOWER. A LinkedIn post containing a detailed summary of the topics discussed is available (DEDALUS EU).
- January 2024: Joining the BRIDGE initiative. Over the course of the first semester of 2024, the coordinator participated in the BRIDGE General Assembly (April 2024; here the [news release](#)) and a several numbers of partners was allocated in the 4 BRIDGE working groups (Regulation; Business Models; Data Management; Citizen and Consumer Engagement).
- August - October 2024: Participation of ICONS, together with ENG, in a several number of multilateral meetings with the representatives of the Smart Energy Cluster to organize the participation of DEDALUS in ENLIT 2024 and develop a concept for the joint workshop.
- October 2024: participation in ENLIT with a stand shared with other 6 EU-funded projects within the Smart Energy Cluster and participation of the coordinator, as a panelist, in the joint workshop **“Unlocking the Power of Smart Energy: A New Era in Efficiency, Flexibility, and Innovation”**.

1.2.7.1. Task 7.1 Dissemination and communication strategy

[M1-M36] [Leader: **ICONS**, Participants: ENG, ICONS]

Task 7.1 is addressed at the development of an integrated Plan for Dissemination and Communication which outlines the strategy for the project to engage its target audience and address key stakeholders, to reach the expected impact goals, supporting the uptake and replication of the DEDALUS results. The Plan is strictly interconnected with the exploitation pathways defined in WP6 and will be first delivered at M6 (D7.1).

The Plan outlines the channels and formats that will be used to ensure broad coverage of all audiences targeted by the project, including key messages for each identified target group, communication and dissemination measures to be implemented, timelines and monitoring methodologies to assess the effectiveness of the strategy.

From M1 to M6, when the doc has been finalised and delivered, ICONS has worked to define the targets and the strategy to reach them, as well as identifying the key messages and the formats. The first step has been the identification of the stakeholders that have been selected according to the KERs' validation delivered in D6.1 “Preliminary Exploitation Plan”. 13 Stakeholders' groups have been identified and profiled through desk research. To match the stakeholders with the most suitable dissemination and communication format, it has been fundamental to elaborate the key messages that could better address and promote the project goals and



challenges. Over the course of the DEDALUS journey, these key messages will help us convey the highlights of the project in the best possible way through a collection of tailor-made communications that will occasionally be optimized and updated as needed and as the project progresses.

To elaborate the most effective messages, in M6 a dedicated online workshop with all WP leaders has been organised to collect feedback and inputs; the outputs generated in this occasion have been integrated in the D&C Plan.

Between M7 and M18 ICONS also created bilateral contacts with the demos' representatives so as to create a sort of C&D desks' network that could operate locally. Each desk will update ICONS with reference to the activity implemented and will get support for the communication tools that can be used to engage target subjects properly.

1.2.7.2. *Task 7.2. Enable public outreach and wide communication*

[M1-M36] [Leader: **ICONS**, Participants: ENG, NTUA, CARTIF, UNIVPM, ARC, COMS, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, CIMNE, PEUSA, ROMUR, CENTRICA, TUC, ICONS, SIN]

In Task 7.2, public communication has been launched with the design of the project's visual identity at M4, including logo, graphic elements and templates for reporting and presentations. Graphic materials and guidelines are included in a brandbook, as part of a main communication kit (D7.2) that also includes a project leaflet, poster and presentation video (M8). Online communication channels have been designed and launched following the visual identity: i) project website (M6, anticipated by a landing page at M4); ii) social media channels (LinkedIn and Twitter), supported by a dedicated social media strategy (outlined in D7.1). Public communication tools have been developed to address the general public and mainstream media, including: i) press releases; ii) blog articles and news items on the project website; iii) journalistic articles, written by professional journalists; iii) a 4-episodes audio podcast series, highlighting the main challenges and benefits behind DR for residential buildings; iv) social media regular animation and campaigns, leveraging EU/international celebrations and the media agenda.

In particular, between M1 and M4 the main efforts have been addressed at the creation of the **project's visual identity** that has been officially delivered in M2, which has been functional for the development of the project website and all other DEDALUS materials and documents. The process started with a **brand personality analysis exercise**, which highlighted the main characteristics attributed to the project. This helped the ICONS team in defining the project's peculiar traits to be conveyed through its brand identity and creating therefore a graphic proposal, complete with logo design.

The project logo, visual identity and guidelines on their correct use were collected in the DEDALUS Brandbook that has been distributed among partners to allow them to implement a



correct communication strategy. The graphic elements designed were also propaedeutic to the elaboration of the communication kit (i.e. flyer, roll up. Poster and project templates).



Figure 3 - A frame of the DEDALUS brand-book

In October 2024, ICONS launched a DEDALUS new pay off to make more understandable the demand response to the pilots' residents.

The new payoff, **“Bringing comfort and energy connection in everyday living”** aims to communicate that DEDALUS does not merely provide a technological solution but offers an energy connection that enhances the daily comfort of residents.

The concept is to transform energy into something accessible and seamlessly integrated into everyday life, delivering well-being and peace of mind at home, regardless of the residents' technical expertise.



In M4 the project landing page has been launched so as to start creating a digital space to spread the project's activities and results; the landing page, that operated mainly as a window site, was replaced in M7 by the proper website **www.dedalus-horizon.eu**.

The website, which is structured in several sections (see Figure 3), has been developed with WordPress and is a **living tool** in continuous evolution that will be regularly updated in terms of contents, new graphic layouts, tools, and features, according to the needs of the project.

The table below summarises the website structure.

Table 8 - DEDALUS website's structure

Section	Description
HOMEPAGE	The official entry point of the website. It features a brief introduction and links to the most relevant sections. Starting from December 2023 (M8), the page will embed the project presentation video to quickly provide the user with the key elements of DEDALUS.
ABOUT	This section includes the main objectives, the expected results, the presentation of the consortium. For each project partner there is a link to the company website.
PILOTS	The section presents the seven DEDALUS pilots. For each one, its main characteristics and the specific solutions being implemented are featured.
NEWS	The section will contain the news published by the project, such as press releases, journalistic articles and interviews, pageflows, newsletters, videos, internal publications like podcasts, info-sheets and Best Practice books.
EVENTS	This section lists both events organized by the project and others such as those organized by sister projects to support their C&D campaigns and allow the whole projects' cluster to achieve larger impacts.
RESOURCES	This page acts as a repository for official documents and allows users to download public materials produced by the project, such as deliverables, reports, publications and so on.
COMMUNICATION	This page acts as a repository for all the documents about the communication activities and the graphic materials (leaflets; roll-up poster, brandbook, for example).



CONTACTS

Contact details of the project Communication secretariat

The website follows the visual identity guidelines of DEDALUS and has been designed to be compatible with PC, tablet, and phone use.

The website is a living tool in continuous evolution as it has been regularly updated with new contents, tools, and features, according to the needs of the project. During the first 18 months, regular updates mainly concerning the "News" and "Events" pages have been published mainly in the form of posts created by ICONS with support from project partners.

The website is accessible to all viewers with no restrictions due to its **inclusive design**. It removes bias and assumptions from the website so that users do not feel excluded due to an impairment, demographics, or other circumstances.

The **MATOMO software** is used to draw detailed statistics about the website's traffic and the traffic sources.

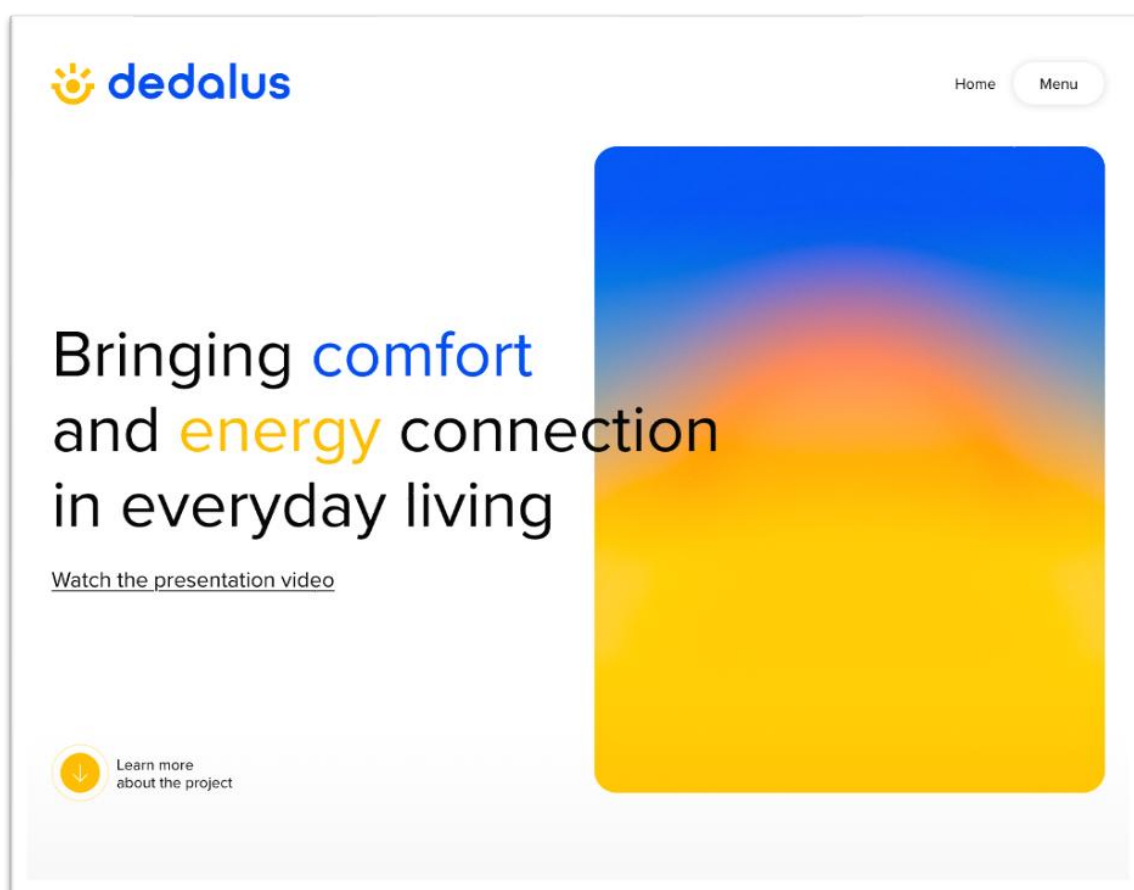


Figure 4 - The DEDALUS website's homepage



The DEDALUS social media strategy has been launched in June 2023 (M2) and has constantly continued during the entire reporting period.

This activity was managed by ICONS, in collaboration with partners who have posted about DEDALUS via their corporate channels.

The objective of the social media strategy is twofold: social networks serve both as an additional communication channel and as a participatory tool to foster dialogue and enhance public understanding, acceptance, and participation of end-users through dedicated messages.

Moreover, during the first 18 months of the projects, the social media activity was focused on the promotion of the project and its objectives; the presentation of each pilot, and the dissemination of the first three scientific papers produced within the Consortium

The project is on three **social media platforms**: LinkedIn, Mastodon, and YouTube.

In M2 the project's social media channels LinkedIn and Twitter have been created and launched; in M7 also the YouTube channel has been activated.

The project's Twitter/X account has been suspended in July 2024 (M15) by the X Administrator due to a technical issue experienced on July 2nd. Since the service has been discontinued, the ICONS social media team has taken action to resolve the issue by asking the Twitter/X administration team for prompt clarification and a swift resolution.

Due to the persistence of the issue, a contingency plan was developed by ICONS, so in M18 (October 2024) the Dedalus Mastodon Account has been created. Mastodon is a social media platform that is part of the “fediverse”, which essentially means that it is part of a decentralized and interconnected social network, where even independent servers can communicate with each other. An optimal server for the DEDALUS project that we have identified is <https://mastodon.energy/>.

The Mastodon and LinkedIn accounts are animated weekly with different kinds of posts to both expand the DEDALUS community and boost its engagement.

Until July 2024 (15) also the Twitter account was animated weekly.

Finally, to track online conversations about DEDALUS, the hashtag **#DEDALUS_EU** was launched at the beginning of the project (June 2023). It has been regularly monitored by ICONS.



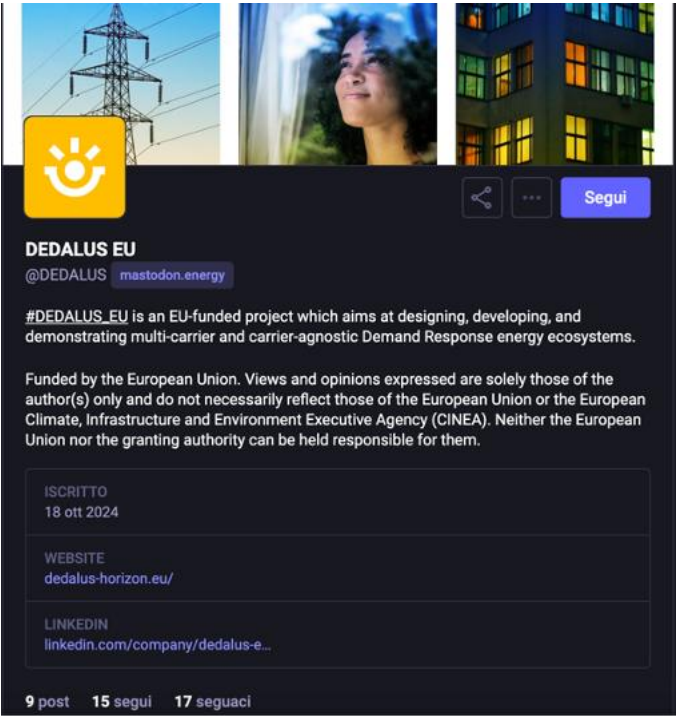
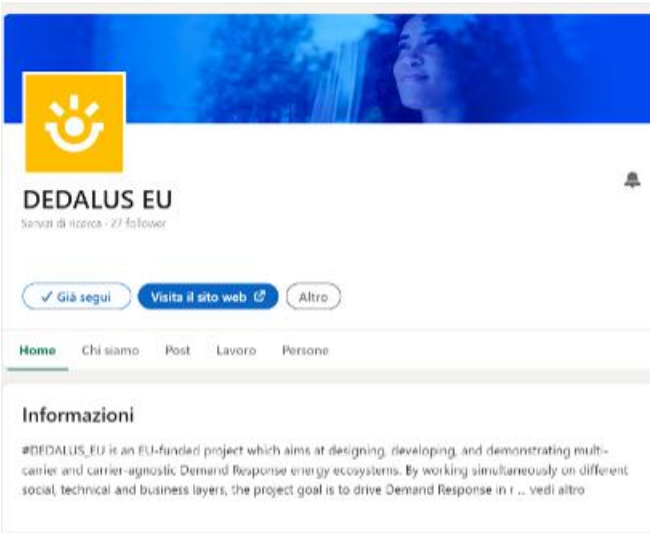




Figure 5 - DEDALUS social media presence (LinkedIn; X; YouTube)

In particular, during the first 18 months of the projects, the social media activity was focused on the promotion of the project and its objectives; the presentation of each pilot, and the dissemination of the first three scientific papers produced within the Consortium and the communication materials.

With reference to the **project's templates**, ICONS created three formats to be used by the partners, both for internal and external purposes: the deliverable template, to produce the project reports; the PowerPoint template, to be used during events, meeting, etc.; the minutes' templates to collect the minutes during project's meetings.

Between M7 and M9 ICONS has worked mainly on the preparation and delivery of the **project's leaflet** that includes main objectives, expected results, and approach, along with a brief overview of the consortium. The graphic layout was developed by the ICONS team following the project's graphic identity, while the texts were agreed between ICONS and the coordinating team.

In addition to the English version, 5 other versions in local languages have been designed; these versions are used by the local C&D desk to approach the target more efficiently.





Figure 6 - Composition of project's leaflet

In M8 the project roll up has been created and printed; this will be used every single time the project will be presented (i.e. event, exhibition, fair, etc.); a printable version has been created and sent to all partners so that everyone can print one and use for own needs related to the project,

The last product delivered in M8 was the **project video**: its aim is to **present the project, its objectives and the smart energy concepts**. The tone of voice used is **easy-to-understand**, so that its comprehension may be easy for a wide range of public audience.



Figure 7 - A framework of the DEDALUS presentation video

The DEDALUS video stands out for its ability to play on two distinct visual planes. On the one hand, it works on emotionality, through everyday situations it allows viewers to emotionally immerse themselves in family and social situations. On the other hand, through more landscape and abstract images it tells the context across the board, making it accessible to a wider audience. Graphic interventions emphasize key points, clarifying crucial elements that might not otherwise emerge clearly. In short, DEDALUS offers a complete visual experience, combining emotion, narrative, and graphic interventions.

The video is 1:07 minutes long and accessible from the project's website and the DEDALUS YouTube channel. It has been distributed on the project's social media channels and the main media multipliers. At the time of writing, it obtained **112 views**.

To watch the video, click [here](#).



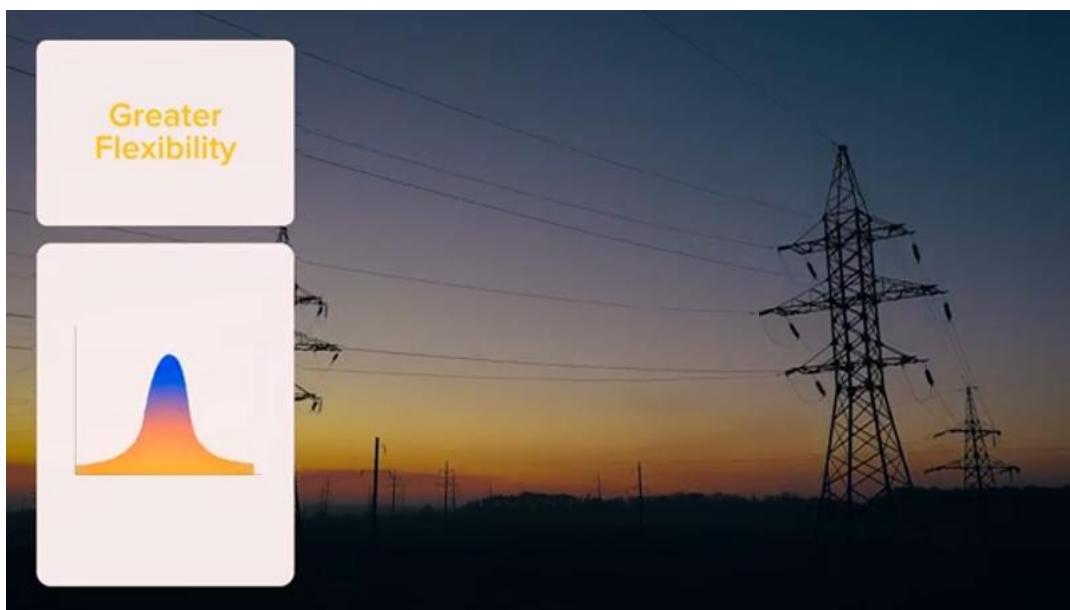


Figure 8 - A framework of the DEDALUS presentation video.



Figure 9 - A framework of the DEDALUS presentation video with project's logo

1. 1.2.7.2.6 Editorial production

Over the course of the first 18 months, a total amount of 8 **editorial content** has been published and distributed by the ICONS teams on the project's website and the main media multipliers.

The table below summarizes the types of publications:



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

Table 9 - Publications and distribution channels.

Type of publication	Number of publications	Distribution channels
Press releases	4	- Project website - DEDALUS social media channels - Media Multipliers - Social media channels of partners' organizations/companies
News releases	2	- Project website - DEDALUS social media channels - Social media channels of partners' organizations/companies
Project's updates	1	- Project website - DEDALUS social media channels - Social media channels of partners' organizations/companies

The ICONS teams also launched two specific **editorial formats**:

- **DEDALUS Discoveries**, that collects the press released focused on the explanation of the scientific papers produced by the partners in the field of the project.
- **DEDALUS Pilots' chronicles**, that tells the project's progress in each pilot from a social and technological point of view.

The table below summarises the DEDALUS editorial production during the first 18 months of the project.

Table 10 – Publications in detail.

Type of publication	Title	Link
Press release	The Dedalus project: harnessing demand response for Europe's energy transition	https://dedalus-horizon.eu/the-dedalus-project-harnessing-



		demand-response-for-europes-energy-transition/
Press Release	DEDALUS Discoveries Unlocking blockchain tokens: key valuation factors and new applications	https://dedalus-horizon.eu/unlocking-blockchain-tokens-key-valuation-factors-and-new-applications/
News Release	DEDALUS enters the Smart Energy Cluster	https://dedalus-horizon.eu/dedalus-enters-the-smart-energy-cluster/
News Release	DEDALUS joins BRIDGE initiative	https://dedalus-horizon.eu/dedalus-joins-bridge-initiative/
Project's Update	DEDALUS Pilots' Chronicles Engaging the elderly in demand response: first insights from our Italian pilot	https://dedalus-horizon.eu/dedalus-pilots-chronicles-engaging-the-elderly-in-demand-response-first-insights-from-our-italian-pilot/
News Release	United voices for an energy-efficient and sustainable building sector in Europe	https://dedalus-horizon.eu/united-voices-for-an-energy-efficient-and-sustainable-building-sector-in-europe/
Press Release	DEDALUS Discoveries Machine learning framework boosts residential electricity clustering for demand-response	https://dedalus-horizon.eu/new-machine-learning-framework-revolutionises-clustering-of-residential-electricity-usage-to-boost-demand-response-programmes/
Press Release	DEDALUS Discoveries Revolutionising building energy management: AI-driven demand response for a sustainable future	https://dedalus-horizon.eu/dedalus-papers-revolutionising-building-energy-management-ai-driven-demand-response-for-a-sustainable-future/

1.2.7.3. Task 7.3 Knowledge transfer and stakeholders dialogue

[M6-M36] [Leader: **ICONS**, Participants: ENG, NTUA, CARTIF, UNIVPM, ARC, COMS, OurPower, GreenPoint, BLUEPRINT, FællesBo, NEOGRID, GREEN CITIES, HERON, DOM-X, ISRAA, DYNAME, CIMNE, PEUSA, ROMUR, CENTRICA, TUC, ICONS, SIN]



This project has received funding from European Union's Horizon Europe Research and Innovation programme under the Grant Agreement No 101103998

This task is aimed at enabling a smooth transfer of the results and knowledge produced by the project, keeping the key stakeholders regularly informed throughout the different phases of project implementation while creating effective engagement patterns and dissemination pipelines. Actionable knowledge formats and content packaging will be developed through a plurality of tools and channels.

Between M1 and M18 the main efforts have been addressed at the planning of the most effective actions to guarantee a proper engagement of the key stakeholders.

ICONS is mapping the activities delivered by the partners in terms of participation to events and technical events; also, the promotion of papers and scientific articles prepared by the partners has been managed to improve the website navigation by users and interested subjects.

The newsletter has been designed and will be distributed at the end of November 2024.

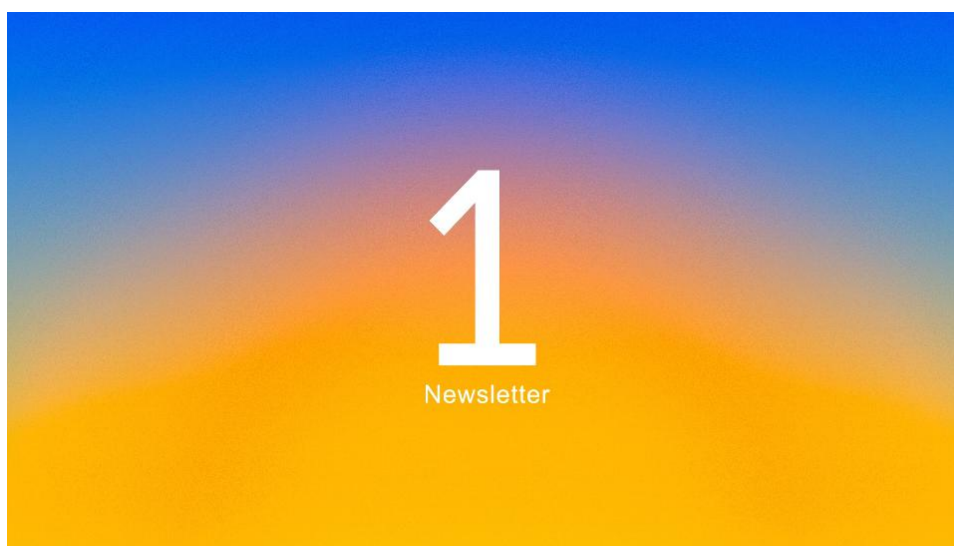


Figure 10 - The cover of the first Dedalus newsletter

Over the course of the first 18 months of the DEDALUS projects, the following scientific papers have been published:

- **Review of blockchain tokens creation and valuation**
[DOI: <https://doi.org/10.3390/fi15120382>]
- **A machine learning-based framework for clustering residential electricity load profiles to enhance demand response programs**
[DOI: <https://doi.org/10.1016/j.apenergy.2024.122943>]
- **Methodological approach for optimizing demand response in building energy management through AI-enhanced comfort-based flexibility models**
[DOI: <https://doi.org/10.23919/SpliTech61897.2024.10612382>]



All these scientific papers are available for download in the Resources section of the project's website.

During the first 18 months of the projects, the Consortium partners participated in a several number of European and international events.

The table summarizes the events attended by the partners between 2023 and 2024.

Table 11 - Events

Event	Type of event	Partners involved	Partners' role	Date
DigiBUILD Workshop "Innovation & Usability – Designing Tomorrow's Energy Efficiency Services for Building Users"	Workshop	Technical University of Cluj-Napoca	Panelist	October 2023
IEA ECB Annex 83 – 8th Meeting	International meeting	CIMNE	Panelist	February 2024
Smart Energy Cluster Second collaborative workshop	Workshop	ENG	Panelist	April 2024
BRIDGE Initiative – General Assembly	European Conference	ENG	Attendee, and responsible of Dedalus stand	April 2024
SpliTech 2024 – 9th International Conference on Smart and Sustainable Technologies	International Conference	UNIVPM	Panelist	June 2024
IISA 2024 4th International	International workshop	ENG; CARTIF; NTUA; UNIVPM	Attendees (on the occasion)	July 2024



workshop on data analytics in the Energy sector	of this event, they also produced a scientific paper)			
Climate People Meeting	International conference	European Green Cities	Presenter	August 2024
Sustainable Places 2024	International Conference	ENG; CARTIF	Panelists	September 2024
ENLIT 2024 (*)	International Conference	ENG (with the support of ICONS)	Panelist	October 2024

(*) During ENLIT 2024, two video interviews have been produced with an Italian trade media outlet, Canale Energia.

The partners interviewed by the Canale Energia's Editor-in -chief have been Engineering I.I. S.p.A, represented by Diego Arnone, and NTUA, represented by Elissaios Sarmas.

The interviews, available at [this link](#), were also disseminated on the DEDALUS social media channels.

1.2.7.4. *Task 7.4. Networking and collaboration with fellow projects and initiatives and cooperation with BRIDGE*

[M6-M36] [Leader: **ENG**, Participants: ENG, NTUA, CARTIF, UNIVPM, ICONS]

The task involves establishing collaborations and synergies with EU-level projects and initiatives focused on demand response (DR) and energy efficiency in residential buildings. It entails active participation in the BRIDGE initiative, particularly in data exchange and interoperability, and fostering connections with other EU-funded projects for joint dissemination activities, knowledge exchange, and event participation.

Since September 2023 (M5), the DEDALUS project has become part of the Smart Energy Cluster (Smart Energy Cluster, s.d.), managed by IEECP and NTUA, which encompasses 27* -funded projects operating in the energy field (*The data is updated to October 2024). This cluster focuses on merging different energy services and incorporating non-energy benefits while overcoming market fragmentation and fostering cooperation. This inclusive approach bridges gaps and creates a common ground for business development across different segments.

The cluster also aims to communicate the projects' activities and success stories to groups interested in smart energy solutions to amplify the impact of innovative service offerings, reduced energy costs for end-users, and faster payback periods for sustainable energy investments. A dedicated website is now available for more information (Smart Energy Cluster, s.d.).



ICONS, as a work package leader, and ENG, as task leader, have participated in several meetings within the #SmartEnergyCluster to better plan joint activities with the other cluster's EU-funded projects. Among these, *DigiBUILD*, *BUILDON*, *ProBono*, and *TwinEU* have been identified as sister projects, fostering deeper collaboration and knowledge exchange to address shared challenges and leverage synergies.

Moreover, in the field of clustering activities, in October 2023, the DEDALUS partner University of Cluj-Napoca participated in a technical workshop organized by DigiBuild. Dr. Tudor Cioara from the University of Cluj-Napoca conducted a panel titled "Unlocking Building Flexibility: Blockchain-Driven DR in Electricity and District Heating Networks". (DEDALUS EU)

Another joint technical workshop was organized by Blueprint Energy Solutions (Blueprint Energy Solutions) and OurPower Energiegenossenschaft SCE mbH (OurPower Energiegenossenschaft SCE mbH) with the EU-funded project ENPOWER. A LinkedIn post containing a detailed summary of the topics discussed is available (DEDALUS EU).

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Since September 2023 (M5), the DEDALUS project has become part of the Smart Energy Cluster (Smart Energy Cluster, s.d.), managed by IEECP and NTUA, which encompasses 26* -funded projects operating in the energy field (* The data is updated to October 2024). This cluster focuses on merging different energy services and incorporating non-energy benefits while overcoming market fragmentation and fostering cooperation. This inclusive approach bridges gaps and creates a common ground for business development across different segments.

The cluster also aims to communicate the projects' activities and success stories to groups interested in smart energy solutions to amplify the impact of innovative service offerings, reduced energy costs for end-users, and faster payback periods for sustainable energy investments. A dedicated website is now available for more information (Smart Energy Cluster, s.d.).

ICONS, as a work package leader, and ENG, as task leader, have participated in several meetings within the #SmartEnergyCluster to better plan joint activities with the other cluster's EU-funded projects. Among these, *DigiBUILD*, *BUILDON*, *Probono*, and *TwinEU* have been identified as sister projects, fostering deeper collaboration and knowledge exchange to address shared challenges and leverage synergies.

Additionally, the project team actively participated in the *Horizon Results Booster (HRB)*, a free service of the European Commission designed to maximize the impact of research and innovation projects. Through participation in HRB Module A, the team contributed valuable insights into clustering and dissemination strategies.

Dedalus participated in this free service within the project group also including the sister projects *DigiBUILD* (as applicant) and *BuildON*.



At the time of writing DEDALUS is waiting for the starting of the BOOSTER, the new free initiative launched by the European Commission.

Furthermore, as aforementioned, the DEDALUS project team participated in *Enlit Europe 2024 - Energy Transition Event*, held from 22 to 24 October 2024 in Milan. This event provided a platform to present the project's outcomes, strengthen collaborations, and showcase its role in shaping innovative solutions in the energy transition domain.

In particular, on the occasion of ENLIT 2024, Dedalus joined forces with other EU-funded projects within the Smart Energy Cluster. Together with 6 other European projects, Dedalus co-organized the joint workshop **“Unlocking the Power of Smart Energy: A New Era in Efficiency, Flexibility, and Innovation”**.

A specific session, titled **“Harnessing Data and Predictive Analytics for Smarter Energy Management: From Demand Response to Digital Efficiency in Building Renovations”** was held by DEDALUS and its sister project DigiBUILD.

Finally, since April 2024 ICONS – as WP7 leader - has been entering the BRIDGE Working Group “Citizen and Consumer Engagement”, participating in monthly meetings and collaborating with the other projects in a several number of ongoing activities.

1.2.7.5. Task 7.5 DEDALUS capacity building programme

[M12-M36] [Leader: UNIVPM, Participants: NTUA, CARTIF, UNIVPM, GREEN CITIES, ICONS]

The task involves disseminating knowledge and expertise developed within the DEDALUS project through a series of blended online and face-to-face training activities. Content developed by scientific partners, has been identified for the Digital Educational Kit hosted on the project website.

A technical workshop for technicians, scientific and academic audience will be organized on the framework of the IEEE Metrology for Living Environment conference that will be held in Venice in June 2025. A policymaker workshop will be organized by Green Cities during 2025. to promote DEDALUS innovations in Demand Response for residential sectors, fostering skills development and stakeholder engagement.

1.2.7.6. Work carried out per partner per task and effort spent

Table 12 – WP7 Effort per partner

Partner	Work carried out in the WP within the period	PMs
1 - ENG	Participation in internal meetings to strategize and establish clear goals and objectives for the task 7.4.	2.57



Partner	Work carried out in the WP within the period	PMs
	• Participation and planning in conferences and meetings related to efficiency and sustainable built environments. • Writing and presentation of the paper called “Methodology for assuring interoperability in demand-response services” for the IISA24 “The Fifteenth International Conference on Information, Intelligence, Systems and Applications”. The paper was written in collaboration with CARTIF. • Writing and submission of a paper to SP2024 (Sustainable Places, 2024). This paper is currently under publication on Open Research Europe. The paper was written in collaboration with CARTIF. • In April 2024, participation in the BRIDGE General Assembly, either physically or remotely. • In June 2024, remote participation in BRIDGE Data Management WG kick-off. • Remote Participation in BRIDGE Regulation WG kick-off meeting in September 2024. • Remote participation in the BRIDGE Data Management WG Plenary Meeting in November 2024. • Participation in ENLIT 2024 in Milan.	
2 - NTUA	General: • Participation in WP7 meetings • Contribution to all WP7 deliverables Task 7.2: • Regular public outreach and wide communication of project’s technical and scientific outcomes through EPU NTUA’s and individual researchers’ online communication channels. • Provision of input for the official DEDALUS social media and the official DEDALUS website. • Participation in interviews for dissemination purpose (content already available on the DEDALUS social media accounts). • Material provision for blog articles and news items on the project website Task 7.3: • Internal meetings to plan synergies and dependencies between T7.1 & 7.3	2.85



Partner	Work carried out in the WP within the period	PMs
	- Contribution to T7.3 activities related to dissemination (events planning and stakeholder engagement). - Drafting, submission and publication of the following scientific papers: - Papias Flexibility IISA - Michalakopoulos Applied Energy - Michalakopoulos Energy Reports - Michalakopoulos Energy (Pending Peer Review) - Michalakopoulos Applied Soft Computing (Pending Peer Review) Task 7.4: - Regular participation in BRIDGE activities (Elissaios Sarmas – EPU NTUA actively contributing in WG1 – Data Management and WP3 – Regulation) - Participation in ENLIT 2024 in Milan, Italy being part of the joint BRIDGE-ETIP SNET panel hosted as part of the EU Projects Zone. - Participation in the BRIDGE General Assembly 2024 in Brussels. - Active contribution to the ETIP SNET WG4 - Digitalization of the electricity system and customer participation Task 7.5: - Identified the content to be provided in the Digital Educational Kit - Contributed to the ToC for the Digital Educational Kit	
3 - CARTIF	- Writing and submission of a paper to IISA24 (The Fifteenth International Conference on Information, Intelligence, Systems and Applications) in collaboration with ENG. - Writing and submission of a paper to SP2024 (Sustainable Places, 2024) in collaboration with ENG. This paper is currently under publication on Open Research Europe. - Meeting attendance.	1.97
4 - UNIVPM	- WP7 meetings - Internal meetings to plan the work for the T7.5 - Contribute to T7.2 and T7.3 activities	0.80



Partner	Work carried out in the WP within the period	PMs
5 - ARC	- WP7 meetings - Contribute to T7.2 activities	0.91
6 - COMS	- Attending WP7 meetings - Dissemination on events and communication channels	0.78
7 - OurPower	- Planning and drafting communication concept and plan - Dissemination on events, communication channels and through stakeholder communication - Definition of key messages - Participation at meetings	0.70
8 - GreenPoint	- Participation in internal meetings - Dissemination activities on events - Participation at WP7 meetings - Participation at joint technical workshop	0.88
9 - BLUEPRINT	- Contribute to T7.2 and T7.3 activities - Presented WP2 results on the Social Science Framework at the Sustainable Places Conference 2024 - Posted several LinkedIn updates about the DEDALUS project including blogs on company's website - Co-wrote an article on the Social Science Framework	0.80
10 - FællesBo	Contribute to T7.2 and T7.3 activities	0.51
11 - NEOGRID	Contribute to T7.2	0.02
12 - GREEN CITIES	- Public outreach about the project (T7.2 communications) at KLIMAFOLKEMOEDET (30.000 visitors) - Leaflet & roll up for the KLIMAFOLKEMOEDE - Posted several LinkedIn updates about the DEDALUS project - Article on its way	2.82



Partner	Work carried out in the WP within the period	PMs
13 - HERON	• Participation in WP7 meetings. • Participation in Bridge Initiative meetings of 'Regulation' and 'Business Models' Working Groups • Dissemination activities in HERON's newsletter and social media • Contribution on key messages preparation and public outreach activities	0.88
14 - DOM-X	• Dissemination activities on expos /events (Energia.tec, INSEC-INDELEX/SECUREXPO) • Active dissemination on social media such as LinkedIn • Participation in WP meetings	0.58
15 - ISRAA	• Meetings attendance • Definition on key messages and collaboration with ICONS on the communication strategy to adopt for the specific target of Italian pilot • Translation of the leaflet and definition of future storytelling strategy for Italian pilot	0.95
16 - DYNAM	• Participation in project meetings • Contribution to target profiling, messages, users' subjects, strategy for communications	0.25
17 - CIMNE	• Developing web documentation for the DEDALUS web page. • Actively interacting in social media social media • Translation of several documents to Spanish and Catalan • Participating in events to disseminate about the project.	0.57
18 - PEUSA	• Translation of several documents to Spanish and Catalan • Participating in events to disseminate about the project.	0.80
19 - ROMUR	• Coordination with PEUSA.	0.79



Partner	Work carried out in the WP within the period	PMs
	- Translation of several documents to Spanish and Catalan - Participating in events to disseminate about the project	
20 - CENTRICA	No relevant activity was done as a part of this WP within the first 9 months of project.	0.00
21 - TUC	In WP7 TUC contributed to the communication and dissemination of the project through the participation in conferences and events and creating awareness by publishing scientific articles in conference proceedings and journals. TUC published as main author or contributor 7 Web of Science indexed and open access journal articles and 2 international conferences proceedings paper. Also, TUC has participated to different scientific and industrial events for disseminating the project, gathering feedback and engaging with potential stakeholders.	1.00
22 - SIN	Disseminating DEDALUS on SIN's website and social media channels.	0.01
23 - ICONS	- Coordination of WP - Elaboration of visual identity and graphic formats (i.e. templates, ppt) - Design and development of project website - Preparation and delivery of project video - Creation and animation of social media channels - Coordination of partners to collect inputs and contents for dissemination and communication purposes - Release of D&C Plan	14.11

1.2.7.7. Deviations from the Grant Agreement

The project lacked coordination in the first months and this also WP7. Starting from December 2024 (M8) the coordinator has properly planned the release of the missing/pending WP7 deliverables. At the time of writing delays have been caught up. Concerning the WP7 deliverables and activities, the following ones are the most relevant delays:



- Submission of D7.1. at M10 instead of M6
- Submission of D7.2 at M10 instead of month 8.

1.2.7.8. *Corrective actions taken/suggested*

In order to catch up the accumulated delays, ENG prepared a plan together with WP7 leader. The plan was discussed with the PO and a new deadline was established to release the pending documents. Many dedicated audio conferences have been organized, all the open issues discussed, all the documents drafted at that time were shared and discussed with all the partners involved. In the period between M1 and M18 all the expected deliverables were drafted. D7.1 and D7.2 were both submitted in month 10.



2. Impact

The first 18 months constitute a first phase of the project lifetime during which the Consortium has discussed all the innovative ideas expressed in the proposal and shaped the scope of the entire project. It's an important phase since all the decisions taken have an impact on all the activities and, as consequence, on the expected outcomes. To evaluate the impact of the project on the building environment, since the time of the proposal preparation, a set of Expected Outcomes (EO) of HORIZON-CL5-2022-D4-01-01, as well as the relevant Long-term Expected Impacts (LEI) have been defined, as specified in the Destinations in the work programme. The expected Key Exploitable Results (KER) of the project implementation are presented in Figure 11, detailing the potential end-users as well as the validation to ensure the TRL desired. In order to achieve this challenging maturity level, different results will exploit and adapt existing solutions developed in previous European projects and other initiatives where consortium partners are involved.

Key Exploitable Results (KERs)	Target groups & end users	WP	Obj	TE	TRL
 KER1- Social Science Framework, innovative multi-dimensional incentives and nudging interventions for residential DR	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Building/facility managers; Energy communities; Local authorities; Researchers	WP2	O1/5/7	All, especially TE-9	6-7
 KER2- AI-based individual and building-level consumers clustering and market segmentation algorithms	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Researchers	WP2	O1/5/7	TE-1	7
 KER3- Open APIs for DR-ready smart appliance IoT platform	Data Marketplaces Operators; Data Space Operators; Data Value Chain Stakeholders; Smart appliances manufacturers; Suppliers & operators of DR equipment; Providers of local business platforms and assets/group of assets; Researchers	WP3	O2/5/7	TE-2	7
 KER4- Energy Data Space adaptation for extended DR interoperability, and privacy-preserving DLT/Blockchain Data Governance and flexibility exchange		WP3	O2/5/7	TE-3, TE-5, TE-6, TE-7	6-7
 KER5- Digital Twins for optimal assets vs consumers flexibility planning and decentralised DR management	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders), and other Energy Value Chain & Data Value Chain Stakeholder; Energy communities; Local authorities; Researchers	WP4	O3/5/7	TE-4, TE-8	6
KER6- Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments		WP4	O3/5/7	TE-4, TE-9	6-7
 KER7- Electrical flexibility management tools for building-level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets		WP4	O4/5/7	TE-4, TE-11, TE-12, TE-13	7
KER8- Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralised cross-commodity electricity vs heat management		WP4	O4/5/7	TE-4, TE-10, TE-14	7
 KER9- Business Sandbox with novel sharing economy and social innovation -based DR business models	Residential energy consumers; Energy stakeholders (DR aggregators, energy suppliers, DSOs/electricity traders); Local authorities; Energy communities; SMEs; Policy makers; Researchers; Regulators	WP6	O5/6/7	All	6-7

Figure 11 - DEDALUS Key Exploitable Results (KER)

In the two sections that follow, two tables report on the advancement toward the achievement of the Expected Outcomes (EO) and of the Long-term Expected Impact (LEI).

2.1. Toward the achievement of the expected outcomes

EO	Brief description	WP	Progress towards the achievement of the EO
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1	Increased potential benefits, trust and acceptability of DR solutions for residential consumers	2 - 5	We participated in workshops with demos to identify what initiatives and solutions so far were done in pilots related to acceptability of DR solutions for residential consumers. Increasing benefits, trust or acceptability of DR will be properly evaluated in the second half of the project when the final version of the technologies developed by the project will be deployed on Pilots.
2	Advanced asset control and aggregation approaches that enable the participation of residential buildings in commercial DR	2,4,5, 7	We collected and updated data about infrastructure/assets in demos, in particular in Austria, that can participate in DR. In the second half of the project tools for asset control will be released in their final version.
3	Expanded pool of assets relevant for DR in the residential sector	3, 5, 6	In the first 18 months of the project no new asset has been developed yet but the related development has been designed in the preliminary requirements elicitation process.

2.2. Toward the Long-Term Expected Impacts (LEI)

LEI	Brief description	WP	Progress towards the achievement of the EO
1	Technological and socio-economic breakthroughs for achieving climate neutrality and the transition to zero pollution of the building stock by 2050, based on inclusive and people-centric R&I	2 - 7	18 months is still a time not yet enough to evaluate how and to what extent the project is achieving this LEI. This section will be updated during the second half of the project.
2	Building stock that effectively combine energy efficiency, renewable energy sources and digital and smart technologies to support the transformation of the energy system towards climate neutrality	5 and 6	18 months is still a time not yet enough to evaluate how and to what extent the project is achieving this LEI. This section will be updated during the second half of the project.



3. Update of the plan for exploitation and dissemination of results

Within DEDALUS, 9 broad Key Exploitable Results (KERs) and 34 sub-KERs have been identified (refer to the table below). For each of the identified KERs tailored exploitation strategies and business cases have been preliminary defined, in the following phases detailed exploitation roadmaps will be designed. An overview of the anticipated exploitable results at the project level is presented in the following table.

#	KERs	TRL at the project's end	Type
1	Social Science Framework, innovative multidimensional incentives and nudging interventions for residential DR	6-7	Framework
	1.1 SSH-based tools and methodologies	N/A	Methodologies
	1.2a Methodologies for multidimensional incentive mechanisms	N/A	Framework
	1.2b Toolkit for multidimensional incentive mechanisms	N/A	Tools
	1.3a Nudging app intervention (electricity)	6	Technology
	1.3b Nudging app intervention (gas heating)	7	Technology
2	AI-based individual and building level-consumer clustering and market segmentation algorithms	7	Technology, Algorithms
	2.1 SSH customer segmentation	6	Methodologies
	2.2 AI based electricity consumption clustering	7	Algorithms
	2.3 AI-based heating consumption clustering	7	Algorithms
3	Open APIs for DR-ready smart appliance IoT platform	7	Technology, data models
	3.1 DR ready electric appliances	7	Technology
	3.2 DR ready HVAC	7	Technology



4	Energy Data Space adaptation for extended DR interoperability, and privacy preserving DLT/Blockchain Data Governance and flexibility exchange	6-7	Technology, Data models
	4.1 DEDALUS ontology	N/A	Tools
	4.2 Interoperability building blocks	N/A	Technology
	4.3a Blockchain-based flexibility tokenization & DR settlement	6	Technology
	4.3b Blockchain-based privacy preserving DR (T3.5)	7	Technology
5	Digital Twins for optimal assets vs. consumers flexibility planning and decentralized DR manage	6	Technology, data models
	5.1a ML estimation of total electrical flexibility	7	Algorithms
	5.1b ML estimation of disaggregated electrical flexibility	7	Algorithms
	5.1b ML for disaggregation of electrical loads	7	Algorithms
	5.1c ML estimation of flexibility in Heat pumps	7	Algorithms
	5.2a Digital Twin as a framework	N/A	Tools
	5.2b Digital Twin Storage and DR	8	Tools
	5.2c Digital Twin physics driven heat pump	6	Tools
6	Comfort-based flexibility models and DR optimal management tools for individual residential consumers and/or serviced apartments	6-7	Technology/ tools, data models
	6.1a Sensor driven comfort-based flexibility	6	Algorithms
	6.1b Link DR with comfort-based flexibility	6	Technology
7	Electrical flexibility management tools for building level energy communities, energy districts and virtual cluster of buildings, based on pre-aggregation and shared DR assets	7	Technology /tools
	7.1 Disaggregation of electricity consumption	7	Algorithms
	7.2a Appliance specific implicit DR	7	Tools
	7.2b Appliance specific implicit DR	6	Algorithms



	7.3a Flexibility exchange / marketplace p2p digital platform	6	Technology
	7.3c Flexibility exchange / marketplace p2p methodology	N/A	Methodologies
	7.3b Building-level flexibility aggregator	7	Technology
	7.4a Algorithms for optimizing the energy dispatch and the flexibility in district-based energy communities	8	Algorithms
	7.4a Flexibility management of shared renewable generation, storage and shifting-loads in energy communities	8	Algorithms
	7.4.b. District level battery for collective self-consumption communities	9	Technology
8	Optimal aggregation of flexible energy (heat, electrical) assets for district heating and decentralized cross-commodity electricity vs. heat management	7	Technology, Service
	8.1 Large scale control of district heating	6	Technology
	8.2 Co-optimization of electricity based and district heating	6	Advancement in knowledge
9	Business Sandbox with the novel sharing economy and social innovation-based DR business models	6-7	Business tools

The DEDALUS project engages a diverse range of stakeholders from both energy and non-energy sectors, including energy market players, technology providers, social housing organizations, building owners, residential consumers, policymakers, and regulatory authorities. These stakeholders are crucial for the successful post-project exploitation and uptake of DEDALUS results. They can leverage the project's Key Exploitable Results (KERs) to replicate solutions, launch enhanced consumer services, unlock appliance and energy asset flexibility, and design new local or national demand response (DR) policies for the residential sector, among other impactful actions.

To sustain the exploitation, communication, and dissemination strategies, the analysis has identified 13 stakeholder groups and 51 sub-groups, as detailed in the following table.



Table 13 – Stakeholder groups

Stakeholder Groups	Sub-groups	KERs
Policy makers and authorities	- European institutions and agencies - National and regional authorities - Regulatory Agencies - Standardization organizations	1, 5, 6, 7, 8, 9
Energy service providers and water utilities	- Energy producers - Grid operators (TSOs and DSOs) - Balancing Responsible Parties - Energy market operators (retailers, traders, etc.) - DR aggregators - ESCOs - EV charging service providers - District heating & cooling network operators - Water network operators	1, 2, 3, 4, 5, 6, 7, 8, 9
Service and solution providers	- Energy management solution providers - DLT, Blockchain based solution providers - Big data and analytics solution providers - Managers and supporters of dataspace and datahub platforms - Cybersecurity service providers - Other service providers - AI, ML models developers and providers - IoT platform developers and providers - BIM and Digital Twins' solution developers - BMS platform developers - Developers and providers of Energy Conservation Measure (ECM) software - API and Gateway developers - Start-ups and spin-offs Apps developers	1, 2, 3, 4, 5, 6, 7, 8, 9
Smart appliances' manufacturers and asset integrator providers	- Appliances manufacturer and integrators - Home automation manufacturers and providers - HVAC and heat pump manufacturers - PV panels and RES technology providers - Residential battery storage system manufacturers and providers	3, 4



Building construction industries	• Architecture and engineering companies • Construction companies and contractors	•
Building owners and managers	• Building owners (public and private) • Building operators and managers • Real state agencies • Social housing operators • Smart co-housing operators	1, 2, 5, 6, 7, 8, 9
Consumers	• Residential energy consumers	1, 2, 5, 6, 7, 8, 9
Energy communities and cooperatives	• Local and district energy communities • Energy Cooperatives	1, 5, 6, 7, 8, 9
Academic and Research communities	• Research and Technology Organizations (RTOs) • Research centres • Universities • R&D departments of industrial players	All KERs
EU partnerships, projects, and initiatives	• EU funded projects • European initiatives and partnerships	All KERs
Technology clusters, industry associations, and organizations	• Industrial associations and clusters • Interest groups	All KERs
Financial investors	• Public/ private investors	5, 6, 7, 8, 9
Media	• Specialized press and local/international media	All KERs

To lay the basis of the DEDALUS exploitation strategy, five main pillars have been identified based on the nature of the preliminary KERs, the Consortium partners, and the value chain target groups:

- **Commercialization of products and services:** The commercialization roadmap for DEDALUS results includes three phases: **Pilot and Business Validation** (months 31-36) to refine offerings, **Market Launch**, to attract initial customers and expand the ecosystem, and **Global Expansion** beyond month 60 to establish a global presence. Leveraging related project communities, the results of the project aim for a sustainable market presence with direct market entry, partnerships, and diverse revenue models like subscriptions and pay-per-transaction.



- **TRL upscaling:** The DEDALUS project advances technologies for demand response (DR), moving them from TRL 5-6 to TRL 6-7. It focuses on data and flexibility models using ML, AI, and blockchain. Post-project, partners will push these technologies to TRL 9 for market readiness, requiring further R&D, funding, and pilot testing over 2-3 years. This collaborative effort ensures alignment with market and regulatory demands for successful commercialization and replication.
- **Research exploitation:** The project will facilitate further research, integrate results and findings into university courses, and support knowledge transfer through publications, conferences, and training. Both academia and industry R&D can benefit from this knowledge.
- **Upscaling and replication in pilots:** After the DEDALUS project, the upscaling strategy will enhance user experience by integrating new technologies and smart devices, expanding to other smart buildings and residential areas. End-users will continue to refine services. Replication will extend successful solutions beyond initial pilots, testing them in diverse contexts to ensure broader adaptation. Key pilots in Austria, Denmark, Greece, Italy, Spain, Ireland, and Romania will showcase concepts like digital twins, nudging apps, and blockchain frameworks, facilitating district-level DR and energy management. Pilot site owners will lead replication efforts, leveraging their expertise to implement and scale these technologies globally, ensuring market readiness and widespread adoption.
- **Advisory to policymakers and development of standards:** DEDALUS aims to foster the adoption of its tools by advocating for supportive legislation and providing policymakers with insights on the benefits of residential DR systems. The project will recommend standards to enhance reliability, customer trust, and reduce costs across Europe. By engaging with relevant Standard Development Organizations (SDOs), DEDALUS will contribute to standards for smart grids, building digitalization, appliance interoperability, and open APIs. This ensures the compatibility and scalability of its innovations, promoting broader adoption and impact.



4. Update of the data management plan

The data management plan is reported in D1.2 “DEDALUS data management plan” whose due date was M6 (October 2023) but, because of the delays explained in the section related to WP1, it was released on month 10 (end of February – beginning of March 2024).

In the last few months, some minor changes have been considered for the DEDALUS Data Management Plan (DMP), especially in the section “Project datasets”. The DEDALUS team is currently working on a draft of the updated version of the DMP. Below is a summary of the updates, which will be extensively detailed in the updated version of D1.2.

- WP1 pertains to Coordination and Management. Its datasets include administrative and financial management documentation, project reports and presentations, and identified project risks, all of which are available through the MS Teams repository hosted by ENG. There are no changes in the datasets identified in the first version of the DMP. The repository remains accessible to all consortium partners, who have been added as members of the DEDALUS team by ENG. Team members are periodically updated in response to requests from consortium partners and any changes in project responsibilities.
- WP2 - Socio-economic enablers for stimulating participatory, multi-value and energy carrier-agnostic residential DR – retains the dataset related to the consumers’ answers about their socioeconomic status, as well as their behaviour, literacy, attitude and willingness regarding energy and demand response.
- WP3 focuses on technology enablers for the data-driven 'activation' of residential energy consumers. In the scope of Task 3.4, CARTIF is using data from the SP pilot within its algorithm. Specifically, a dataset collects electricity consumption per dwelling (including PV production) and weather conditions. A dataset covering this need has already been reported in the current version of the DMP (D1.2) as Multi-energy flexibility data; thus, no additional information will be added to the table at this time. However, the naming convention will likely be updated in the next version of the DMP to make this dataset compliant to the DEDALUS data model. CARTIF plans to populate the data model with the necessary information, which will then be used by the algorithm, ensuring compliance with the DEDALUS data model. Additional datasets will include “Dwelling asset” which consists of a list of available electrical assets per dwelling, allowing for the characterization of the electricity consumption curve. CIMNE will interview homeowners to identify the available assets, such as washing machines, dryers, etc. Additionally, “users' behavior” dataset will focus on understanding how users interact with the previously listed assets. CIMNE will compile this information based on interviews.
- In the context of WP4, "Value-added energy and non-energy data-driven services for DR in energy-efficient residential buildings," two datasets related to the data from expected services will be added to the next version of the DMP.
- WP5, which focuses on the demonstration of multi-value and energy carrier-agnostic residential DR and social acceptance, also includes updates regarding data management. The dataset dedicated to monitoring data from Pilot 4 (Italy) has been updated. This dataset collects energy consumption data from two buildings (at the



apartment level) using 3-phase sensors, along with real-time indoor air quality (IAQ) conditions and light intensity data. The data will be utilized by the following DEDALUS technology enablers: TE-8: Digital Twins for consumer-aware optimal flexibility planning; TE-9: Comfort Flexibility Model for residential DR. Additionally, the dataset will support monitoring purposes. The updated version of the DMP will also involve an update of the data sets to be managed in the Spanish pilot site. These data sets combine open data sources and private data (from residents and housings) and are processed following the privacy rules defined in the GDPR. Moreover, in the new DMP, datasets from the Greek pilot have been included, containing smart meter data for residential consumers (energy, power, voltage & power factor) and smart plug data for monitoring of specific appliances (state, energy & power).

- WP6 and WP7: no changes will be included in the updated version of the DMP for the datasets identified by ICONS in relation to its dissemination and communication activities and by HERON for what concerns the People-powered Go2Market and exploitation strategies.



5. Deviations from Annex 1 and Annex 2

5.1. Use of Resources

5.1.1. Overview of the Person-Months spent in M1-M18 of the Project

The table below illustrates the period actual effort in person-months (PM), at Work Package (WP) level, with respect to the period planned (calculated as linear trend according to the duration of each WP). The % of Advancement vs. Total Planned effort (as in the DoA) is also shown, as well as the remaining effort. The actual effort claimed in the current period is **467.98** PMs, representing **49%** of the total planned effort.

Table 14 - RP1 Actual vs. Planned Effort per WP

Work Package	Actual	Planned	% Actual vs Period Planned	Deviation vs Period Planned	Total Planned	% Advancement vs Total Planned	Remaining
WP1	21.68	30.50	71%	-8.82	61.00	36%	39.32
WP2	117.11	125.00	94%	-7.89	125.00	94%	7.89
WP3	80.88	97.80	83%	-16.92	163.00	50%	82.12
WP4	79.39	97.44	81%	-18.05	174.00	46%	94.61
WP5	104.64	135.53	77%	-30.89	256.00	41%	151.36
WP6	28.73	45.50	63%	-16.77	91.00	32%	62.27
WP7	35.55	44.50	80%	-8.95	89.00	40%	53.45
TOTAL DEDALUS	467.98	576.27	81%	-108.29	959.00	49%	491.02

For better understanding of the effort situation per WP, a graphic representation of the actual effort spent on each WP compared with the DoA total planned effort is provided here below.

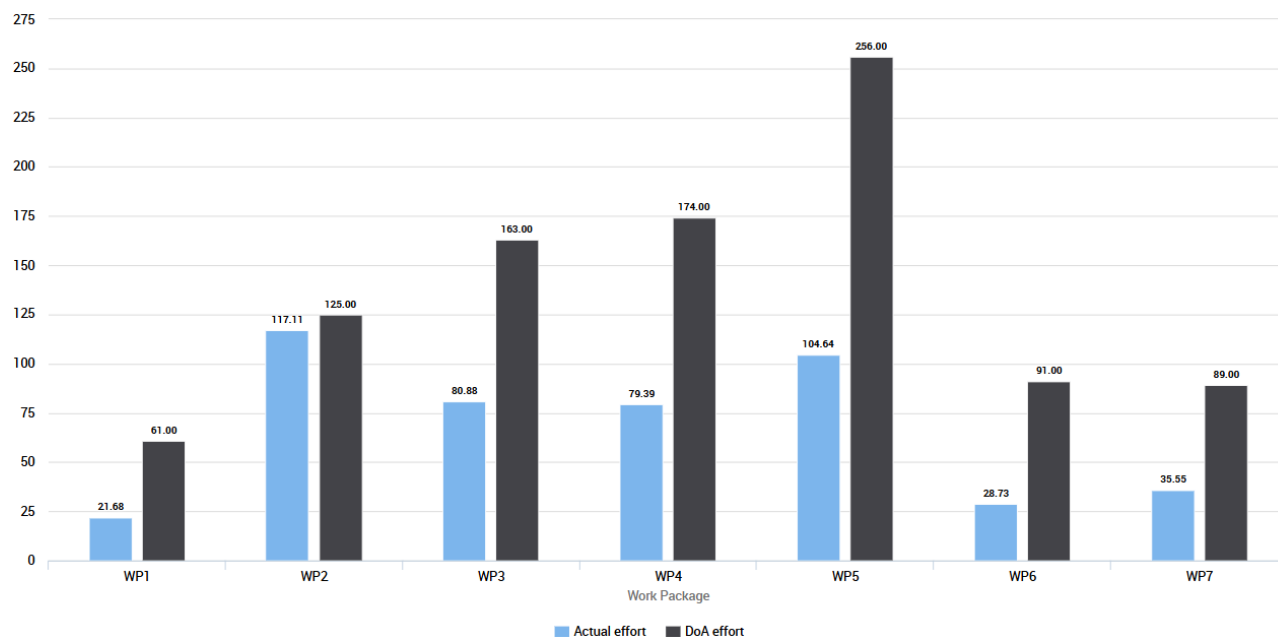


Figure 12 - RP1 Actual vs. Total Planned Effort per WP



On the other hand, the following table shows the effort spent by each Beneficiary per work package compared to the planned effort. The information regarding the % of Advancement and the remaining planned figures are also included in the tables. Explanations on the major deviations are included in the Section “5.1.3 Explanations on deviations” of this Report.

Table 15 - RP1 Actual vs. Planned Effort per Beneficiary

	Actual	Planned	% Actual vs Period Planned	Deviation vs Period Planned	Total Planned	% Advancement vs Total Planned	Remaining
WP1	10.93	15.00	73%	-4.07	30.00	36%	19.07
WP2	8.75	12.00	73%	-3.25	12.00	73%	3.25
WP3	9.18	12.60	73%	-3.42	21.00	44%	11.82
WP4	15.92	21.84	73%	-5.92	39.00	41%	23.08
WP5	4.24	9.53	44%	-5.29	18.00	24%	13.76
WP6	2.41	3.50	69%	-1.09	7.00	34%	4.59
WP7	2.57	4.00	64%	-1.43	8.00	32%	5.43
TOTAL ENG	54.00	78.47	69%	-24.47	135.00	40%	81.00
WP1	1.45	3.50	41%	-2.05	7.00	21%	5.55
WP2	12.92	12.00	108%	0.92	12.00	108%	-0.92
WP3	5.58	8.40	66%	-2.82	14.00	40%	8.42
WP4	1.87	3.36	56%	-1.49	6.00	31%	4.13
WP5	10.58	12.71	83%	-2.13	24.00	44%	13.42
WP6	0.95	2.00	48%	-1.05	4.00	24%	3.05
WP7	2.85	3.00	95%	-0.15	6.00	48%	3.15
TOTAL NTUA	36.20	44.97	81%	-8.77	73.00	50%	36.80
WP1	0.82	1.00	82%	-0.18	2.00	41%	1.18
WP2	2.99	3.00	100%	-0.01	3.00	100%	0.01
WP3	14.20	13.20	108%	1.00	22.00	65%	7.80
WP4	4.64	4.48	104%	0.16	8.00	58%	3.36
WP5	1.42	3.18	45%	-1.76	6.00	24%	4.58
WP6	0.76	1.00	76%	-0.24	2.00	38%	1.24
WP7	1.97	2.50	79%	-0.53	5.00	39%	3.03
TOTAL CARTIF	26.80	28.36	95%	-1.56	48.00	56%	21.20
WP1	0.60	1.00	60%	-0.40	2.00	30%	1.40
WP2	6.00	6.00	100%	0.00	6.00	100%	0.00
WP3	0.00	0.00	0%	0.00	0.00	0%	0.00
WP4	4.10	7.84	52%	-3.74	14.00	29%	9.90
WP5	0.90	3.71	24%	-2.81	7.00	13%	6.10
WP6	0.70	1.00	70%	-0.30	2.00	35%	1.30
WP7	0.80	4.50	18%	-3.70	9.00	9%	8.20
TOTAL UNIVPM	13.10	24.05	54%	-10.95	40.00	33%	26.90
WP1	0.50	0.50	100%	0.00	1.00	50%	0.50
WP2	4.00	4.00	100%	0.00	4.00	100%	0.00
WP3	10.50	10.80	97%	-0.30	18.00	58%	7.50
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	2.10	2.65	79%	-0.55	5.00	42%	2.90
WP6	2.00	2.00	100%	0.00	4.00	50%	2.00
WP7	0.91	1.00	91%	-0.09	2.00	46%	1.09
TOTAL ARC	20.01	20.95	96%	-0.94	34.00	59%	13.99
WP1	0.40	0.50	80%	-0.10	1.00	40%	0.60
WP2	2.49	2.00	125%	0.49	2.00	125%	-0.49
WP3	9.67	11.40	85%	-1.73	19.00	51%	9.33
WP4	3.83	4.48	85%	-0.65	8.00	48%	4.17
WP5	1.24	3.18	39%	-1.94	6.00	21%	4.76
WP6	1.47	2.00	74%	-0.53	4.00	37%	2.53
WP7	0.78	1.00	78%	-0.22	2.00	39%	1.22
TOTAL COMS	19.88	24.56	81%	-4.68	42.00	47%	22.12
WP1	0.49	0.50	98%	-0.01	1.00	49%	0.51
WP2	4.17	4.00	104%	0.17	4.00	104%	-0.17
WP3	1.08	1.20	90%	-0.12	2.00	54%	0.92
WP4	8.08	8.96	90%	-0.88	16.00	51%	7.92
WP5	4.64	7.41	63%	-2.77	14.00	33%	9.36
WP6	1.25	1.50	83%	-0.25	3.00	42%	1.75
WP7	0.70	1.00	70%	-0.30	2.00	35%	1.30
OURPOWER	20.41	24.57	83%	-4.16	42.00	49%	21.59



	Actual	Planned	% Actual vs Period Planned	Deviation vs Period Planned	Total Planned	% Advancement vs Total Planned	Remaining
WP1	0.45	0.50	90%	-0.05	1.00	45%	0.55
WP2	3.59	4.00	90%	-0.41	4.00	90%	0.41
WP3	2.24	1.80	124%	0.44	3.00	75%	0.76
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	3.67	5.29	69%	-1.62	10.00	37%	6.33
WP6	0.60	1.00	60%	-0.40	2.00	30%	1.40
WP7	0.88	1.00	88%	-0.12	2.00	44%	1.12
GREENPOINT	11.43	13.59	84%	-2.16	22.00	52%	10.57
WP1	0.50	0.50	100%	0.00	1.00	50%	0.50
WP2	12.00	13.00	92%	-1.00	13.00	92%	1.00
WP3	0.80	1.20	67%	-0.40	2.00	40%	1.20
WP4	9.50	8.96	106%	0.54	16.00	59%	6.50
WP5	4.50	5.29	85%	-0.79	10.00	45%	5.50
WP6	1.30	2.00	65%	-0.70	4.00	33%	2.70
WP7	0.80	1.00	80%	-0.20	2.00	40%	1.20
BLUEPRINT	29.40	31.95	92%	-2.55	48.00	61%	18.60
WP1	0.21	0.50	42%	-0.29	1.00	21%	0.79
WP2	2.99	3.00	100%	-0.01	3.00	100%	0.01
WP3	0.99	1.20	83%	-0.21	2.00	50%	1.01
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	4.76	6.35	75%	-1.59	12.00	40%	7.24
WP6	0.45	1.00	45%	-0.55	2.00	23%	1.55
WP7	0.51	1.00	51%	-0.49	2.00	26%	1.49
FAELLESBO	9.91	13.05	76%	-3.14	22.00	45%	12.09
WP1	0.48	0.50	96%	-0.02	1.00	48%	0.52
WP2	1.14	2.00	57%	-0.86	2.00	57%	0.86
WP3	0.63	1.80	35%	-1.17	3.00	21%	2.37
WP4	7.21	5.60	129%	1.61	10.00	72%	2.79
WP5	4.45	5.82	76%	-1.37	11.00	40%	6.55
WP6	0.00	2.00	0%	-2.00	4.00	0%	4.00
WP7	0.02	1.00	2%	-0.98	2.00	1%	1.98
TOT. NEOGRID	13.93	18.72	74%	-4.79	33.00	42%	19.07
WP1	0.36	0.50	72%	-0.14	1.00	36%	0.64
WP2	3.44	4.00	86%	-0.56	4.00	86%	0.56
WP3	0.00	0.00	0%	0.00	0.00	0%	0.00
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	0.43	2.12	20%	-1.69	4.00	11%	3.57
WP6	0.81	4.00	20%	-3.19	8.00	10%	7.19
WP7	2.82	3.00	94%	-0.18	6.00	47%	3.18
GREEN CITIES	7.86	13.62	58%	-5.76	23.00	34%	15.14
WP1	0.46	0.50	92%	-0.04	1.00	46%	0.54
WP2	7.00	7.00	100%	0.00	7.00	100%	0.00
WP3	3.81	4.20	91%	-0.39	7.00	54%	3.19
WP4	7.99	8.40	95%	-0.41	15.00	53%	7.01
WP5	5.18	8.47	61%	-3.29	16.00	32%	10.82
WP6	4.34	4.50	96%	-0.16	9.00	48%	4.66
WP7	0.88	1.00	88%	-0.12	2.00	44%	1.12
TOTAL HERON	29.66	34.07	87%	-4.41	57.00	52%	27.34
WP1	0.55	0.50	110%	0.05	1.00	55%	0.45
WP2	12.46	10.00	125%	2.46	10.00	125%	-2.46
WP3	4.66	6.00	78%	-1.34	10.00	47%	5.34
WP4	5.65	6.72	84%	-1.07	12.00	47%	6.35
WP5	5.75	6.35	91%	-0.60	12.00	48%	6.25
WP6	0.82	1.50	55%	-0.68	3.00	27%	2.18
WP7	0.58	1.00	58%	-0.42	2.00	29%	1.42
TOTAL DOM-X	30.47	32.07	95%	-1.60	50.00	61%	19.53
WP1	0.56	0.50	112%	0.06	1.00	56%	0.44
WP2	7.98	7.00	114%	0.98	7.00	114%	-0.98
WP3	1.90	1.80	106%	0.10	3.00	63%	1.10
WP4	3.80	3.92	97%	-0.12	7.00	54%	3.20
WP5	8.32	7.94	105%	0.38	15.00	55%	6.68
WP6	1.00	1.00	100%	0.00	2.00	50%	1.00
WP7	0.95	1.00	95%	-0.05	2.00	48%	1.05
TOTAL ISRAA	24.51	23.16	106%	1.35	37.00	66%	12.49
WP1	0.28	0.50	56%	-0.22	1.00	28%	0.72
WP2	1.51	3.00	50%	-1.49	3.00	50%	1.49
WP3	0.92	1.80	51%	-0.88	3.00	31%	2.08
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	2.34	4.76	49%	-2.42	9.00	26%	6.66
WP6	0.92	1.50	61%	-0.58	3.00	31%	2.08
WP7	0.25	0.50	50%	-0.25	1.00	25%	0.75
TOT. DYNAM	6.22	12.06	52%	-5.84	20.00	31%	13.78



	Actual	Planned	% Actual vs Period Planned	Deviation vs Period Planned	Total Planned	% Advancement vs Total Planned	Remaining
WP1	1.14	1.00	114%	0.14	2.00	57%	0.86
WP2	3.01	3.00	100%	0.01	3.00	100%	-0.01
WP3	4.00	4.80	83%	-0.80	8.00	50%	4.00
WP4	3.00	2.24	134%	0.76	4.00	75%	1.00
WP5	15.29	12.18	126%	3.11	23.00	66%	7.71
WP6	0.99	1.00	99%	-0.01	2.00	50%	1.01
WP7	0.57	1.00	57%	-0.43	2.00	29%	1.43
TOTAL CIMNE	28.00	25.22	111%	2.78	44.00	64%	16.00
WP1	0.30	0.50	60%	-0.20	1.00	30%	0.70
WP2	0.50	1.00	50%	-0.50	1.00	50%	0.50
WP3	1.00	1.20	83%	-0.20	2.00	50%	1.00
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	6.50	6.35	102%	0.15	12.00	54%	5.50
WP6	1.50	1.50	100%	0.00	3.00	50%	1.50
WP7	0.80	1.00	80%	-0.20	2.00	40%	1.20
TOTAL PEUSA	10.60	11.55	92%	-0.95	21.00	50%	10.40
WP1	0.50	0.50	100%	0.00	1.00	50%	0.50
WP2	1.00	1.00	100%	0.00	1.00	100%	0.00
WP3	2.74	4.80	57%	-2.06	8.00	34%	5.26
WP4	1.51	2.24	67%	-0.73	4.00	38%	2.49
WP5	12.75	9.00	142%	3.75	17.00	75%	4.25
WP6	1.01	1.00	101%	0.01	2.00	51%	0.99
WP7	0.79	1.00	79%	-0.21	2.00	40%	1.21
TOTAL ROMUR	20.30	19.54	104%	0.76	35.00	58%	14.70
WP1	0.17	0.50	34%	-0.33	1.00	17%	0.83
WP2	0.59	4.00	15%	-3.41	4.00	15%	3.41
WP3	1.98	3.60	55%	-1.62	6.00	33%	4.02
WP4	0.29	5.04	6%	-4.75	9.00	3%	8.71
WP5	1.66	4.76	35%	-3.10	9.00	18%	7.34
WP6	0.16	3.50	5%	-3.34	7.00	2%	6.84
WP7	0.00	1.00	0%	-1.00	2.00	0%	2.00
TOTAL CENTRICA	4.85	22.41	22%	-17.55	38.00	13%	33.15
WP1	0.00	0.50	0%	-0.50	1.00	0%	1.00
WP2	9.00	9.00	100%	0.00	9.00	100%	0.00
WP3	5.00	6.00	83%	-1.00	10.00	50%	5.00
WP4	2.00	3.36	60%	-1.36	6.00	33%	4.00
WP5	2.53	4.76	53%	-2.24	9.00	28%	6.47
WP6	0.00	1.00	0%	-1.00	2.00	0%	2.00
WP7	1.00	1.00	100%	0.00	2.00	50%	1.00
TOTAL TUC	19.53	25.62	76%	-6.09	39.00	50%	19.47
WP1	0.49	0.50	98%	-0.01	1.00	49%	0.51
WP2	9.24	10.00	92%	-0.76	10.00	92%	0.76
WP3	0.00	0.00	0%	0.00	0.00	0%	0.00
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	1.39	3.71	38%	-2.32	7.00	20%	5.61
WP6	0.01	1.00	1%	-0.99	2.00	1%	1.99
WP7	0.01	1.00	1%	-0.99	2.00	1%	1.99
TOTAL SIN	11.14	16.21	69%	-5.07	22.00	51%	10.86
WP1	0.04	0.50	8%	-0.46	1.00	4%	0.96
WP2	0.34	1.00	34%	-0.66	1.00	34%	0.66
WP3	0.00	0.00	0%	0.00	0.00	0%	0.00
WP4	0.00	0.00	0%	0.00	0.00	0%	0.00
WP5	0.00	0.00	0%	0.00	0.00	0%	0.00
WP6	5.28	5.00	106%	0.28	10.00	53%	4.72
WP7	14.11	11.00	128%	3.11	22.00	64%	7.89
TOTAL ICONS	19.77	17.50	113%	2.27	34.00	58%	14.23
TOT. DEDALUS	467.98	576.27	81%	-108.29	959.00	49%	491.02



For better understanding of the effort situation per Beneficiary, a graphic representation of the **effort spent by each Beneficiary** compared with the **DoA total planned effort** is provided here below:

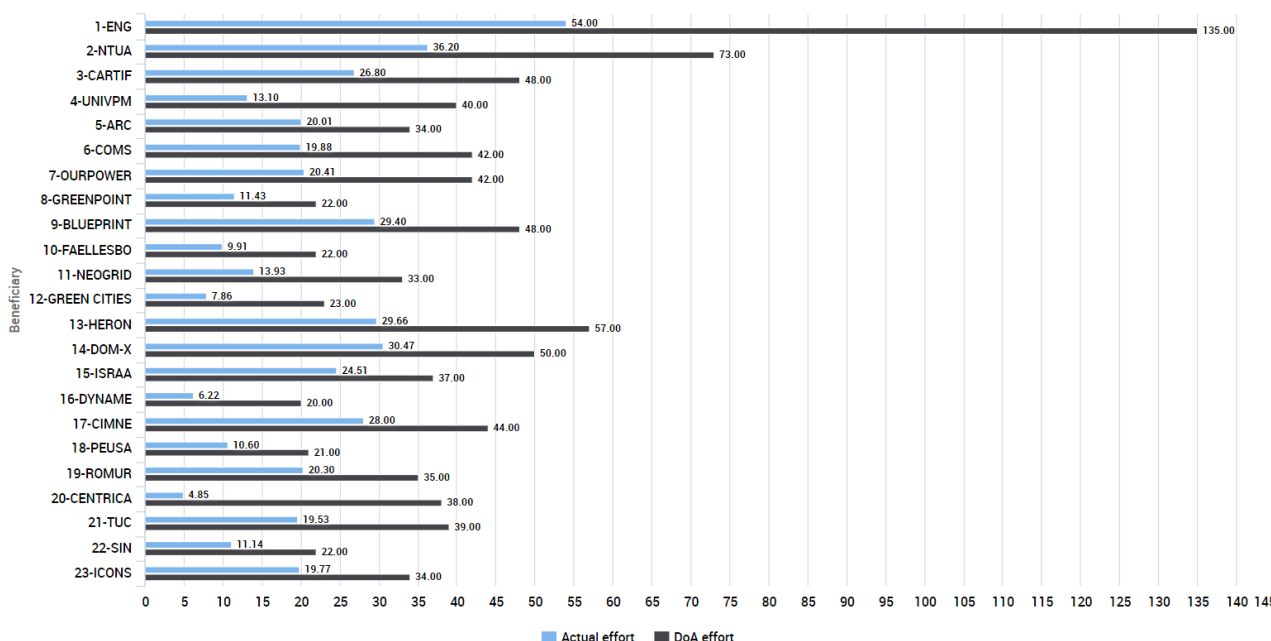


Figure 13 - RP1 Actual vs. Total Planned Effort per Beneficiary

5.1.2. Personnel, Subcontracting and Other major direct costs items for each Beneficiary for the Period M1-M18

Here below there is an overview of the period costs and funding in comparison with the DoA total planned figures. The % advancement vs. total planned and the remaining costs and funding are also shown. Please note that the first period represents 50% of the Project lifetime (month 18 out of 36). All amounts are declared in Euro. The actual costs claimed in the first reporting period amount to **3,463,865.31** euro, representing the **47%** of the total planned costs.



Table 16 - RP1 Costs and Funding Overview

		Actual	Total Planned	% Advancement vs Total Planned	Remaining
1 ENG	Total Costs	417,292.85	785,875.00	53%	368,582.15
	EU Requested contribution	292,105.00	550,112.50	53%	258,007.50
2 NTUA	Total Costs	213,380.08	494,500.00	43%	281,119.92
	EU Requested contribution	213,380.08	494,500.00	43%	281,119.92
3 CARTIF	Total Costs	160,932.70	251,875.00	64%	90,942.30
	EU Requested contribution	160,932.70	251,875.00	64%	90,942.30
4 UNIVPM	Total Costs	99,473.70	283,125.00	35%	183,651.30
	EU Requested contribution	99,473.70	283,125.00	35%	183,651.30
5 ARC	Total Costs	166,603.34	238,375.00	70%	71,771.66
	EU Requested contribution	116,622.34	166,862.50	70%	50,240.16
6 COMS	Total Costs	111,149.79	280,875.00	40%	169,725.21
	EU Requested contribution	77,804.85	196,612.50	40%	118,807.65
7 OURPOWER	Total Costs	199,402.46	438,650.00	45%	239,247.54
	EU Requested contribution	139,581.72	307,055.00	45%	167,473.28
8 GREENPOINT	Total Costs	86,988.75	175,625.00	50%	88,636.25
	EU Requested contribution	60,892.13	122,937.50	50%	62,045.37
9 BLUEPRINT	Total Costs	213,925.01	370,625.00	58%	156,699.99
	EU Requested contribution	149,747.51	259,437.50	58%	109,689.99
10 FAELLESBO	Total Costs	107,853.95	250,037.50	43%	142,183.55
	EU Requested contribution	107,853.95	250,037.50	43%	142,183.55
11 NEOGRID	Total Costs	183,764.26	371,750.00	49%	187,985.74
	EU Requested contribution	128,634.98	260,225.00	49%	131,590.02
12 GREEN CITIES	Total Costs	67,251.25	214,125.00	31%	146,873.75
	EU Requested contribution	67,251.25	214,125.00	31%	146,873.75
13 HERON	Total Costs	215,192.21	430,250.00	50%	215,057.79
	EU Requested contribution	150,634.55	301,175.00	50%	150,540.45
14 DOM-X	Total Costs	203,745.46	421,375.00	48%	217,629.54
	EU Requested contribution	142,621.82	294,962.50	48%	152,340.68
15 ISRAA	Total Costs	122,295.65	238,750.00	51%	116,454.35
	EU Requested contribution	122,295.65	238,750.00	51%	116,454.35
16 DYNAMIE	Total Costs	90,743.95	160,500.00	57%	69,756.05
	EU Requested contribution	63,520.77	112,350.00	57%	48,829.23
17 CIMNE	Total Costs	186,158.75	278,375.00	67%	92,216.25
	EU Requested contribution	186,158.75	278,375.00	67%	92,216.25
18 PEUSA	Total Costs	70,966.31	145,762.50	49%	74,796.19
	EU Requested contribution	49,676.42	102,033.75	49%	52,357.33
19 ROMUR	Total Costs	113,673.30	207,500.00	55%	93,826.70
	EU Requested contribution	79,571.31	145,250.00	55%	65,678.69
20 CENTRICA	Total Costs	41,690.61	503,750.00	8%	462,059.39
	EU Requested contribution	29,183.43	352,625.00	8%	323,441.57
21 TUC	Total Costs	151,186.84	330,000.00	46%	178,813.16
	EU Requested contribution	151,186.84	330,000.00	46%	178,813.16
22 SIN	Total Costs	101,595.19	194,875.00	52%	93,279.81
	EU Requested contribution	101,595.19	194,875.00	52%	93,279.81
23 ICONS	Total Costs	138,598.90	292,500.00	47%	153,901.10
	EU Requested contribution	138,598.90	292,500.00	47%	153,901.10
Total Costs		3,463,865.31	7,359,075.00	47%	3,895,209.69
Total EU requested contribution		2,829,323.84	5,999,801.25	47%	3,170,477.41



The following table instead shows the **costs incurred in the first period, per cost item and Beneficiary**, compared to the **DoA total planned costs**. The information regarding the % of Advancement and the remaining costs is also included in the table. Explanations on deviations are included in the Section “5.1.3 Explanations on deviations” of this Report.

Table 17 - RP1 Actual vs. Total Planned Costs per Beneficiary

		Actual	Total Planned	% Advancement vs Total Planned	Remaining
1 ENG	Personnel costs	320,624.96	607,500.00	53%	286,875.04
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,557.51	14,500.00	31%	9,942.49
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	8,651.81	6,700.00	129%	-1,951.81
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	83,458.57	157,175.00	53%	73,716.43
	Total 1 ENG	417,292.85	785,875.00	53%	368,582.15
2 NTUA	Personnel costs	147,370.32	379,600.00	39%	232,229.68
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	13,343.00	6,500.00	205%	-6,843.00
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	9,990.74	9,500.00	105%	-490.74
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	42,676.02	98,900.00	43%	56,223.98
	Total 2 NTUA	213,380.08	494,500.00	43%	281,119.92
3 CARTIF	Personnel costs	120,612.58	192,000.00	63%	71,387.42
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	7,290.33	6,500.00	112%	-790.33
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	843.25	3,000.00	28%	2,156.75
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	32,186.54	50,375.00	64%	18,188.46
	Total 3 CARTIF	160,932.70	251,875.00	64%	90,942.30
4 UNIVPM	Personnel costs	63,921.66	200,000.00	32%	136,078.34
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	6,975.30	7,500.00	93%	524.70
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	8,682.00	19,000.00	46%	10,318.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	19,894.74	56,625.00	35%	36,730.26
	Total 4 UNIVPM	99,473.70	283,125.00	35%	183,651.30
5 ARC	Personnel costs	130,750.27	170,000.00	77%	39,249.73
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	2,502.68	9,700.00	26%	7,197.32
	Equipment	0.00	10,000.00	0%	10,000.00
	Other goods works and	29.72	1,000.00	3%	970.28
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	33,320.67	47,675.00	70%	14,354.33
	Total 5 ARC	166,603.34	238,375.00	70%	71,771.66



		Actual	Total Planned	% Advancement vs Total Planned	Remaining
6 COMS	Personnel costs	84,389.18	214,200.00	39%	129,810.82
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,530.65	7,500.00	60%	2,969.35
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	0.00	3,000.00	0%	3,000.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	22,229.96	56,175.00	40%	33,945.04
	Total 6 COMS	111,149.79	280,875.00	40%	169,725.21
7 OURPOWER	Personnel costs	147,520.88	325,920.00	45%	178,399.12
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	3,247.31	6,500.00	50%	3,252.69
	Equipment	7,843.21	15,000.00	52%	7,156.79
	Other goods works and	910.57	3,500.00	26%	2,589.43
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	39,880.49	87,730.00	45%	47,849.51
	Total 7 OURPOWER	199,402.46	438,650.00	45%	239,247.54
8 GREENPOINT	Personnel costs	62,791.00	132,000.00	48%	69,209.00
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	2,200.00	6,500.00	34%	4,300.00
	Equipment	2,450.00	n.a.	n.a.	-2,450.00
	Other goods works and	2,150.00	2,000.00	108%	-150.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	17,397.75	35,125.00	50%	17,727.25
	Total 8 GREENPOINT	86,988.75	175,625.00	50%	88,636.25
9 BLUEPRINT	Personnel costs	165,375.00	288,000.00	57%	122,625.00
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	5,198.51	6,500.00	80%	1,301.49
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	566.50	2,000.00	28%	1,433.50
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	42,785.00	74,125.00	58%	31,340.00
	Total 9 BLUEPRINT	213,925.01	370,625.00	58%	156,699.99
10 FAELLESBO	Personnel costs	84,865.84	189,530.00	45%	104,664.16
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	678.66	7,500.00	9%	6,821.34
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	738.66	3,000.00	25%	2,261.34
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	21,570.79	50,007.50	43%	28,436.71
	Total 10 FAELLESBO	107,853.95	250,037.50	43%	142,183.55
11 NEOGRID	Personnel costs	128,555.46	273,900.00	47%	145,344.54
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,473.30	7,500.00	60%	3,026.70
	Equipment	13,982.65	15,000.00	93%	1,017.35
	Other goods works and	0.00	1,000.00	0%	1,000.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	36,752.85	74,350.00	49%	37,597.15
	Total 11 NEOGRID	183,764.26	371,750.00	49%	187,985.74



		Actual	Total Planned	% Advancement vs Total Planned	Remaining
12 GREEN CIT	Personnel costs	51,856.00	151,800.00	34%	99,944.00
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	1,754.00	9,500.00	18%	7,746.00
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	191.00	10,000.00	2%	9,809.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	13,450.25	42,825.00	31%	29,374.75
	Total 12 GREEN CITIES	67,251.25	214,125.00	31%	146,873.75
13 HERON	Personnel costs	165,754.58	313,500.00	53%	147,745.42
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	3,174.26	6,500.00	49%	3,325.74
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	3,224.93	24,200.00	13%	20,975.07
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	43,038.44	86,050.00	50%	43,011.56
	Total 13 HERON	215,192.21	430,250.00	50%	215,057.79
14 DOM-X	Personnel costs	140,427.90	300,000.00	47%	159,572.10
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,795.57	6,500.00	74%	1,704.43
	Equipment	388.36	6,000.00	6%	5,611.64
	Other goods works and	13,577.26	24,600.00	55%	11,022.74
	Internally invoiced goods	4,759.10	n.a.	n.a.	-4,759.10
	Indirect costs	39,797.27	84,275.00	47%	44,477.73
	Total 14 DOM-X	203,745.46	421,375.00	48%	217,629.54
15 ISRAA	Personnel costs	90,153.38	166,500.00	54%	76,346.62
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,857.49	6,500.00	75%	1,642.51
	Equipment	2,795.65	15,000.00	19%	12,204.35
	Other goods works and	30.00	3,000.00	1%	2,970.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	24,459.13	47,750.00	51%	23,290.87
	Total 15 ISRAA	122,295.65	238,750.00	51%	116,454.35
16 DYNAMÉ	Personnel costs	46,812.87	120,000.00	39%	73,187.13
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	4,182.29	6,400.00	65%	2,217.71
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	21,600.00	2,000.00	1080%	-19,600.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	18,148.79	32,100.00	57%	13,951.21
	Total 16 DYNAMÉ	90,743.95	160,500.00	57%	69,756.05
17 CIMNE	Personnel costs	143,788.31	211,200.00	68%	67,411.69
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	5,138.69	6,500.00	79%	1,361.31
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	0.00	5,000.00	0%	5,000.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	37,231.75	55,675.00	67%	18,443.25
	Total 17 CIMNE	186,158.75	278,375.00	67%	92,216.25



		Actual	Total Planned	% Advancement vs Total Planned	Remaining
18 PEUSA	Personnel costs	53,055.52	105,110.00	50%	52,054.48
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	799.98	6,500.00	12%	5,700.02
	Equipment	2,917.55	5,000.00	58%	2,082.45
	Other goods works and	0.00	n.a.	n.a.	0.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	14,193.26	29,152.50	49%	14,959.24
	Total 18 PEUSA	70,966.31	145,762.50	49%	74,796.19
19 ROMUR	Personnel costs	77,938.64	140,000.00	56%	62,061.36
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	3,250.00	6,500.00	50%	3,250.00
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	9,750.00	19,500.00	50%	9,750.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	22,734.66	41,500.00	55%	18,765.34
	Total 19 ROMUR	113,673.30	207,500.00	55%	93,826.70
20 CENTRICA	Personnel costs	31,489.61	342,000.00	9%	310,510.39
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	1,862.88	9,500.00	20%	7,637.12
	Equipment	0.00	45,000.00	0%	45,000.00
	Other goods works and	0.00	6,500.00	0%	6,500.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	8,338.12	100,750.00	8%	92,411.88
	Total 20 CENTRICA	41,690.61	503,750.00	8%	462,059.39
21 TUC	Personnel costs	115,357.05	253,500.00	46%	138,142.95
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	5,592.42	7,500.00	75%	1,907.58
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	0.00	3,000.00	0%	3,000.00
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	30,237.37	66,000.00	46%	35,762.63
	Total 21 TUC	151,186.84	330,000.00	46%	178,813.16
22 SIN	Personnel costs	77,929.91	147,400.00	53%	69,470.09
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	3,242.04	7,500.00	43%	4,257.96
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	104.20	1,000.00	10%	895.80
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	20,319.04	38,975.00	52%	18,655.96
	Total 22 SIN	101,595.19	194,875.00	52%	93,279.81
23 ICONS	Personnel costs	106,799.76	221,000.00	48%	114,200.24
	Subcontracting	0.00	n.a.	n.a.	0.00
	Travel costs	3,704.31	8,300.00	45%	4,595.69
	Equipment	0.00	n.a.	n.a.	0.00
	Other goods works and	375.05	4,700.00	8%	4,324.95
	Internally invoiced goods	0.00	n.a.	n.a.	0.00
	Indirect costs	27,719.78	58,500.00	47%	30,780.22
	Total 23 ICONS	138,598.90	292,500.00	47%	153,901.10
TOTAL DEDALUS		3,463,865.31	7,359,075.00	47%	3,895,209.69



In addition to the information included in the table above and for better understanding of the costing situation per Beneficiary, a graphic representation of the **costs spent by each Beneficiary** for the Project compared with the **DoA total planned costs** is provided below:

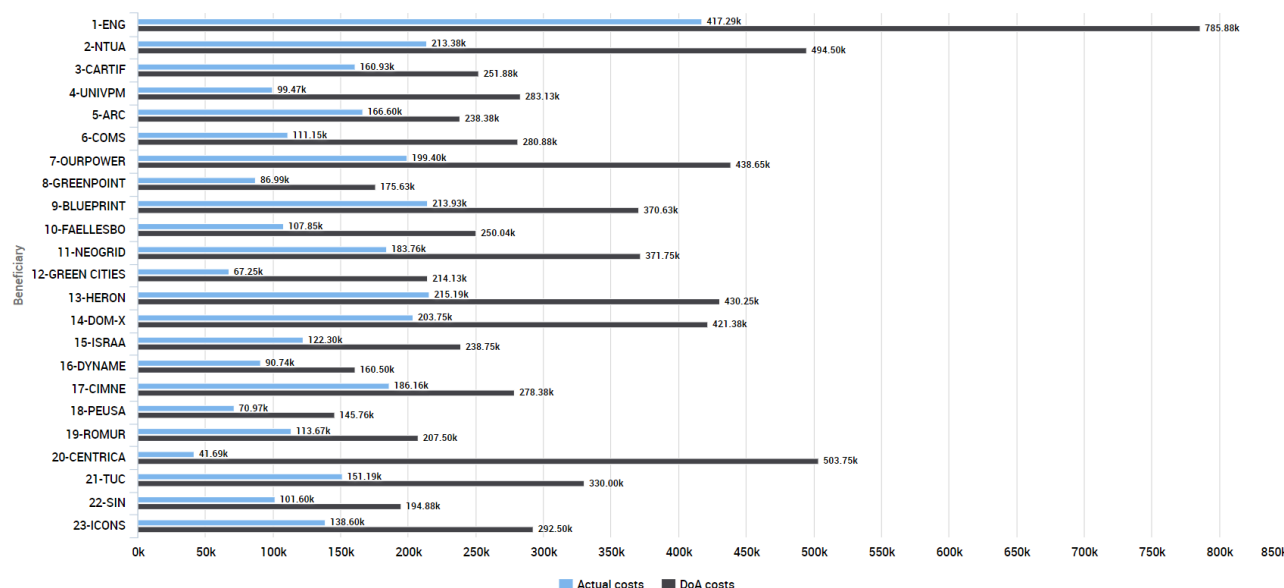


Figure 14 - RP1 Actual vs. Total Planned Costs per Beneficiary

5.1.3. Explanations on deviations

1_ENG

“Other Goods, Works and Services” cost overspending: the overspending related to the other goods, works and services was mainly due to two main reasons: a reduced budget for this cost items established during the proposal preparation phase, since, after the COVID Pandemic, it was quite common to think that no or few events in presence would have been organized; an unplanned but strictly necessary technical meeting. Indeed, because of an initial missing coordination of the project (caused by internal organizational issues), the scope of the project, at month 8 was still under definition. So, a technical meeting was required to fix several issues that could be blocking for the project activities.

Deviation in relation to the weighted average salary foreseen under the GA and the weighted average salary that is claimed under RP1: because of the delays accumulated in the first months of the project, skilled and experienced personnel has been involved to catch up with delays and make the project stick on the plan again. Of course, the cost of this personnel is higher than the average cost computed for the budget of the proposal. The objectives were achieved as planned in the reporting period, even though later than in the initial plan. The critical path was not affected.

Travel cost underspending: it is related to the fact that less employees than planned travelled and some of the meetings were attended remotely during the first reporting period. ENG confirms that the travel cost underspending did not have any negative impact on the project. The budget not spent for travels was partially used to cover the “Other Goods, Works and Services” cost overspending in the first reporting period.



2_NTUA

Travel costs overspending: NTUA has actively participated in a plethora of dissemination and communication events representing the DEDALUS project. More specifically, NTUA has organized a physical plenary meeting in Athens with the attendance of all the partners, it has represented DEDALUS at ENLIT Europe 2024 and at Sustainable Places 2024 demonstrating the early results of the project in front of thousands of people. The aforementioned activities are the reason for purchase costs overspending.

Other Goods, Works and Services: NTUA has published several scientific publications at high-impact, prestigious international conferences and journals. The costs for these publications are the reason for overspending.

Mitigation: NTUA is planning to adjust their budget breakdown and transfer funds from the "Personnel" budget category to the "Travel" budget category, as per Article 5.5 on budget flexibility.

3_CARTIF

Travel costs have been higher than planned and correspond to the actual costs for the attendance to two Project Meetings plus two dissemination events (Sustainable Places 2023 and IISA 2024). CARTIF has already attended online a Plenary Meeting plus a technical meeting to save on travel costs. In any case CARTIF confirms that this will be balanced in the end with other cost lines (personnel, OGWS) if needed to stay within budget. Also, for the remaining parts of the project CARTIF will favour online attendance to meetings whenever possible.

CARTIF can confirm that **Personnel Costs** are in line with the actual planning of efforts for the project and that they will stay within their planned budget implementing all tasks as expected. Note that for CARTIF the WP with higher efforts reported is WP3 -ending in M28- including a higher effort required during RP1 in the task that they are involved in.

Costs reported so far under the **Other Goods, Works and Services** cost category correspond to the conference fees for two conferences (Sustainable Places September 2024 + IISA July 2024). They confirm this is in line with their planning and won't have a negative impact to the project. As also indicated, travel costs have been higher than planned so they expect to eventually balance Purchase Costs as needed to stay within budget.

4_UNIVPM

Travel cost overspending: this is related to the fact that UNIVPM have already participated in several project meetings with 2/3 people of staff in order to better discuss with project partners the research activities in which they are involved, and in conference abroad for project dissemination. In the second reporting period they will participate in other project meetings and in conference in order to disseminate project activities and results. The overspending will be balanced with the underspending in "Other goods, Works and Services" because for the project implementation they will mainly use equipment and materials already available in their laboratories.

Effort overall underspending with a focus on WP4, WP5, WP6 and WP7: The PMs spent are lower than expected because the bulk of the technical development, in particular related to WP4, will take place in the second reporting period, as it was necessary to wait for data from the pilots to be analyzed. In relation to WP7, in which UNIVPM is task leader, also in this case, the majority of the effort will be done in the second reporting period with the development of the Educational Kit and the organization of workshops.



5_ARC

Average personnel rate: when ARC prepared their budget during the proposal stage, their average PM cost was 5K Eur. However, inflation and TL-EUR exchange rate fluctuations experienced during the project caused unforeseen increases in their personnel costs. When TL-EUR conversion is completed with the average exchange rate of the reporting periods, personnel costs are high on a EUR basis. As a result, within the first Periodic report ARC average PM rate is calculated as approx 6.3K Eur.

Travel cost underspending: Although ARC participated in 3 meetings physically (the kick-off meeting in Athen, the first technical meeting in Palermo, the second plenary meeting in Rome), the third plenary meeting was attended online. Because of the EU visa problem, only one person was able to attend, so they couldn't spend their travel costs as they planned.

No "Equipment" cost claimed, compared to the ones foreseen in the DoA: their equipments will be spent in the second reporting period because these integrations are related to pilot implementation and demonstrations.

"Other Goods Works and Services" cost underspending: ARC's other goods works and services costs are related to equipment transfers to the pilots. So this cost will be spent mostly in the second reporting period.

6_COMS

No "Other Goods, Works and Services" costs claimed compared to the ones foreseen in the DoA: no such costs have occurred yet as these costs are associated with the piloting and therefore planned to take place early in the second reporting period.

COMS confirms that two of the SME co-owners involved in the project implementation are participating via standard employment contracts. Therefore, no SME unit costs are claimed.

7_OURPOWER

"Other Goods, Works, and Services" cost underspending: workshops and stakeholder engagement are primarily scheduled for the second reporting period.

8_GREENPOINT

Unforeseen equipment costs claimed: These funds relate to tasks WP2 and WP3, where energy monitoring and electricity consumption measurements were carried out. GREENPOINT confirms that the claimed costs only include depreciation costs. For the second reporting period, GREENPOINT has not planned additional expenses for equipment.

"Other Goods, Works and Services" cost overspending: For the second reporting period, GREENPOINT has not planned any additional expenses for "Other Goods, Works and Services".

Their Travel expenses turned out to be lower because at the second meeting in Milan, only one person from their company was present and the rest of the team was online. Also, at one more meeting, they attended online. However, at the next meetings, they will be present, so they can use the remaining funds. This will not negatively affect the project.

SME receives a salary as normal personnel costs (category A.1). GREENPOINT included the SME owner's salary in the total salary expenses.

9_BLUEPRINT

Travel and Subsistence cost overspending: BLUEPRINT had increased travel effort not only focused on technical workshop and all participation required to support WP2 efforts, for which it



was leader and that were performed in first 18 months, but also coordinated setup of Austrian demo and on site visits. Given that WP2 is completed, and other Austrian partners have taken over technical development, travel will be adjusted and is not expected to exceed forecasted budget.

Other Goods, Works and Services underspending: only one conference was participated, and other works and goods were planned for pilot site activities that started later into the project; further conferences participation for dissemination are planned in second half of the project.

SME unit costs claimed, which were not foreseen in the DoA: Prior to current project manager of Blueprint onboarding the new role in the company and joining the project, Elena Boskov-Kovacs and Ales Hvala – SME co-owners were performing project management and early tasks in WP1 and WP2 as well as WP6 and supported these activities to ensure smooth transition and reduce delays on the project.

10_FAELLESBO

Travel cost underspending: FællesBo could unfortunately not participate in the first consortium meeting. For another consortium meeting in Rome with focus on energy management technical issues FællesBo estimated that FællesBo as pilot project was well represented by Neogrid being the technical partner for the FællesBo pilot project. Therefore, FællesBo did not participate in the Rome meeting. FællesBo expects to participate in future consortium meetings.

Effort underspending on WP5, WP6 and WP7: FællesBo pilot project is carried out in building blocks, which are being renovated prior to installation of measurement and energy management equipment. Installation is being finalized, but the DEDALUS activities have been delayed waiting for finalization of installation.

As far as the **Other Goods, Works, and Services Costs** are concerned, FællesBo has carried out communication activities with tenants as a mini-workshop with the tenants representatives being the board for department 24, hosting the DEDALUS demo-project. The cost for this activity has been lower than anticipated in the budget. There has been no negative impact from this.

11_NEOGRID

Equipment cost overspending: equipment is mainly spent on upgrading the Fællesbo buildings, which have been done in the first part of the project.

No “Other Goods, Works and Services” costs claimed compared to the ones foreseen in the DoA: NEOGRID confirms that this is not a mistake, and these costs are planned for the remaining part of the project.

Effort overall underspending on WP2: NEOGRID has effort related to interaction with users which includes GUI design which for them will be used in task T4.5 in WP4. More effort will be spent on WP2 in the second reporting period, when NEOGRID will have something that should be tested towards the tenants.

Effort underspending on WP3: NEOGRID is participating in the ontology discussions, where they are still discussing the extend of what is required to be implemented. Else they are invited to comment and follow tasks.

Effort underspending on WP5: NEOGRID is a little behind. In the first reporting period they spent time in preparing the buildings and they expect to ramp this up during the next phase of the project.

No effort spent on WP6: NEOGRID is waiting to be involved. The work will mainly be in the last part of the project when evaluating technical performance and therefore working on the business models.



Effort underspending on WP7: not much communication activities have been logged on their main tasks in the first reporting period.

Moreover, NEOGRID confirms that **SME owners** are participating with a normally employment contract.

12_GREEN CITIES

Travel cost underspending: GREEN CITIES has participated in all meetings and events, but some have been on Teams. They also try to keep a low budget, travelling by public transportation (train, bus etc.) if possible.

Other Goods, Works and Services cost underspending: costs for consumables and the organisation of meetings, co-creation process, capacity building workshops and other project events incl. folders, publications, is expected in the second reporting period.

Average personnel rate: GREEN CITIES has unit wages - same for all personnel in EGC. Therefore, GREEN CITIES can budget and report approximately the same amount.

Effort overall underspending, with a focus on WP5 and WP6: GREEN CITIES is responsible for the roadmaps for replication strategies, and the capacity building programme. Furthermore, GREEN CITIES participates in developing the Danish pilot site towards energy efficiency and DR optimal interactions between district heating and electricity networks. These elements are rather late in the timeline of the project.

GREEN CITIES hereby declares that no **SME owner** is working without an employment contract. Furthermore, all employees in EGC are documenting use of hours in time sheets.

13_HERON

Other Goods, Works and Services costs underspending: according to the project progress, the actual project needs for OGS costs are not linear but rather back-loaded, i.e. more costs are expected during the 2nd reporting period when HERON's tasks will be fully initiated. Such costs foreseen concern pilot expansion and further app development to accommodate dynamic tariffs and nudging interventions.

Average personnel rate: HERON confirms that actual costs have been declared (not budgeted ones).

14_DOM-X

Travel cost overspending: this deviation is reported due to the fact that the initial budget allocated was only 6.5k euros. Notably, DOM-X have already performed 4 physical meetings for the project needs so far, where in some cases more than one person from their company attended. Moreover, in the DEDALUS project it is agreed to perform a physical meeting every 6 months, which means that extra budget will be required in this category. They deem necessary to transfer some budget from another category to be able to address the needs for travelling.

Equipment cost underspending: DOM-X confirms that the reported equipment costs correspond only to depreciation costs. According to the DoW, they had planned the budget of € 6,000 to acquire and install a heat-pump for testing the interoperability and demand management capabilities through the DOMX heating controller. However, during the project's execution they managed to get access to relevant testing setup from an end-user of the Greek pilot, thus they refrained from acquiring this initially planned equipment. They are planning to transfer the underspent budget from this category to the travel costs category that experiences overspending.



Unforeseen Internally invoiced goods and services: according to the DoW, they had initially foreseen charging the costs of the DOMX heating controllers and Indoor Air Quality sensors required for the execution of the Greek pilot under the Other Goods, Works and Services category. However, during the project execution and after extensive discussions with the project coordinator, they decided to charge the aforementioned device costs under the "internally invoiced goods and services" category. The decision has been documented and communicated to the PO on 14th August 2024.

Unforeseen SME owners unit cost: DOM-X confirms that according to the DoW no SME owner unit costs were foreseen. However, given the high research expertise of both SME owners of DOMX company, their involvement was deemed necessary for the successful execution of the project. Both SME owners have been involved in the execution of the DEDALUS project since the early stage, while their involvement has increased as the project's development is progressing. More specifically, their extensive research background due to their participation in the NUDGE EU research project was crucial for their contribution in the WP2 and WP4. DOM-X confirms that both SME owners do not receive a salary and are compensated for their work for the SME by other means than a salary.

15_ISRAA

Travel cost overspending: major cost incurred by ISRAA for travel is determined by: - Increased costs caused by inflation at European level; - Participation of ISRAA team in the event: "Fit for the Future - AI for healthy aging - AAA ISO Leaders Forum 03/11/23". The decision to participate in the event was due to the need to learn benchmark experiences related to urban regeneration projects in which technologies for green management of energy production and consumption related to nursing homes residents' engagement were used (SHAPES project).

Equipment cost underspending: the remaining depreciation of equipment purchased in the first part of the project will be included in future reporting.

"Other Goods, Works and Services" cost underspending: this is due to the fact that purchases of this type are planned for the second part of the project.

16_DYNAME

Unforeseen "Other Goods, Works and Services" costs claimed: Dyname is claiming costs for the services of Dr. Ing. Renzo Taffarello into this cost category since the project activities required competences that were only partially available in the Dyname's organizational system, such as Digital Twin, Big Data and Analytics, Use Case Design and digitalization of energy. Dr. Ing. Renzo Taffarello has been identified for the services and competences needed by Dyname in support to all the Dedalus project activities. In general, Dr. Ing. Renzo Taffarello is going to support Dyname on the topics concerning Digital Twin for Consumer- aware flexibility systems and business models, hardware-software solutions and integration in processes, business assessments of models and optimizations options, use of Big Data, approach for building-community scale. Due to the transversal role of Renzo's competencies, Dyname decided to use his expertise with a contract as 'temporary manager for technical-scientific support' for an identified average workload of 1 day/consultant per month (8 hr/day).

The Other Goods, Works, and Services Costs (Category C3) exceeded the amount foreseen in Annex 2 because DYNAME included the services of Dr. Ing. Renzo Taffarello. By the end of the project DYNAME expects to cover the exceeding "Other Goods, Workks and Services" costs using the personnel costs initially foreseen that they reduced accordingly, with no impact on the performance and results of the project.



DYNAME decided to use Dr. Ing. Renzo Taffarello's expertise with a contract as 'temporary manager for technical-scientific support'. DYNAME opted for allocating his costs in the category C3 because of the transversal role of Renzo's competencies. His contribution is not related to the implementation of a specific action task, helping DYNAME in several areas.

Unforeseen SME owner unit costs: The SME owner of Dyname, Adriano Frezza, is fundamental for the project activities in charge to Dyname because his extensive knowledge about the energy markets, his deep working experience in the field of the customer/user acceptance of new innovative services, and his unique capability in the analysis of the services cases upon which to plan the development of future business. Adriano has participated to activities and costs in the first reporting period and he will participate to the activities and costs also in the second reporting period of the project. As a SME owner, Adriano is not an employee of the company and he does not receive a salary.

The **Travel costs** exceeded the amount foreseen in Annex 2 due to the fact that the first general meeting (Athens) was a very important moment for having a detailed overview of the project and 3 persons attended instead of 1, to gain a better understanding of the overall project. DYNAME expects to keep the Travel costs within the amount foreseen in Annex 2 without compromising their contribution.

Their **Average Personnel Costs** per month exceeded the amount foreseen in Annex 2 due to:

- the real costs of Fabrizio (Employee) have been calculated accordingly the updated accounting/work contract up today.
- the SME Owner calculation has been included and it was NOT foreseen at the submission time.

DYNAME expects to keep the overall costs within the amount foreseen in Annex 2 without compromising their contribution to the project.

17_CIMNE

Travel cost overspending: A total of 8 trips have been made by the DEDALUS project research staff members, including participation in the KoM (3 researchers) and 3 project meetings (WP1), meetings for the visit to the PEUSA pilot (WP5) (3 trips for two researchers) and 2 participations in events for dissemination (WP7). The cost of travel has been higher than initially expected due to the general increase in the costs of accommodation (hotels) and flights. It is expected that during the second half of the project (RP2) these costs will be compensated and there will be no major deviations in the final cost of travel.

No "Other Goods, Works and Services" costs claimed compared to the ones foreseen in the DoA: the invoices for these planned services have not been received at the time of closing the period. During the second reporting period CIMNE will receive the first invoice for 3,240 € corresponding to the API development services for the implementation of communications with the centralised battery management company.

The deviation in terms of **Personnel Costs** is due to:

- the increase in public wages in the Spanish Public Administration due to the high inflation rates of recent years both in Spain and Europe in general, which has caused an annual increase in the salaries of public workers compared to what was foreseen during the preparation of the proposal and the budget.
- the workloads during the first months of execution of some WPs, more PMs have been executed during these first months which CIMNE hopes will be adjusted during the second half of the project. CIMNE coordinates the works of the Spanish Pilot that



integrates partners that collaborate with the project, but are not funded partners. One of them is the city hall of La Seu d'Urgell that during the firsts months of the project changed it's Major (there were local elections) and all the technical staff was replaced. CIMNE had to start new communication, re-establish new connections with technical partners, this has required a huge effort to engage the new team with the project and to get from them the necessary cooperation.

These changes, however, do not affect the objectives of the project and will allow the expected results to be achieved. However, it has required to ship the workload and also to put extra efforts not covered by the funding of the project, in order to fulfil the activities of the project.

18_PEUSA

Travel cost underspending: out of the 3 meetings, PEUSA could only attend 2, and only one person. Moreover, on the first one, issues with the hotel reservation website resulted in a free stay for them, as they could not provide them with an invoice.

Personnel rate: PEUSA confirms that the claimed costs are based on actual records.

19_ROMUR

Unforeseen SME owners' unit costs: ROMUR confirms that SME owners are participating in the project. ROMUR is a small company where owners are 100% involved in the project in every task. The claiming of unforeseen SME owners unit costs is due to an initial oversight in the first phase of the project. These costs are also expected to be claimed for the second reporting period.

20_CENTRICA

Travel cost underspending: due to the issues with the Irish pilot, all technical development for this pilot has been halted on Centrica's side. Currently, only one person is actively involved in the project to address the pilot-related issues. In the future, when technical work resumes, more people from Centrica will be involved in the project. This will also mean increased participation in project meetings and technical workshops, requiring additional travel from Centrica's side.

No Equipment cost claimed: due to issues with the Irish pilot, Centrica had to identify a new pilot for the DEDALUS project. The new equipment will be purchased and installed at the new pilot site later in the project.

No "Other Goods, Works and Services" cost claimed: this budget was allocated to cover costs related to publications, pilot events, and participation in conferences. Due to the issues with the Irish pilot and the subsequent halt in technical development, Centrica did not incur any expenses in this category during the first reporting period. This budget is expected to be utilized during the second half of the project.

Deviation in relation to the weighted average salary foreseen under the GA and the weighted average salary that is claimed under RP1: due to the issues with the Irish pilot and the subsequent halt in technical development, CENTRICA did not have to use a high salaried resource to undertake the work. Therefore, there was only one person actively involved in addressing the pilot-related issues. In the future, when technical work resumes, more people higher salaried will be engaged in the project.

Effort overall underspending: due to above-mentioned issues with the Irish pilot, all technical development (WP3, WP4, WP5) for this pilot has been halted on Centrica's side. In the future, when technical work resumes, more people will be engaged in the project.

Moreover, CENTRICA confirms that no **SME owners** were participating in the action.



21_TUC

Travel cost overspending: the reported costs covered 3 consortium plenary meetings for two participants from TUC. This cost is correlated to the increase in flights prices in Europe. The remaining budget for travel will be increased by using the other direct cost budget for travel costs, thus will suffice to cover the travel costs by the end of the project.

No “Other Goods, Works and Services” costs claimed compared to the ones foreseen in the DoA: no purchases for goods or services have been made. Initially these costs were planned for acquiring pilot related infrastructure, but the equipment has been covered from other university sources. The amount for other direct costs will be used for travel costs.

Effort underspending on WP4: the PMs reported cover the work done for the design of the “Apartment building Digital Twin for blockchain integration” in the reporting period. The rest of the development efforts for this component are planned for the 2nd reporting period since it will be evaluated in the final evaluation stage that starts on M26.

Effort underspending on WP5: TUC had allocated 1 PM on Task 5.1 that started on M1, which has been spent 100%. The rest of the PM in WP5 (8 PM) is allocated to Tasks 5.7 and 5.8 that started both on M14. TUC has spent partial effort on Task 5.7 in the 4 months covered by the reporting period for activities related to pre-pilot stage in Pilot 7 (Romania). The rest will be spent in the 2nd reporting period since most of the technology evaluation activities in this pilot are planned for the full and large-scale evaluation stages (M17-M34).

22_SIN

Other Goods, Works and Services cost underspending: the budget will be used in the next stages of the project to assist strategy and evaluation. For example, Doodle for planning workshops and developing strategies with other partners. For evaluating the social acceptance as part of WP5, Alchemer to be used for data collection through questionnaires. Data analysis software such as SPSS may also be required depending on the output from external stakeholder engagement activities.

Effort underspending on WP5: The work in progress as part of WP5 was strongly connected to activities in WP2 which required carrying out workshops and analysing data with all pilots. This was a challenge to process all the data and there were some delays which meant the timeline shifted such that the subsequent planning of the engagement strategies & impact assessment in WP5 began later. SIN expects that work will intensify for WP5 in the second reporting period and compensate for the underspend.

Effort underspending on WP6 and WP7: Because these WPs relate to exploitation and communication, it would be more beneficial to use the allocated PMs when more results and impact assessments are completed in the next stages of the project.

23_ICONS

Other Goods, Works and Services cost underspending: In the 1st reporting period, ICONS focused its effort on the website creation and the related costs for domain acquisition and maintenance. The remaining budget will be mainly used in the following months for the article preparation and delivery. For what concerns the printing costs, the coordinator took the responsibility for the printing of materials used in some international events.



Conclusions

This is the Part B of the official Periodic Technical report to be uploaded as a PDF document on the EC portal together with all the information required by the EU portal at the end of the first official reporting period. It explains the progress towards the achievement of the project objectives and reports the effort and the costs that have been spent in the first 18 months.

DEDALUS is sticking on the plan. A relevant delay affected the first 7 months of the project, but all the delays have been properly managed and fixed.

Four Plenary Project Meetings have been held as well as many audio conferences to discuss about the progress, about technical issues, about the scope of the project. A technical meeting was held in presence to fix blocking issues.

Relevant progresses were achieved in all WPs: a project repository has been set up together with an effective communication tool; procedures for production of documentation (presentations, reports, minutes) have been defined, periodic audio conferences, 4 plenary meetings and 1 technical meeting have been held, the Data Management Plan was submitted at the end of month 10. An intermediate progress report, namely D1.3 DEDALUS first activity report, released as per GA. The transdisciplinary advisory board was set up,

Moreover, a preliminary analysis of use cases was done for each pilot site using a bottom-up approach, identifying high-level requirements. 11 High-Level Use Cases (HLUCs) have been defined, and a detailed gap analysis of individual pilot sites was performed. Co-creation workshops and bilateral calls were held with pilot site partners and technical partners to ensure effective input from end-users. A set of technology enablers have been developed that will facilitate the creation of the value-added services within WP4. As a result, the first technology release of those technology enablers was delivered by the end of May 2024. The second version is under development.

An analysis of Technologies Enablers was done with the aim to understand how the current version of available software can be extended in the context of the Project according to the functional specification coming from the WP2. The first wave of DEDALUS Digital Twins and data-driven services for residential DR were released. The second wave of Digital Twin and data-driven services development is ongoing.

The foundation for demonstration activities has been set up. This included evaluation of pilot infrastructure readiness for proper demonstration, understanding of the available datasets and the specific requirements of each pilot. A centralized data repository was created to consolidate sample datasets and documentation for API access. The evaluation methodology and baseline analysis were developed to assess the technological feasibility, user engagement, economic viability, and environmental impact. KPIs were defined.

The pre-pilot phase provided actionable insights, highlighting areas for improvement and aligning technologies with the specific needs of each pilot. Social engagement efforts also advanced significantly, informed by earlier engagement activities, such as questionnaire surveys and co-creation workshops, which shaped the preliminary design of technical workshops.

Detailed updates on infrastructure enhancements were provided as well as early evaluations of deployed TEs, and the progress of KPIs.



The concept of DEDALUS Business Cases was introduced: based on the HLUCs, BCs were linked with Technological Innovations (TEs) identifying the technologies that are required for each BC to materialise.

Social media channels (LinkedIn and X/Twitter) have been launched. The visual identity and the project's landing page (www.dedalus-horizon.eu) have been launched, followed by the first version of the website. The communication kit (roll-up poster; leaflet; presentation video) was created. DEDALUS entered the Smart Energy Cluster. As well as the BRIDGE initiative.

The project participated to ENLIT with a stand shared with other 6 EU-funded projects within the Smart Energy Cluster.

At the time of writing, the project is sticking on the plan.

